RESPIRATORY DISORDERS

ATELECTASIS

Collapse or airless condition of the alveoli caused by hypoventilation, obstruction of airway or compression

- This may results from:
  - Altered breathing patterns
  - Retained secretions, pain
  - Alterations in small airway function
  - Prolonged supine positioning
  - Increased abdominal pressure
  - Reduced lung volumes due to musculoskeletal or neurologic disorders, restrictive defects, and specific surgical procedures (e.g, upper abdominal, thoracic, or open heart surgery)

ATELECTASIS

- **Clinical Manifestations**
  - Cough & sputum production
  - Dyspnea, tachypnea, tachycardia
  - Signs of pulmonary infection may be present
  - Fever
  - Central cyanosis

- **Preventing Atelectasis**
  - Change patient's position frequently, especially from supine to upright position, to promote ventilation and prevent secretions from accumulating.
  - Encourage early mobilization from bed to chair followed by early ambulation.
  - Encourage appropriate deep breathing and coughing to mobilize secretions.
  - Teach/reinforce incentive spirometry.
  - Administer prescribed opioids and sedatives judiciously to prevent respiratory depression.
  - Perform postural drainage and chest percussion, if indicated.
  - Institute suctioning to remove tracheobronchial secretions, if indicated

- **Management**

  First line measures: (turning, early ambulation, coughing, incentive spirometry, breathing exercises)

  If there is no response: (PEEP)

  Bronchoscopy

  Postural Drainage & percussion

  If the cause is compression remove the cause

ACUTE BRONCHITIS

- Acute bronchitis is an infection of the lower respiratory tract that is generally occurs after upper respiratory tract infection.

Pathophysiology and Etiology

- Primarily viral etiology, but may also arise from bacterial agents.
- Airways become inflamed and irritated with increased mucus production.

Clinical Manifestations

- Dyspnea, fever, tachypnea.
- Productive cough, clear to purulent sputum.
- Chest pain, occasionally.
- Diffuse rhonchi and crackles heard on

Diagnostic Evaluation

- Chest X-ray—no evidence of infiltrates or consolidation.

- Sputum for Gram stain, culture, and sensitivity tests.

Management

- Antibiotic therapy for 7 to 10 days may be indicated for patients with bacterial infection.
- Hydration and humidification.
- Bronchodilators for bronchospasm.
- Symptoms management for fever, cough.

Nursing Assessment

- Obtain history of upper airway infection.
- Assess severity of cough and characteristics of sputum production.
- Auscultate chest for diffuse rhonchi and crackles.

Nursing Diagnosis

- Ineffective Airway Clearance related to sputum production.

Nursing Interventions

Establishing Effective Airway Clearance

- Encourage mobilization of secretions through patient mobilization and, possibly, hydration, chest physical therapy, and coughing.
- Bronchodilators to reduce bronchospasm and enhance secretion clearance.
- Caution patients on the use cough suppressants, antihistamines, and decongestants that may cause drying and retention of secretions.
- Cough preparations containing mucolytics may be appropriate.

PNEUMONIA

- Is an inflammatory process, involving the terminal airways and alveoli of the lung, caused by infectious agents.

Pathophysiology and Etiology

- The organism gains access to the lungs through aspiration of oropharyngeal contents, by inhalation of respiratory secretions from infected individuals, by way of the bloodstream, or from direct spread to the lungs as a result of surgery or trauma.

Clinical Manifestations

For most common forms of bacterial pneumonia:

- Sudden onset; shaking chill; rapidly rising fever of (38.3° C to 40.5° C).

- Cough productive of purulent sputum.

- Pleuritic chest pain aggravated by respiration/coughing.

- Dyspnea, tachypnea accompanied by respiratory grunting, nasal flaring, use of accessory muscles of respiration, fatigue.

- Rapid pulse.

Diagnostic Evaluation

- Chest X-ray shows presence/extent of pulmonary disease, typically consolidation.
- Culture and sensitivity tests of sputum may indicate the causative organism.
- Blood culture detects bacteremia.

Management

- Antimicrobial therapy depends on laboratory identification of causative organism and sensitivity to specific antimicrobials.
- Oxygen therapy if patient has inadequate gas exchange.

Complications

- Pleural effusion.
- Sustained hypotension and shock, especially in gram-negative bacterial disease, particularly in elderly patients.
- Superinfection: pericarditis, bacteremia, and meningitis.
- Delirium this is considered a medical emergency.
- Atelectasis due to mucous plugs.

Nursing Assessment

- Take a careful history to help establish etiologic diagnosis.

- Observe for anxious, flushed appearance, shallow respirations, confusion, disorientation.

- Auscultate for crackles overlying affected region.

Nursing Diagnoses

- Impaired Gas Exchange related to decreased ventilation secondary to inflammation and infection involving distal airspaces.
- Ineffective Airway Clearance related to excessive tracheobronchial secretions.
- Acute Pain related to inflammatory process and dyspnea.
- Risk for Injury secondary to complications.

Nursing Interventions

- Improving Gas Exchange (ABG, oxygen, upright position).
- Enhancing Airway Clearance (sputum specimen, encourage cough, fluid intake, humidify air).
- Relieving Pleuritic Pain (semi fowler's position, analgesic, splint the chest while coughing).
- Monitoring for Complications (VS, SaO2, Auscultate lungs and heart, Assess for resistant fever).

PULMONARY EMBOLISM

Definition

- obstruction of one or more pulmonary arteries by a thrombus (or thrombi) originating usually in the deep veins of the legs, the right side of the heart or, rarely, an upper extremity.
- Pulmonary infarction: refers to necrosis of lung tissue that can result from interference with blood supply.

Pathophysiology and Etiology

Obstruction, either partial or full, of pulmonary arteries, which causes decrease or absent blood flow; therefore, there is ventilation but no perfusion.

Pathophysiology and Etiology Cont. ….

- Hemodynamic consequences:
  - Increased pulmonary artery pressure (PAP)
  - Increased right ventricular workload to maintain pulmonary blood flow
  - Right ventricular failure
  - Decreased cardiac output
  - Decreased blood pressure
  - Shock
- Pulmonary emboli can vary in size and seriousness of consequences.

Predisposing factors include:

- Stasis, prolonged immobilization.
- Concurrent phlebitis.
- Previous heart disease (heart failure, myocardial infarction), or lung disease.
- Injury to vessel wall.
- Coagulation disorders.
- Metabolic, endocrine, vascular, or collagen disorders.
- Malignancy.
- Advancing age,
- Estrogen therapy.
- Obesity.

NURSING ALERT

- Be aware of high-risk patients for pulmonary embolism; immobilization, trauma to pelvis and lower extremities (especially hip fracture), obesity, history of thromboembolic disease, varicose veins, pregnancy, heart failure, MI, malignant disease, postoperative patients, elderly patients.

Clinical Manifestations

- Dyspnea, tachypnea, apprehension.
- Chest pain.
- Cyanosis.
- Pleural friction rub.

Diagnostic Evaluation

- ABG levels; decreased Pao2 is usually found, due to perfusion abnormality of the lung.
- Chest X-ray; normal or possible wedge-shaped infiltrate.
- Pulmonary angiogram (most definitive) emboli seen as filling defects.

Management

Emergency Management

- For massive pulmonary embolism, goal is to stabilize cardiorespiratory status.
- Oxygen, I.V. route for drugs and fluids.
- ECG is monitored continuously for right-sided heart failure.
- Morphine.
- Pulmonary angiography, ABG analysis.

Subsequent Management Anticoagulation and Thrombolysis

- **Anticoagulants *(e.g. Heparin, coumadin)*, Thrombolytics *(e.g. Streptokinase)*.**

Surgical Intervention

Complications

BLEEDING AS A RESULT OF TREATMENT.

RESPIRATORY FAILURE.

Nursing Assessment

- Examine the patient's legs carefully, Assess for swelling of leg.
- Monitor respiratory rate and rhythm.
- Monitor cardiac rate and rhythm.
- Monitor ECG and ABG levels.
- Monitor B.P. and U.O.P.
- Assess for pain and anxiety.
- Evaluate results of PT/PTT tests.

Nursing Diagnoses

- Ineffective Breathing Pattern related to acute increase in alveolar dead airspace and possible changes in lung mechanics from embolism.
- Ineffective Tissue Perfusion (Pulmonary) related to decreased blood circulation.
- Acute Pain (pleuritic) related to congestion, possible pleural effusion, possible lung infarction.
- Anxiety related to dyspnea, pain, and seriousness of condition.
- Risk for Injury related to altered hemodynamic factors and anticoagulant therapy.

Nursing Interventions

- Correcting Breathing Pattern
- Improving Tissue Perfusion
- Relieving Pain
- Reducing Anxiety
- Intervening for Complications

Evaluation: Expected Outcomes

- Verbalizes less shortness of breath
- Vital signs stable, adequate urinary output
- Reports freedom from pain
- Appears more relaxed; sleeping at long intervals
- Progresses without complications

CHRONIC OBSTRUCTIVE PULMONARY DISEASE

CHRONIC OBSTRUCTIVE PULMONARY DISEASE

- **COPD**: Refers to a disease characterized by airflow limitation that is not fully reversible.
- The airflow limitation is generally progressive and is normally associated with an inflammatory response of the lungs due to irritants.
- COPD includes chronic bronchitis and pulmonary emphysema.
- Some clinicians consider asthma as part of COPD, but due to its reversibility, it is considered by most to be a separate entity.

CHRONIC OBSTRUCTIVE PULMONARY DISEASE

Chronic bronchitis

- Is a chronic inflammation of the lower respiratory tract characterized by excessive mucous secretion, cough, and dyspnea associated with recurring infections of the lower respiratory tract.

Pulmonary emphysema

- Is a complex lung disease characterized by destruction of the alveoli, enlargement of distal airspaces, and a breakdown of alveolar walls.

Pathophysiology and Etiology

The Pathophysiology of COPD includes:

- Excessive secretion of mucus and chronic infection within the airways (bronchitis).
- Increase in size of airspaces distal to the terminal bronchioles, with loss of alveolar walls and elastic recoil of the lungs (emphysema).

The etiology of COPD includes:

- Cigarette smoking.
- Air pollution, occupational exposure.
- Allergy, autoimmunity.
- Infection.
- Genetic predisposition.
- Aging.

Clinical Manifestations

Chronic Bronchitis

- Usually insidious.
- productive cough lasting at least 3 months a year for 2 successive years.
- Production of thick, gelatinous sputum.
- Wheezing and dyspnea.

Emphysema

- Gradual in onset.
- Dyspnea, decreased exercise tolerance.
- Cough may be minimal.
- Sputum expectoration mild.
- Increased anteroposterior diameter of chest (barrel chest) due to air trapping with diaphragmatic flattening.

Diagnostic Evaluation

- PFTs demonstrate airflow obstruction reduced forced vital capacity (FVC), increased residual volume to total lung capacity (TLC) ratio.
- ABG levels decreased PaO2, pH, and increased CO2.
- Chest X-ray in late stages; hyperinflation, flattened diaphragm

Management

- Smoking cessation
- Inhaled bronchodilators
- Methylxanthines, such as theophylline
- Corticosteroids
- Chest physical therapy
- Supplemental oxygen therapy
- Antimicrobial agents

Complications

- Respiratory failure
- Pneumonia
- Right-sided heart failure, (Corpulmonale)
- Cardiac dysrhythmias
- Skeletal muscle dysfunction

Bronchial Asthma

- Asthma is a chronic inflammatory disease of the airways that causes airway hyperresponsiveness, mucosal edema, and mucus production.
- Asthma differs from COPD in that it is reversible process either spontaneously or with treatment.
- Allergy is the strongest predisposing factor for the development of asthma.

Clinical Manifestations

- The three most common symptoms of asthma are cough, dyspnea, and wheezing.

Prevention

- Avoid dust, dust mites, roaches, certain types of cloth, pets, horses, detergents, soaps, certain foods, molds, and pollens.

Complications

- Status asthmaticus

(Severe and persistent asthma that does not respond to conventional therapy)

- Respiratory failure
- Pneumonia
- Atelectasis
- Airway obstruction

Medical Management

- Corticosteroids: most effective antiinflammatory medication (inhaled form):
  - Pharmacologic Therapy: Methylprednisolone, Prednisolone, Prednisone (long term).
  - Theophylline is a mild to moderate bronchodilator that is usually used in addition to inhaled corticosteroids, mainly for relief of nighttime asthma symptoms.
  - Salmeterol is a long-acting beta2-adrenergic receptor agonist drug that is prescribed for the treatment of asthma and chronic obstructive pulmonary disease.

Nursing Assessment

- Determine smoking history, exposure history, positive family history of respiratory disease, onset of dyspnea.
- Note amount, color, and consistency of sputum.
- Inspect for use of accessory muscles of respiration and use of abdominal muscles during expiration; note increase of anteroposterior diameter of chest.
- Auscultate for decreased/absent breath sounds, crackles, decreased heart sounds.
- Determine oxygen saturation at rest and with activity.

Nursing Diagnoses

- Ineffective Airway Clearance related to bronchoconstriction, increased mucus.
- Ineffective Breathing Pattern related to chronic airflow limitation
- Risk for Infection related to retained secretions.
- Impaired Gas Exchange related to chronic pulmonary obstruction
- Imbalanced Nutrition: Less Than Body Requirements related to increased work of breathing, air swallowing, drug effects with resultant wasting of respiratory and skeletal muscles.
- Activity Intolerance related to compromised pulmonary function.
- Disturbed Sleep Pattern related to hypoxemia and hypercapnia.
- Ineffective Coping related to the stress of living with chronic disease, loss of independence.

Nursing intervention

Improving Airway Clearance

- Eliminate pulmonary irritants, particularly cigarette smoking.
- Administer bronchodilators to control bronchospasm and dyspnea and assist with raising sputum.
- Use postural drainage positions to aid in clearance of secretions.
- Use controlled coughing.
- Keep secretions liquid - high level of fluid intake.
- Avoid dairy products if these increase sputum production.

Nursing intervention

Improving Breathing Pattern

- Breathing retraining exercises to strengthen. diaphragm and muscles of expiration to decrease work of breathing.
- Use pursed-lip breathing at intervals.
- Discuss and demonstrate relaxation exercises to reduce stress, tension, and anxiety.

- **Controlling Infection:** Recognize early manifestations of respiratory infection.
- **Improving Gas Exchange:** ABG, O2 saturation.
- **Improving Nutrition:** Frequent small meals, high-calorie, high-protein snacks; such as cheese & nuts, offer liquid nutritional supplements
- **Increasing Activity Tolerance:** Walking, stationary bicycling, swimming.
- **Improving Sleep Patterns:** A balanced schedule of activity and rest, use nocturnal oxygen therapy, avoid use of sedatives that may cause respiratory depression.

- **Enhancing Coping**

- Understand that the constant shortness of breath and fatigue make the patient irritable, apprehensive, anxious, and depressed, with feelings of helplessness and hopelessness.
- Assess the patient for reactive behaviors (anger, depression, acceptance).
- Demonstrate a positive and interested approach to the patient.
  - Be a good listener and show that you care.
  - Be sensitive to patient's fears, anxiety,
  - Allow the patient to express feelings.
  - Strengthen the patient's self-image.

Controlling Symptoms

To control cough and phlegm

- Drink plenty of water (8 to 10 glasses per day) to keep phlegm (sputum) thin.
- Use inhalers on a regular basis as prescribed:
  - Bronchodilators, Corticosteroids to reduced swelling.
- Avoid irritants to the lungs, such as cigarette smoke, dust, smog, perfumes, cold air, and very hot air.
- Report change in color, amount, or thickness of phlegm that could indicate an infection.
- Try to avoid respiratory infections by limiting contact with people during cold and flu season, and wash hands frequently.
- Get the influenza and pneumonia vaccines.

To control shortness of breath

- Practice pursed-lip breathing by breathing in through the nose and out through pursed lips (like you are whistling), with a long slow expiration.
- Position yourself for better breathing by leaning forward while sitting with elbows on table or resting on knees.
- Use relaxation techniques, such as listening to soft music, imagining you are in a quiet peaceful place.
- If prescribed, use oxygen as directed, especially while performing such activities as bathing, dressing, eating, and walking.

To control fatigue

- Do not stop doing physical activity; instead, learn how to manage by planning activities to conserve energy.
- Start with an exercise program that is easy, and progress slowly to increase your activity.
- Talk to your health care provider about joining a pulmonary rehabilitation program.
- Eat a well-balanced diet.
- Sleep with head elevated using several pillows or in a reclining chair to reduce shortness of breath and increase rest.
- If awakened by cough, sit up, sip fluid, and use inhaler to try to clear lungs of phlegm.
- Avoid overuse of inhalers, which may cause shakiness and insomnia.

NURSING ALERT

- Normally, CO_2 levels in the blood provide a stimulus for respiration. However, in patients with COPD, chronically elevated CO_2 impairs this mechanism and low oxygen levels act as stimulus for respiration. Giving a high concentration of supplemental oxygen to people who retain CO_2 may suppress the hypoxic drive, leading to worsening hypoventilation, respiratory decompensation, and the development of a worsening respiratory acidosis.

DRUG ALERT

- Instruct patient in proper sequence of medications, using bronchodilator first, followed by inhaled corticosteroid.
  Instruct patient to use a spacer device with inhaled corticosteroids, and rinse and spit after using inhaled corticosteroid to prevent oral candidiasis.

Bronchiectasis

- Is a chronic dilatation of the bronchi and bronchioles due to inflammation and destruction of their walls.

Pathophysiology and Etiology

- Damage to the bronchial wall leads to thick sputum, causing obstruction.
- Usually involves the lower lobes.

Clinical Manifestations

- Persistent cough with production of copious amounts of purulent sputum.
- Intermittent hemoptysis.
- Recurrent fever.
- Crackles and rhonchi heard over involved lobes
- Finger clubbing.

Diagnostic Evaluation

- Chest X-ray may reveal areas of atelectasis with widespread dilatation of bronchi.
- Sputum examination may detect offending pathogens.
- High-resolution CT scan is useful in diagnosis of bronchiectasis.

Management

- Infection controlled by:
  - Smoking cessation.
  - Prompt antimicrobial treatment.
  - Immunization against potential pulmonary pathogens (influenza and pneumococcal vaccine).
- Secretion clearance techniques, such as postural drainage and postural drainage.
- Bronchodilators.
- Surgical resection (segmental resection) when conservative management fails.

Complications

- Progressive suppuration (production and discharge of pus).
- Hemoptysis, major pulmonary hemorrhage.
- COPD, emphysema, chronic respiratory insufficiency.

Nursing Assessment

- Obtain history regarding amount and characteristics of sputum produced, including hemoptysis.
- Auscultate lungs for diffuse rhonchi (snoring) and crackles.

Nursing Diagnosis

Ineffective Airway Clearance related to tenacious (not easily loosened) and copious secretions.

Nursing Interventions

Maintaining Airway Clearance

- Chest physical therapy & postural drainage.
- Encourage increased intake of fluids to reduce viscosity of sputum and make expectoration easier.
- Consider vaporizer to provide humidification and keep secretions thin.

TUBERCULOSIS

TUBERCULOSIS

- **TB:** is an infectious disease caused by bacteria (*Mycobacterium tuberculosis*) that are usually spread from person to person through the air.
- It usually infects the lung but can occur at virtually any site in the body.
- HIV-infected patients are especially at risk.

Transmission

- The term Mycobacterium is descriptive of the organism, which is a bacterium that resembles a fungus. The organisms multiply at varying rates and are characterized as acid-fast aerobic organisms that can be killed by heat, sunshine, drying, and ultraviolet light.
- TB is an airborne disease transmitted by droplet nuclei, usually from within the respiratory tract of an infected person who exhales them during coughing, talking, sneezing, or singing.

Clinical Manifestations

- Patient may be asymptomatic or may have insidious symptoms that may be ignored.
- Fatigue, anorexia, weight loss, low-grade fever, night sweats, Some patients have acute febrile illness, chills, and flu-like symptoms.
- Cough progressing in frequency and producing mucoid or mucopurulent sputum.
- Hemoptysis; chest pain; dyspnea.
- Extrapulmonary TB: pain, inflammation, and dysfunction in any of the tissues infected.

Diagnostic Evaluation

- Sputum smear detection of acid-fast bacilli in stained smears is the first bacteriologic clue of TB. Obtain first morning sputum on 3 consecutive days.
- Sputum culture a positive culture for *M. tuberculosis* confirms a diagnosis of TB.
- Chest X-ray to determine presence and extent of disease.
- Tuberculin skin test (purified protein derivative [PPD] or (Mantoux test) inoculation of tubercle

Tuberculin skin test

TABLE 8.5 Classification of the Tuberculin Skin Test Reaction



5 or more millimeters

An induration of **5 or more millimeters** is considered positive in

- HIV-infected persons
- A recent contact of a person with TB disease
- Persons with fibrotic changes on chest radiograph consistent with prior TB
- Patients with organ transplants
- Persons who are immunosuppressed for other reasons (e.g., taking the equivalent of >15 mg/day of prednisone for 1 month or longer, taking TNF-alpha antagonists)



10 or more millimeters

An induration of **10 or more millimeters** is considered positive in

- Recent immigrants (<5 years) from high-prevalence countries
- Injection drug users
- Residents and employees of high-risk congregate settings
- Mycobacteriology laboratory personnel
- Persons with clinical conditions that place them at high risk
- Children <4 years of age
- Infants, children, and adolescents exposed to adults in high-risk categories



15 or more millimeters

An induration of **15 or more millimeters** is considered positive in any person, including persons with no known risk factors for TB. However, targeted skin testing programs should only be conducted among high-risk groups

Retrieved April 15, 2008 from http://www.cdc.gov/tb.

Management

- A combination of drugs to which the organisms are susceptible is given to destroy viable bacilli as rapidly as possible and to protect against the emergence of drug-resistant organisms.

- Current recommended regimen of uncomplicated, previously untreated pulmonary TB is an initial phase of 2 months of bactericidal drugs, including isoniazid (INH), rifampin (Rifadin), pyrazinamide (PZA), and ethambutol (EMB).

TABLE 11-2 Recommended Drugs for the Initial Treatment of Tuberculosis in Adults

DRUG	DOSAGE FORMS	DAILY DOSE	TWICE WEEKLY DOSE	THRICE WEEKLY DOSE	MAJOR ADVERSE REACTIONS
Isoniazid (INH)	▪ Tablets: 50 mg, 100 mg, 300 mg ▪ Syrup: 50 mg/5 mL ▪ Vials: 1 g	5 mg/kg PO or I.M. (maximum 300 mg)	15 mg/kg Maximum 900 mg (can also be given once per week)	15 mg/kg Maximum 900 mg	Hepatic enzyme elevation, peripheral neuropathy, hepatitis, hypersensitivity
Rifampin (Rifadin)	▪ Capsules: 150 mg, 300 mg ▪ Syrup: formulated from capsules, 10 mg/mL	10 mg/kg PO (maximum 600 mg)	10 mg/kg Maximum 600 mg	10 mg/kg Maximum 600 mg	Orange discoloration of secretions and urine; nausea, vomiting, anorexia, hepatitis, febrile reaction, purpura (rare), pruritus and rash
Pyrazinamide (PZA)	▪ Tablets: 500 mg	18.2–26.3 mg/kg PO	36–44 mg/kg	27–39 mg/kg Maximum 3 g	Hepatotoxicity, hyperuricemia, arthralgias, skin rash, GI upset
Ethambutol (Myambutol)	▪ Tablets: 100 mg, 400 mg	14.5–21.1 mg/kg	36–44 mg/kg	21–31 mg/kg	Optic neuritis (decreased red-green color discrimination, decreased visual acuity), skin rash

Complications

- Pleural effusion.
- TB pneumonia.
- Other organs involvement with TB.
- Serious reactions to drug therapy.

Nursing Assessment

- History.
- Assess for symptoms of active disease.
- Auscultate lungs for crackles.
- During drug therapy, assess for liver dysfunction.
  - Question the patient about loss of appetite, fatigue, joint pain, fever, tenderness in liver region, clay-colored stools, and dark urine.
  - Monitor for fever, right upper quadrant abdominal tenderness, nausea, vomiting, rash, and persistent paresthesia of hands and feet.
- Monitor results of periodic liver function studies.

Nursing Diagnoses

- Ineffective Breathing Pattern related to pulmonary infection.
- Risk for Infection related to nature of the disease and patient's symptoms.
- Imbalanced Nutrition: Less Than Body Requirements related to poor appetite, fatigue, and productive cough.
- Noncompliance related to lack of motivation and long-term treatment.

Nursing Interventions

- Improving Breathing Pattern
- Preventing Transmission of Infection
- Improving Nutritional Status
- Improving Compliance

Cancer of The Lung

- Bronchogenic cancer refers to an epithelial cancer, which originates in the bronchial surface epithelium or bronchial mucus glands, arising within the wall or epithelial lining of the bronchus.

- The lung is also a common site of metastasis by way of venous circulation or lymphatic spread.

Predisposing Factors

- Cigarette smoking—amount, frequency, and duration of smoking have positive relationship to cancer of the lung.
- Occupational exposure to asbestos, polyciliac aromatic hydrocarbons (from incomplete combustion of carbon-based fuels, such as wood, coal, diesel, fat, tobacco, incense, or tar), indoor radon, arsenic, chromium, nickel, iron, radioactive substances, isopropyl oil, petroleum oil mists (fog) alone or in combination with tobacco smoke.

Clinical Manifestations

- Usually occur late
- Cough
- Dyspnea, wheezing (suggests partial bronchial obstruction)
- Excessive sputum production, repeated URTI
- Hemoptysis

Clinical Manifestations

- Malaise, fever, weight loss, fatigue, anorexia.
- Symptoms of metastasis—bone pain; abdominal discomfort, nausea and vomiting from liver involvement; pancytopenia from bone marrow involvement; headache from CNS metastasis.
- Usual sites of metastasis—lymph nodes, bones, liver.

Diagnostic Evaluation

- CT scan of upper chest and abdomen.
- Cytologic examination of sputum/chest fluids for malignant cells.
- Fiber-optic bronchoscopy for observation of location and extent of tumor; for biopsy.
- Lymph node biopsy.
- Pulmonary function tests (PFTs).
- Laboratory testing, including complete blood count, calcium level, liver function tests.

Management

- Treatment depends on the cell type, stage of disease, and the physiologic status of the patient.
- It includes a multidisciplinary approach that may be used separately or in combination, including:
  - Surgical resection.
  - Radiation therapy.
  - Chemotherapy.
  - Immunotherapy.

Gastrointestinal Disorders

Gastrointestinal Assessment

COMMON LABORATORY PROCEDURES

FECALYSIS

- Examination of stool consistency, color and the presence of occult blood.
- Special tests for fat, nitrogen, parasites, ova, pathogens and others.

COMMON LABORATORY PROCEDURES

FECALYSIS: Occult Blood Testing

- It is a screening test for colonic cancer.
- Instruct the patient to adhere to a 3-day meatless diet.
- No intake of NSAIDS, aspirin and anti-coagulant.
- Diet should have a high-fiber content.
- Avoid enemas or laxatives before the stool specimen collection.

COMMON LABORATORY PROCEDURES

Upper GIT study: barium swallow

- Examines the upper GI tract
- Barium sulfate is usually used as contrast

COMMON LABORATORY PROCEDURES

Upper GIT study: barium swallow

- Pre-test: Explain procedure to patient, low-residue diet for 2 to 3 days before test, NPO post-midnight.
- Post-test: Laxative is ordered, increase pt fluid intake, instruct that stools will turn white, monitor for obstruction.

The passage of the barium through the esophagus, stomach, and small intestine is monitored



Barium is ingested



ADAM.

COMMON LABORATORY PROCEDURES

Lower GIT study: barium enema

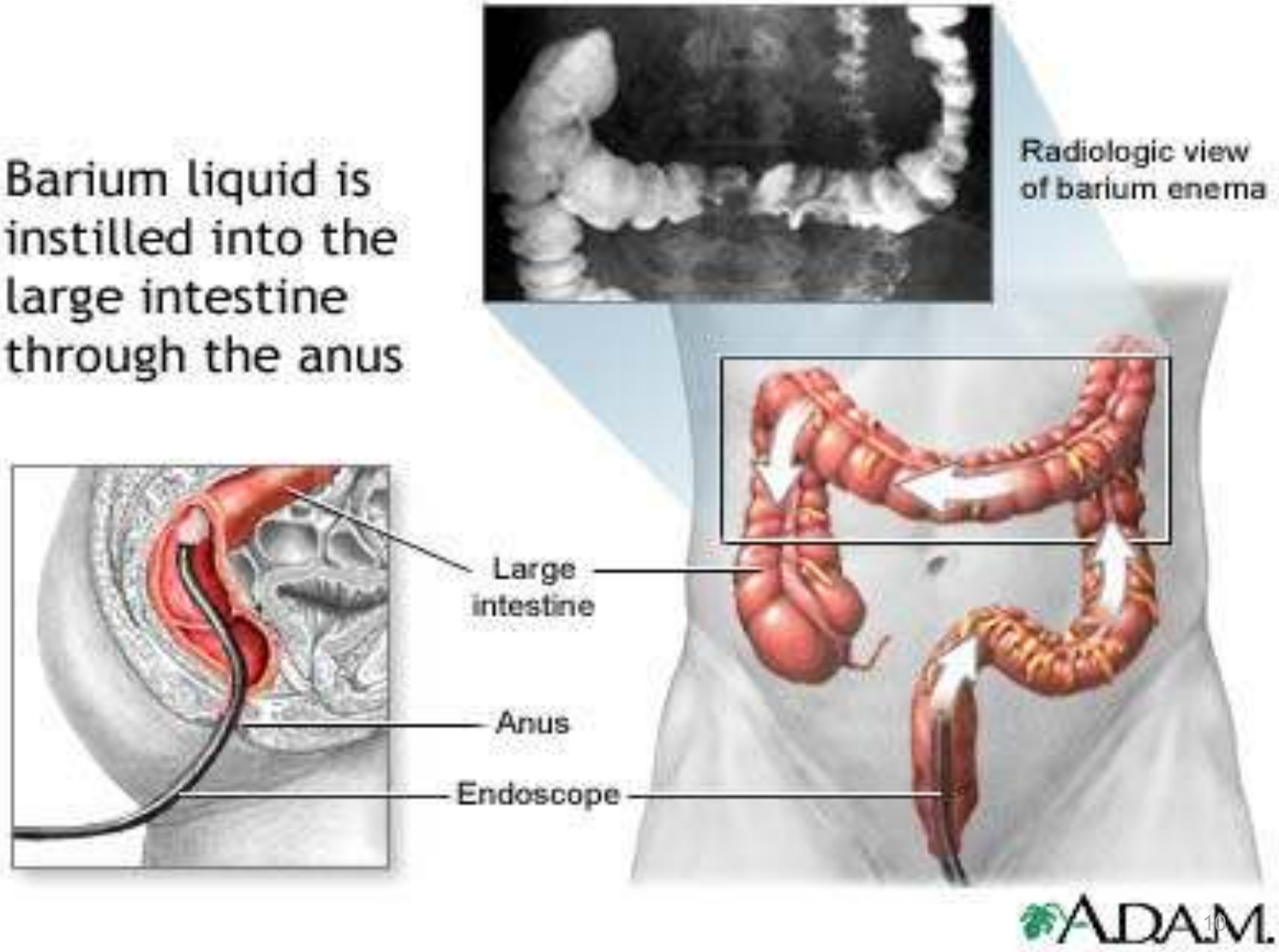
- Examines the lower GI tract, can visualize structural changes, such as tumors, polyps, diverticula, fistulas, obstructions, and ulcerative colitis.
- Pre-test: Clear liquid diet and laxatives, NPO post-midnight, cleansing enema prior to the test.

COMMON LABORATORY PROCEDURES

- Post-test: Laxative is ordered, increase patient fluid intake, instruct that stools will turn white, monitor for obstruction.
- NURSING ALERT

If barium enema and upper GI series are both ordered, the upper GI series is done last so barium traveling down the digestive tract does not interfere with the results of the barium enema.

Barium liquid is instilled into the large intestine through the anus



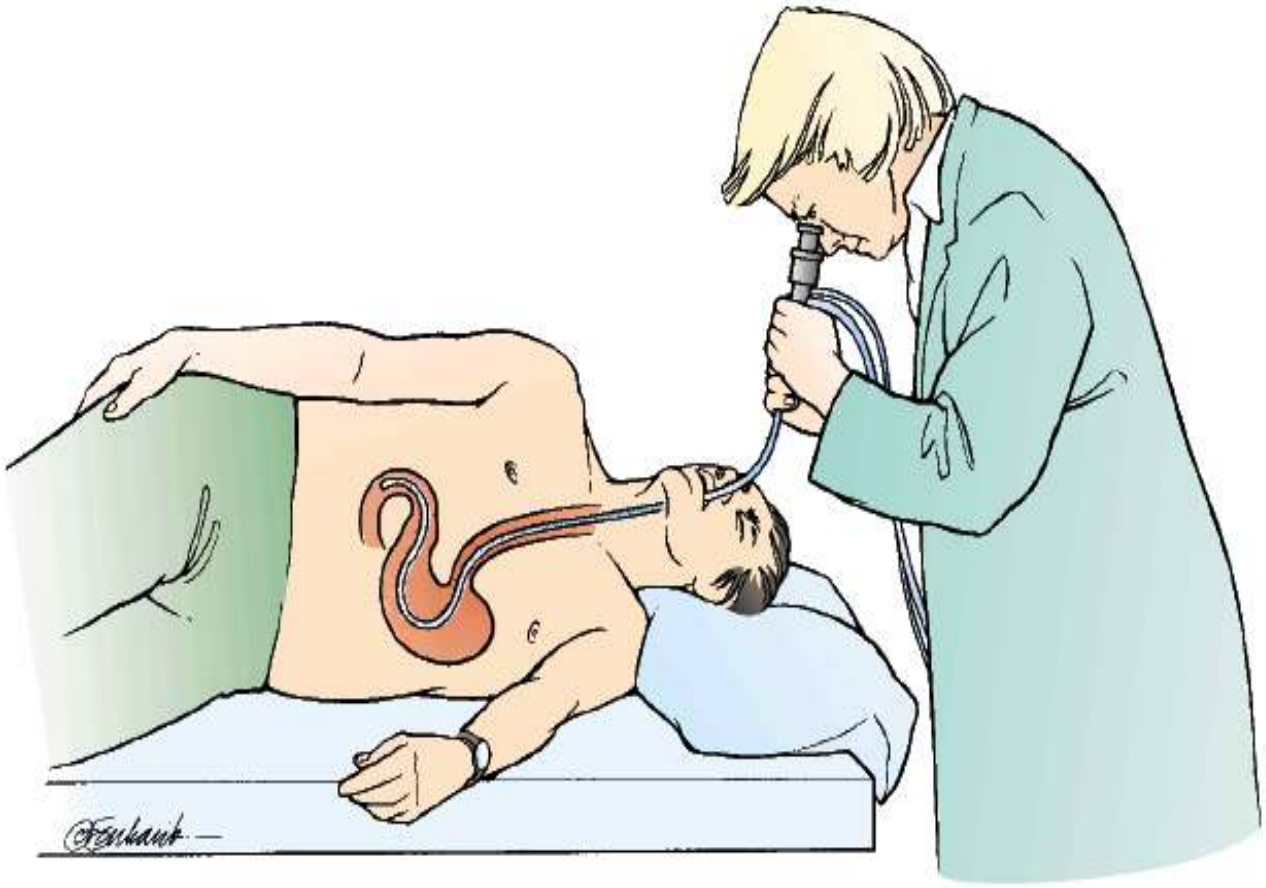
Endoscopic Procedures

- Endoscopy is the use of a flexible tube (the fiber-optic endoscope) to visualize the GI tract and to perform certain diagnostic and therapeutic procedures.
- Endoscopic functions other than visualization include biopsy or cytology of lesions, removal of foreign objects or polyps, control of internal bleeding, and opening of strictures.

COMMON LABORATORY PROCEDURES

EGD (esophagogastroduodenoscopy)

- Visualization of the upper GIT by endoscope, diagnose acute or chronic upper GI bleeding, esophageal or gastric varices, polyps, malignancy, ulcers, gastritis, esophagitis, esophageal stenosis, and gastro esophageal reflux.
- Pre-test: ensure consent, NPO 8 hours, pre-medications like atropine and anxiolytics.



Patient undergoing gastroscopy

COMMON LABORATORY PROCEDURES

EGD (esophagogastroduodenoscopy)

- Intra-test: The throat will be anesthetized with a spray or gargle. An I.V. sedative will be administered.
- position: LEFT lateral to facilitate salivary drainage and easy access.

COMMON LABORATORY PROCEDURES

EGD (esophagogastroduodenoscopy)

- Post-test: NPO until gag reflex returns, place patient in Sims position until he awakens, monitor for complications, saline gargles for mild oral discomfort.

NURSING ALERT

- Perforation of the GI tract is a complication of endoscopy. Assess for abdominal or chest pain, dyspnea, fever, tachycardia, lightheadedness, and distended abdomen. Report immediately.

COMMON LABORATORY PROCEDURES

Lower GI- Scopy

- Use of endoscope to visualize the anus, rectum, sigmoid and colon, malignancy, polyps, inflammation, or strictures.
- Pre-test: consent, NPO 8 hours, cleansing enema until return is clear.

COMMON LABORATORY PROCEDURES

Lower GI- Scopy

- Intra-test: position is LEFT lateral, right leg is bent and placed anteriorly.
- Post-test: bed rest, monitor for complications like bleeding and perforation.

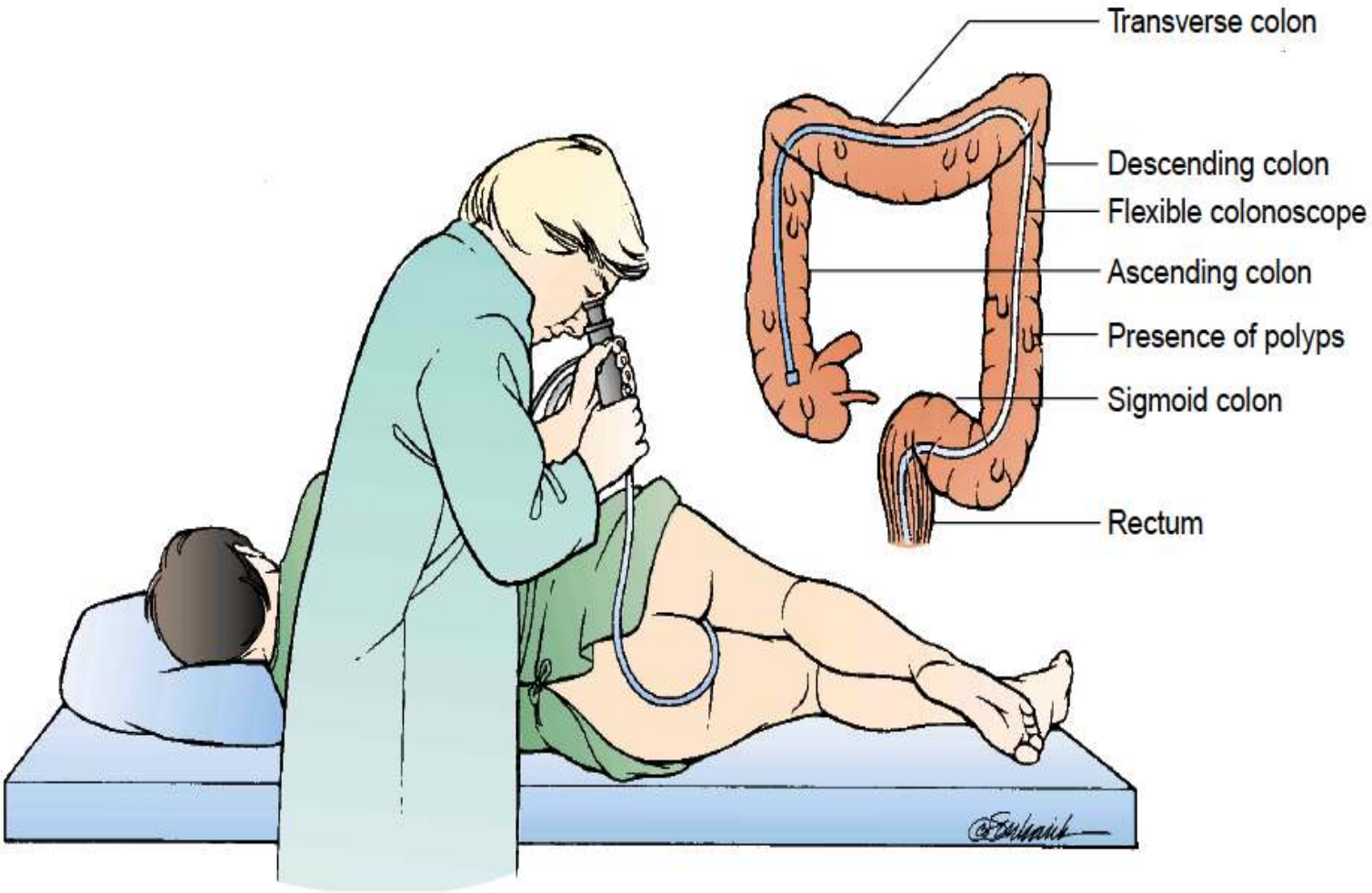


FIGURE 34-7 Colonoscopy. The flexible scope is passed through the rectum and sigmoid colon into the descending, transverse, and ascending colon.

Gastric Analysis

- Aspiration of gastric juice to measure pH, appearance, volume and contents.
- Pre-test: NPO 8 hours, avoidance of stimulants, drugs and smoking.
- Post-test: resume normal activities.

Ultrasonography (Ultrasound)

- A noninvasive test that focuses high-frequency sound waves over an abdominal organ to obtain an image of the structure.

  Ultrasound can detect small abdominal masses, fluid-filled cysts, gallstones, dilated bile ducts, ascites, and vascular abnormalities. Ultrasound with Doppler may be ordered for vascular assessment.

Nursing and Patient Care Considerations

- If indicated, prepare patient before the procedure with a special diet, laxative, or other medication to cleanse the bowel and decrease gas. NPO for at least 6 hours before the procedure.

Esophageal Disorders

Hiatal Hernia

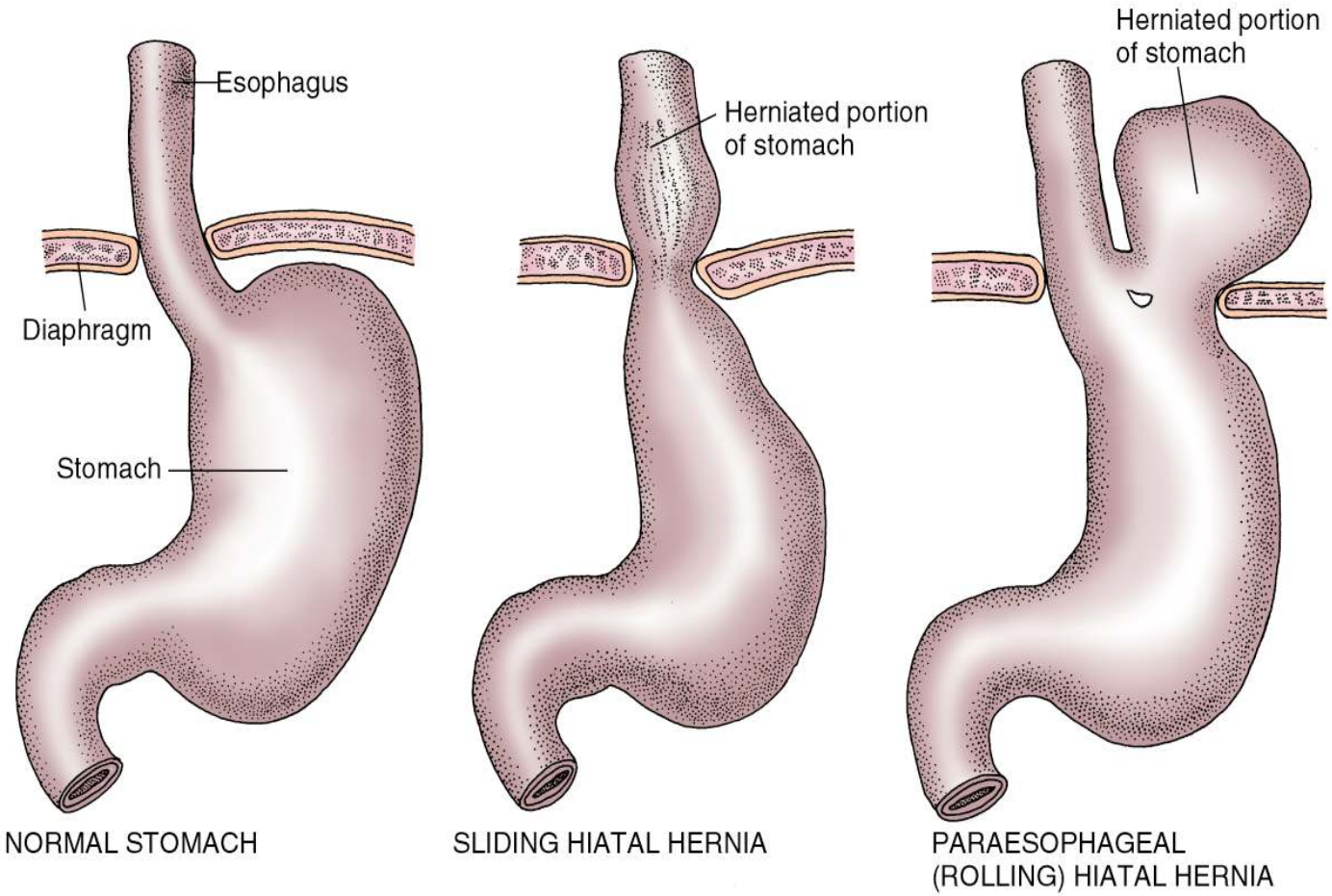
Definition

A. Part of stomach protrudes through the esophageal hiatus of the diaphragm into thoracic cavity.

B. Most cases are asymptomatic; incidence increases with age.

C. Sliding hiatal hernia: gastroesophageal junction and fundus of stomach slide through the esophageal hiatus.

D. Paraesophageal hiatal hernia: the gastroesophageal junction is in normal place but part of stomach herniates through esophageal hiatus; hernia can become strangulated; client may develop gastritis with bleeding.


Esophagus
Diaphragm
Stomach
Herniated portion of stomach
Herniated portion of stomach
NORMAL STOMACH
SLIDING HIATAL HERNIA
PARAESOPHAGEAL (ROLLING) HIATAL HERNIA

Hiatal Hernia

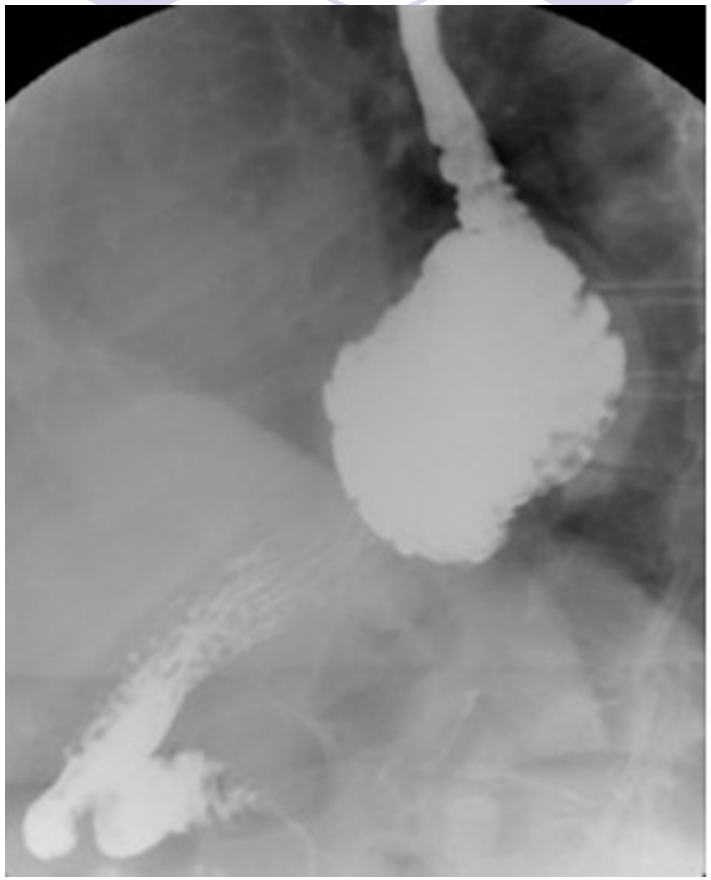
Manifestations

- At least 50% of patients are asymptomatic.
- Sliding hiatal hernia is often implicated in reflux.
- The patient with a paraesophageal hernia usually feels a sense of fullness after eating or may be asymptomatic.
- Heartburn
- Regurgitation
- Dysphagia

Hiatal Hernia

Diagnostic Tests

A. Barium swallow
B. Upper endoscopy



A barium swallow study demonstrates a large hiatal hernia with the majority of the stomach in the chest, not in the abdomen where it belongs.

Hiatal Hernia

Management

A. Frequent, small feedings that can pass easily through the esophagus.

B. To eat a low-fat diet; to avoid caffeine, tobacco, beer, foods containing peppermint or spearmint, and carbonated beverages

C. The patient is advised not to recline for 1 hour after eating, to prevent reflux or movement of the hernia.

D. Elevate the head of the bed to prevent the hernia from sliding upward.

E. Antacids or H2 receptor antagonists, such as famotidine or ranitidine.

F. Proton pump inhibitors (to decrease the release of gastric acid)

Hiatal Hernia

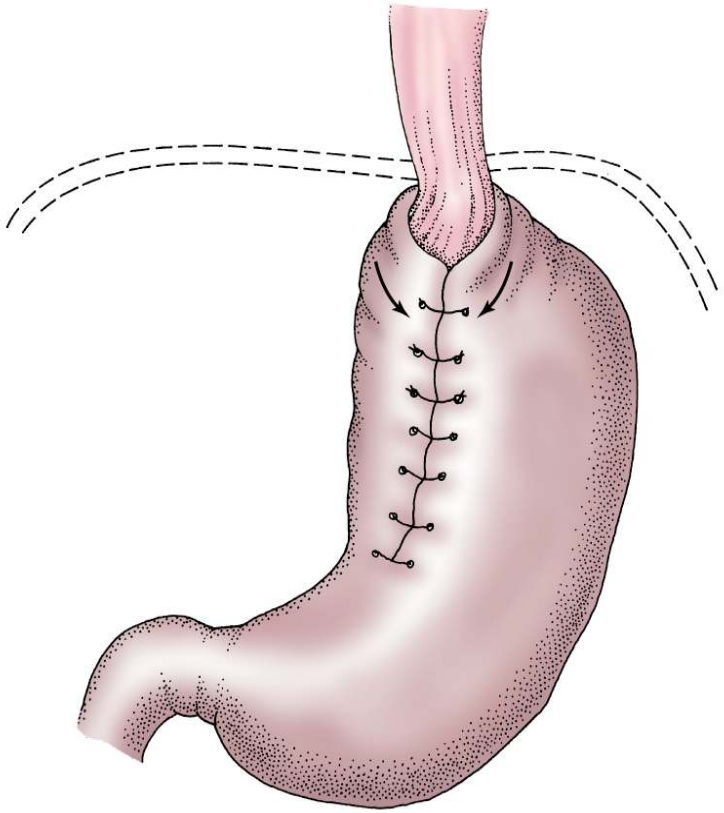
Surgery

A. Surgery is indicated in about 15% of patients; if medical treatment is not effective.

B. May require emergency surgery to correct incarcerated (torsion, twisting), of the stomach that leads to restriction of blood flow.

C. Usually Nissen fundoplication by thoracic or abdominal approach.

Nissen fundoplication



Nursing Interventions

1. Provide small frequent feedings
2. Avoid supine position for 1 hour after eating
3. Elevate the head of the bed on 8-inch block
4. Provide pre-op and post-op care

Esophageal Varices

Definition

Dilation and tortuosity of the submucosal veins in the distal esophagus.

Etiology

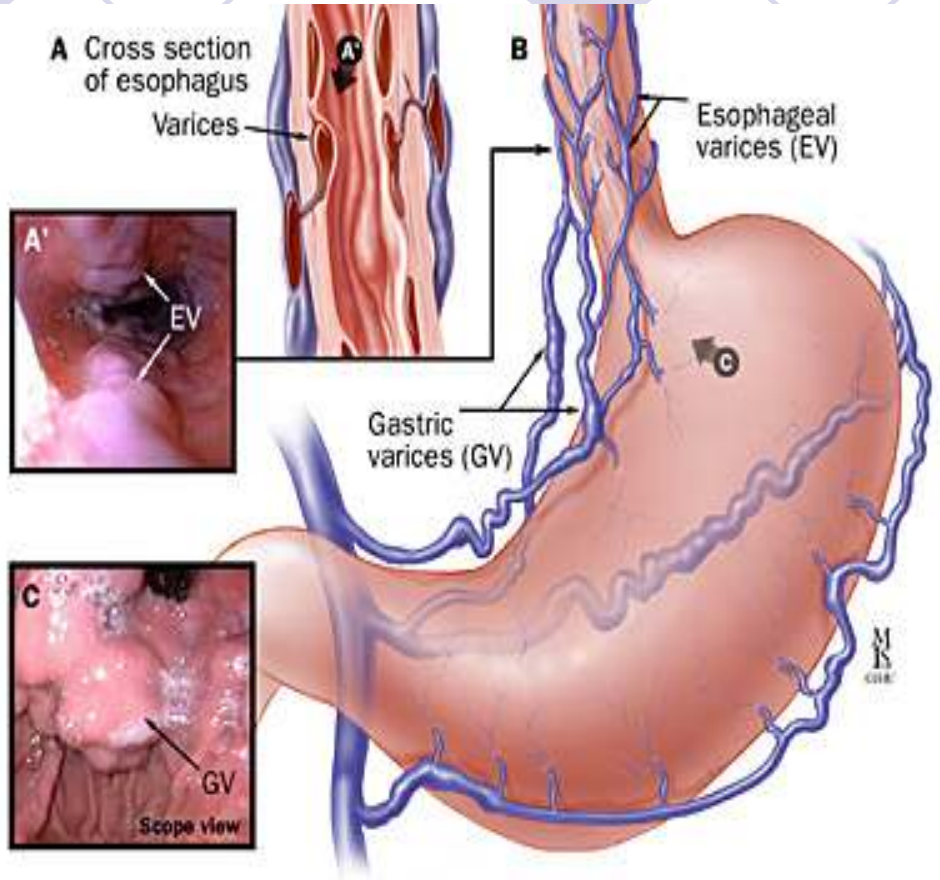
Commonly caused by portal hypertension secondary to liver cirrhosis.

This is an emergency condition!

Assessment findings for EV

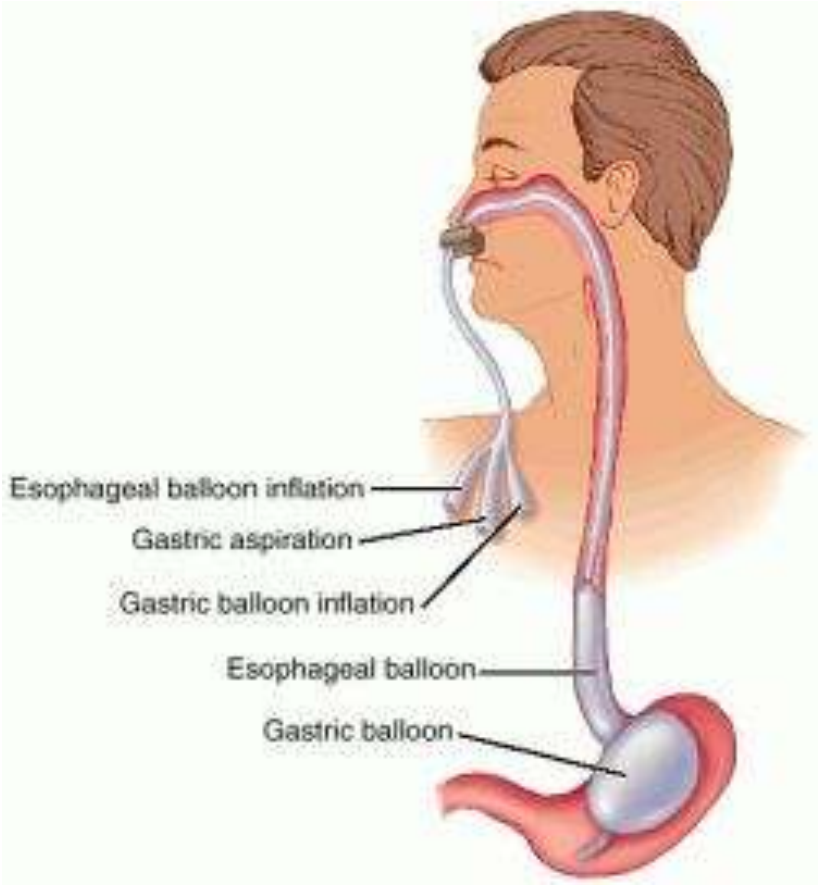
1. Hematemesis
2. Melena
3. Ascites
4. Jaundice
5. Hepatomegaly/splenomegaly
6. Signs of Shock: tachycardia, hypotension, tachypnea, cold clammy skin, narrowed pulse pressure

Diagnostic Procedure: Esophagoscopy



Management and Nursing Interventions

1. Monitor VS strictly. Note for signs of shock
2. Monitor for LOC (level of consciousness)
3. Maintain NPO
4. Monitor blood studies
5. Administer O2
6. prepare for blood transfusion
7. prepare to administer Vasopressin and Nitroglycerin to decrease the


Esophageal balloon inflation
Gastric aspiration
Gastric balloon inflation
Esophageal balloon
Gastric balloon

Gastric Disorders

PEPTIC ULCER DISEASE

DEFINITION

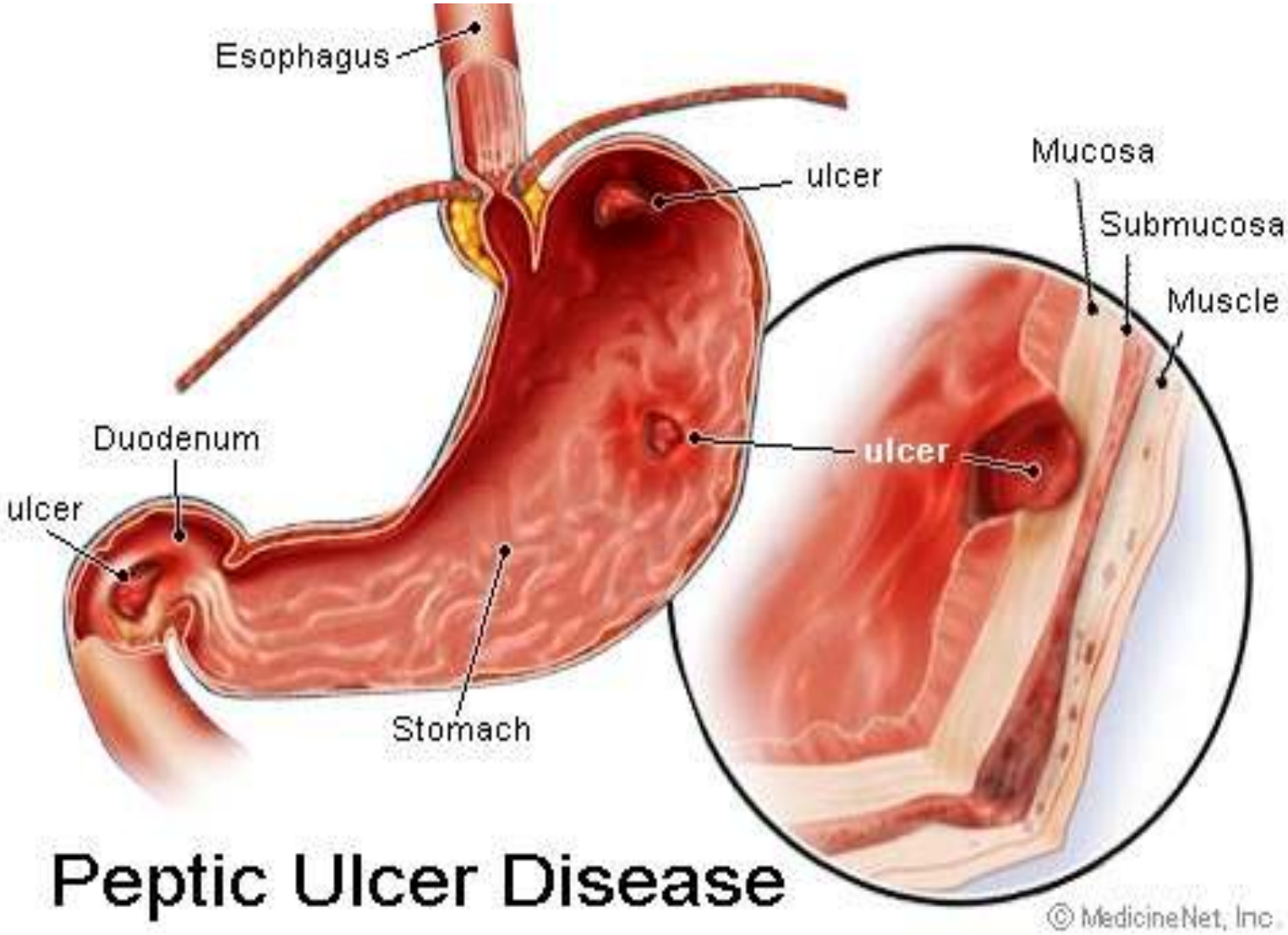
- Peptic ulcer disease refers to ulcerations in the mucosa of the lower esophagus, stomach, or duodenum.

Pathophysiology and Etiology

- Etiology of peptic ulcers disease is multifactorial.
- *Helicobacter pylori* infection present in most patients with peptic ulcer disease.
- NSAIDs-induced injury presents as a chemical gastropathy.
- Acid secretory abnormalities (especially in duodenal ulcers).
- Zollinger-Ellison syndrome (hypersecretory syndrome) should be considered in refractory (resistant to usual treatments) ulcers.

RISK FACTORS

- Drugs (NSAIDs, prolonged high-dose corticosteroids).
- Family history, Zollinger-Ellison syndrome.
- Cigarettes
- Stress
- Blood group O
- Lower socioeconomic status.
- Studies are inconclusive in determining an association between ulcer formation and diet or the intake of alcohol and caffeine.


Esophagus
ulcer
Mucosa
Submucosa
Muscle
Duodenum
ulcer
ulcer
Stomach
Peptic Ulcer Disease
© MedicineNet, Inc.

Clinical Manifestations

- Burning epigastric pain occurring 1½ to 3 hours after a meal.
- Nocturnal epigastric, abdominal pain or burning. May awaken patient at night, usually around midnight to 3 a.m.
- Epigastric tenderness on examination.
- Early satiety (state of being overly full),anorexia, weight loss, heartburn.
- Dizziness, syncope, hematemesis, or melena (may indicate hemorrhage).
- Anemia.

Complications

- GI hemorrhage
- Ulcer perforation
- Gastric outlet obstruction

Management

General Measures

- Eliminate use of NSAIDs or other causative drugs.
- Eliminate cigarette smoking (impairs healing).
- Well-balanced diet with meals at regular intervals. Avoid dietary irritants.

Drug Therapy

- Multiple drug regimens are used to treat *H. pylori*.
- Standard dose (Proton pump inhibitor) PPI twice daily (or esomeprazole once daily) plus

Management

Surgery

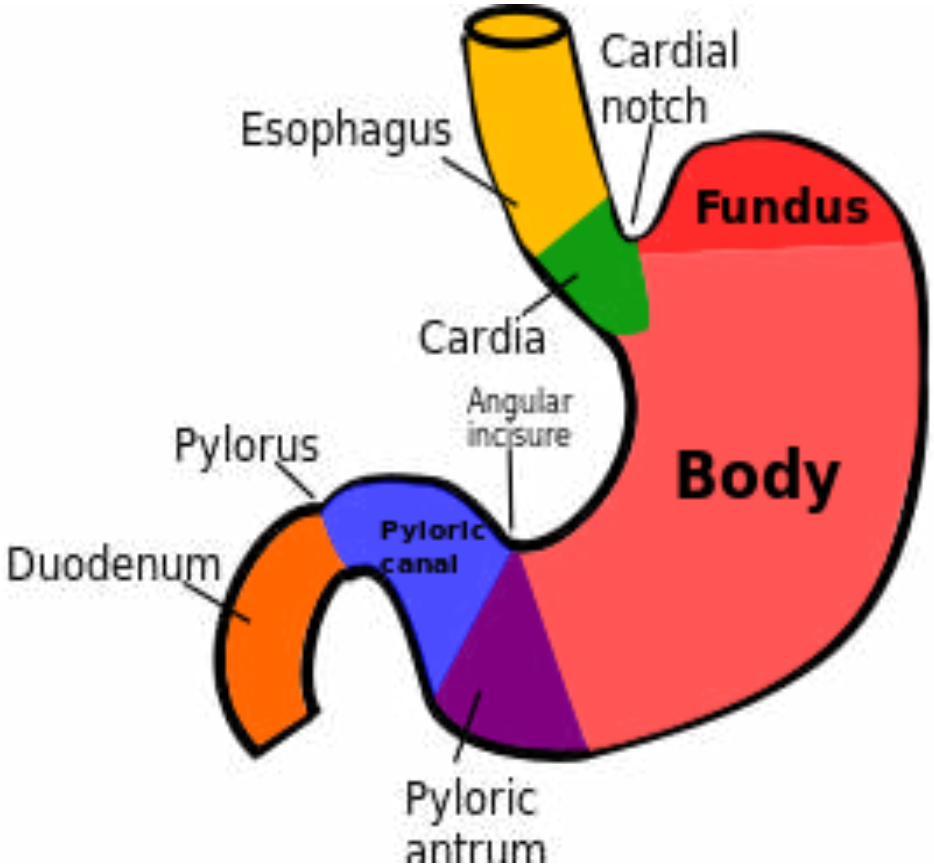
- Surgical interventions may be indicated for hemorrhage, obstruction, perforation, and acid reduction. Surgery may also be indicated with ulcer disease of long duration or severity or difficulty with medical regimen compliance.

Gastroduodenostomy (Billroth I)

- Partial gastrectomy with removal of antrum and pylorus of stomach.
- The gastric stump is anastomosed with the duodenum.

Gastrojejunostomy (Billroth II)

- Partial gastrectomy with removal of antrum and pylorus of stomach.
- The gastric stump is anastomosed with the jejunum.


Cardial notch
Esophagus
Fundus
Cardia
Angular incisure
Pylorus
Body
Pyloric canal
Duodenum
Pyloric antrum

Management Surgery Cont. …

Vagotomy

Pyloroplasty

Total gastrectomy

Post-op complications

- Dumping Syndrome
- Vitamin B12 Deficiency
- Leaking from suture line
- Shock and Hemorrhage
- Dehiscence
- Evisceration

Dumping Syndrome

Assessment

- Weakness, dizziness, tachycardia
- Epigastric fullness, cramping and hyperactive bowel sounds
- 15-20 mins. after eating
- Diaphoresis
- Resolves in 6-12 months

Prevention

- Smaller meals >3hr apart
- High proteins/fiber, low simple carbohydrates

Nursing Assessment

- Determine location, character, radiation of pain, factors aggravating or relieving pain, how long it lasts, when it occurs.
- Ask about eating patterns, regularity, types of food, eating circumstances.
- Ask about medications (especially aspirin, anti-inflammatory drugs, or steroids).
- History of illnesses including previous GI bleeds.
- Obtain psychosocial history.
- Take vital signs, including lying, standing, and

Nursing Diagnoses

- Deficient Fluid Volume related to hemorrhage
- Acute Pain related to epigastric distress secondary to hypersecretion of acid, mucosal erosion, or perforation
- Diarrhea related to GI bleeding
- Imbalanced Nutrition: Less Than Body Requirements related to the disease process
- Deficient Knowledge related to physical, dietary, and pharmacologic treatment of disease

Nursing Interventions

Avoiding Fluid Volume Deficit

- Monitor intake and output continuously to determine fluid volume status.
- Monitor stools for blood and emesis.
- Monitor hemoglobin and hematocrit and electrolytes.
- Administer prescribed I.V. fluids and blood replacement, as prescribed.
- Insert NG tube as prescribed, and monitor the tube drainage for signs of visible and occult blood.

Nursing Interventions

Avoiding Fluid Volume Deficit Cont. ….

- Administer medications through the NG tube to neutralize acidity, as prescribed.
- Prepare patient for saline lavage, as ordered.
- Observe patient for an increase in pulse and a decrease in BP (signs of shock).
- Prepare patient for diagnostic procedure or surgery to determine or stop the source of bleeding.

Achieving Pain Relief

- Administer prescribed medication.
- Provide small, frequent meals to prevent gastric distention if not NPO.
- Advise the patient about the irritating effects of certain drugs and foods.

Decreasing Diarrhea

- Monitor patient's elimination patterns to determine effects of medications.
- Monitor vital signs, and watch for signs of hypovolemia.
- Administer antidiarrheal medication as prescribed.
- Watch for signs and symptoms of impaired skin integrity (erythema, pain, pruritus) around anus to promote comfort and decrease risk of infection.

Achieving Adequate Nutrition

- Eliminate foods that cause pain or distress;
- Provide small, frequent meals.
- Provide high-protein diet with nutritional supplements as ordered.
- Administer parenteral nutrition as ordered if bleeding is prolonged and patient is malnourished.

Patient Education and Health Maintenance

- Explain all tests and procedures to increase knowledge and cooperation; minimize anxiety.
- Teach patient signs and symptoms of bleeding and when to notify the health care provider.
- Promote healthy lifestyle changes to include adequate nutrition, cessation of smoking, decreased alcohol consumption, stress reduction strategies.
- Explain the purpose, dosage, and adverse effects of each medication prescribed.

INTESTINAL Disorders

Abdominal Hernias

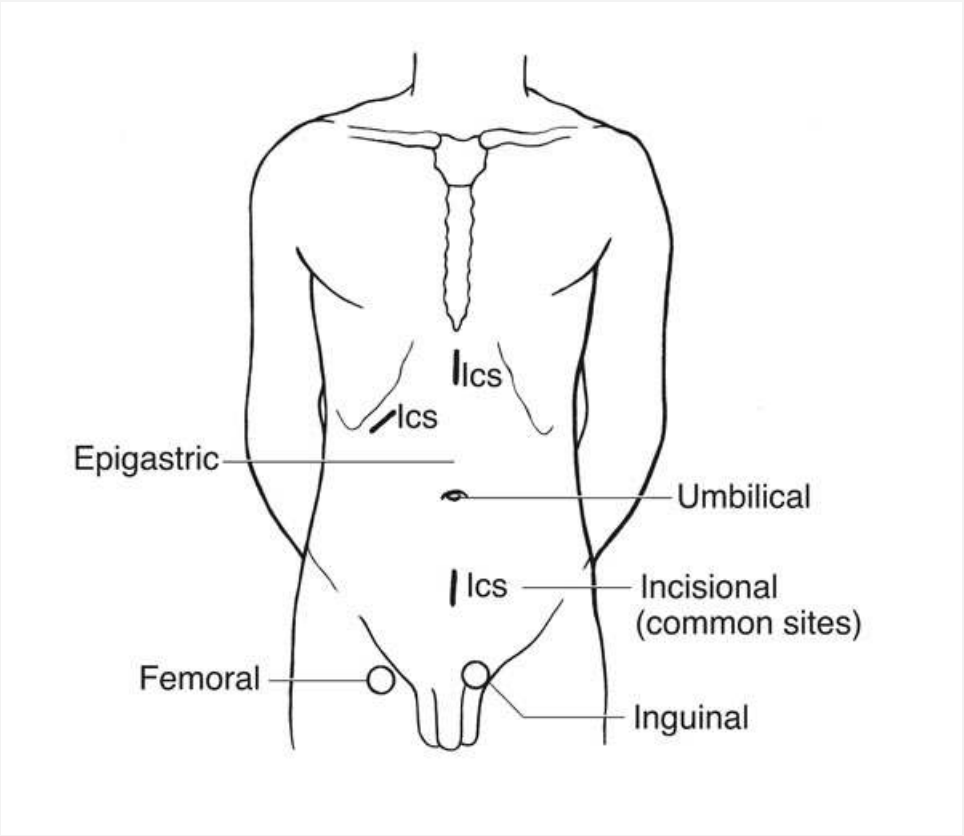
- **A hernia:** is a protrusion of an organ, tissue, or structure through the wall of the cavity in which it is normally contained. It is often called a rupture.

Causes

- Results from congenital or acquired weakness (traumatic injury, aging) of the abdominal wall.
- May result from increased intra-abdominal pressure due to heavy lifting, obesity, pregnancy, straining, coughing, or proximity to tumor.

Classification by Site

- Inguinal hernia into the inguinal canal (more common in males).
- Femoral hernia into the femoral canal, appearing below the inguinal ligament.
- Umbilical intestinal protrusion at the umbilicus due to failure of umbilical orifice to close. Occurs most often in obese women, children, and in patients with increased intra-abdominal pressure from cirrhosis and ascites.
- Ventral or incisional intestinal protrusion due to weakness at the abdominal wall.


lcs
lcs
Epigastric
Umbilical
lcs
Incisional
(common sites)
Femoral
Inguinal

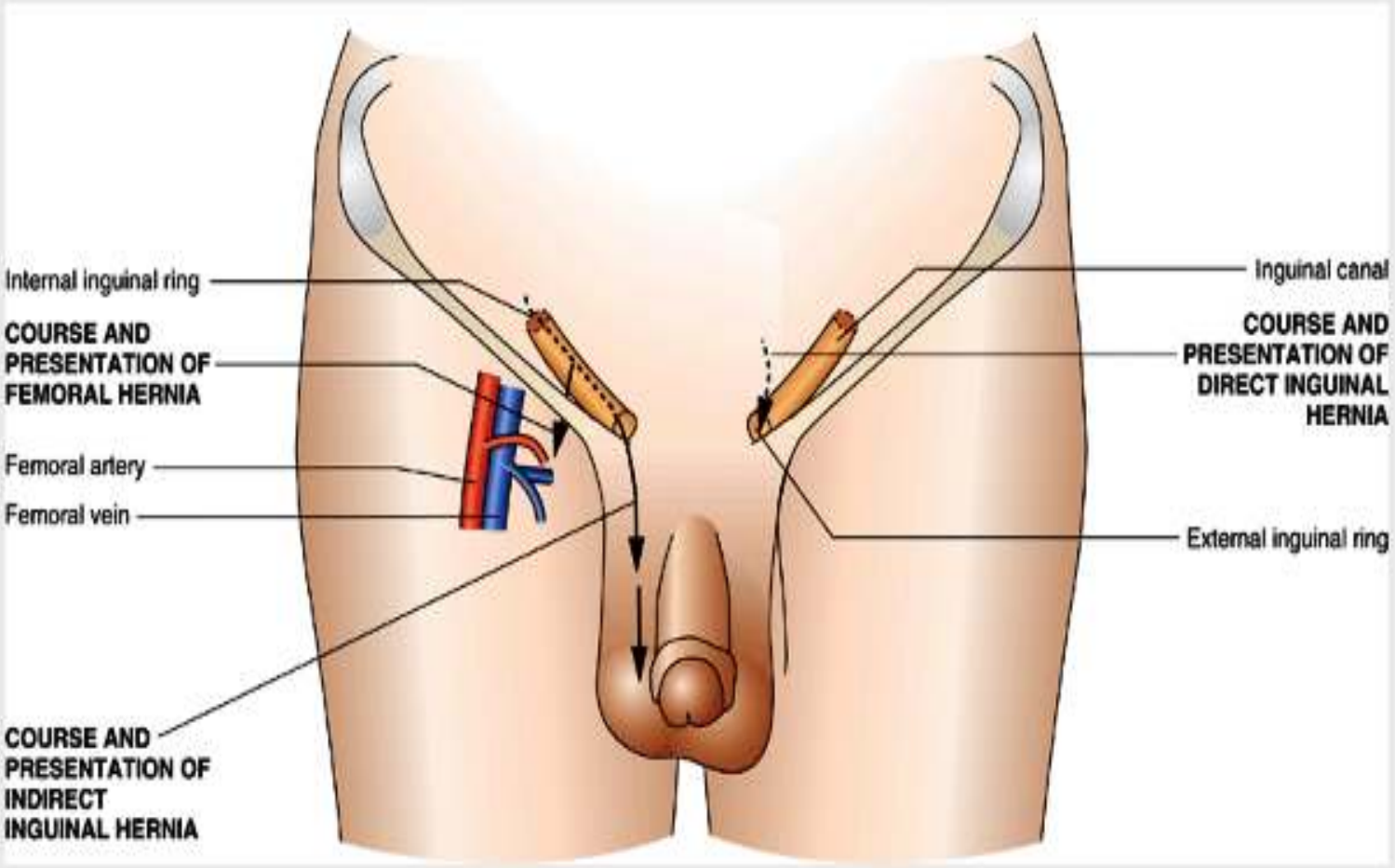
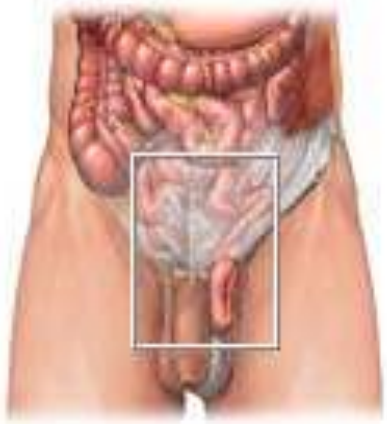


FIGURE 18-6 Direct and indirect inguinal hernia.

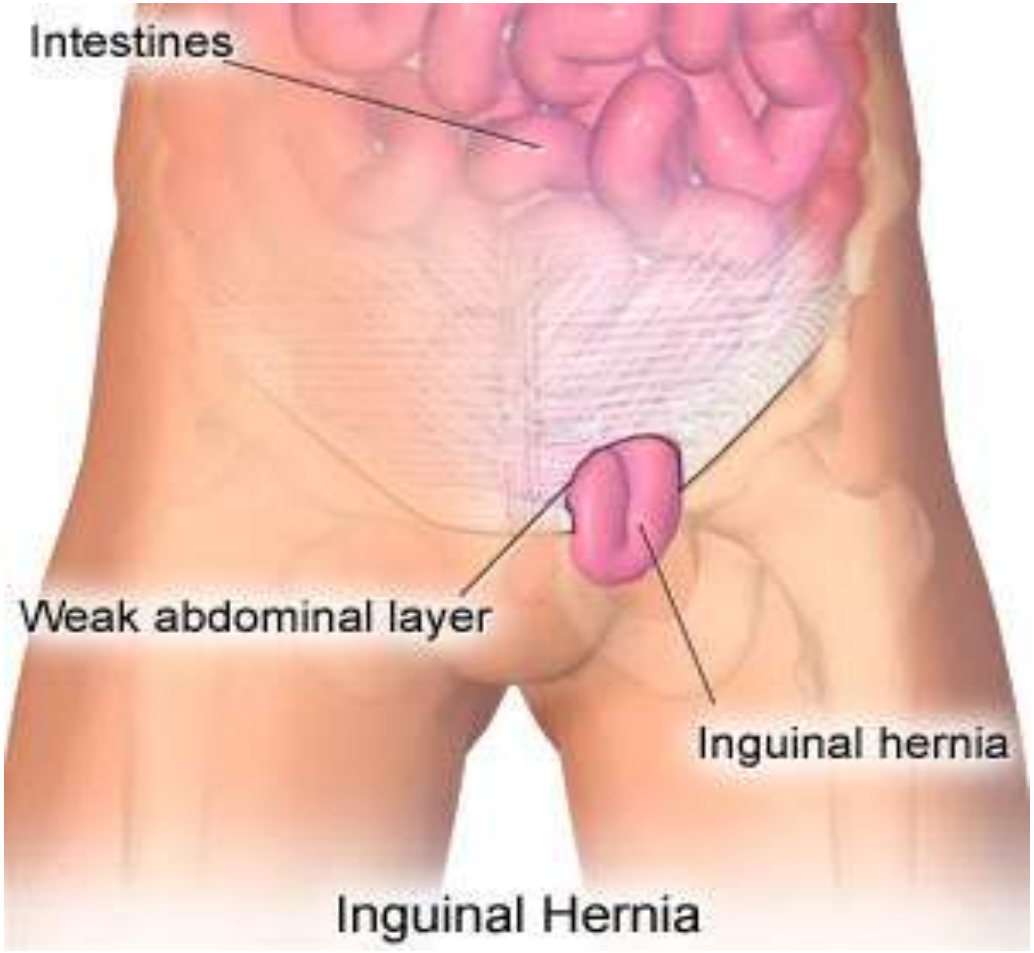


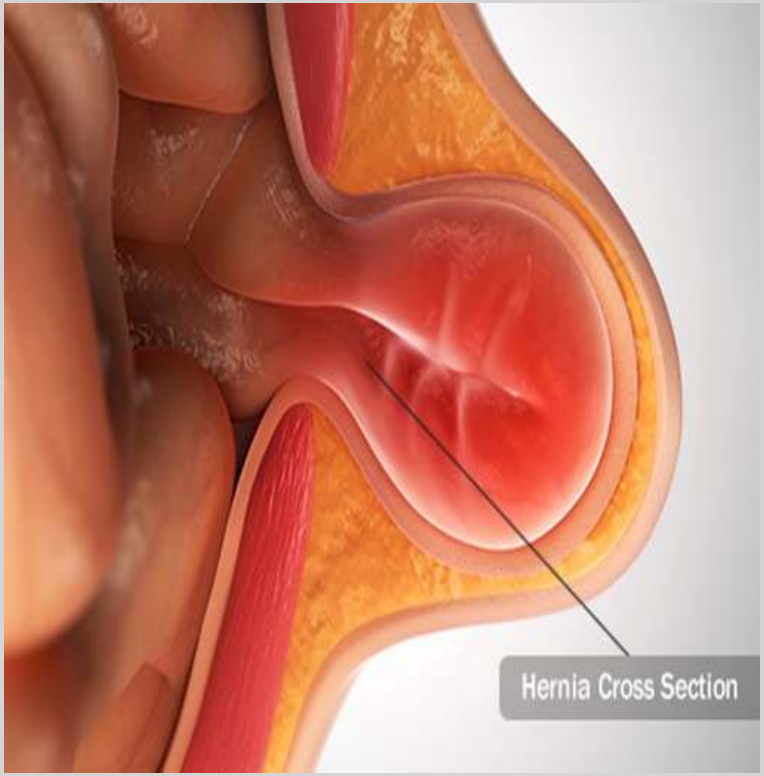


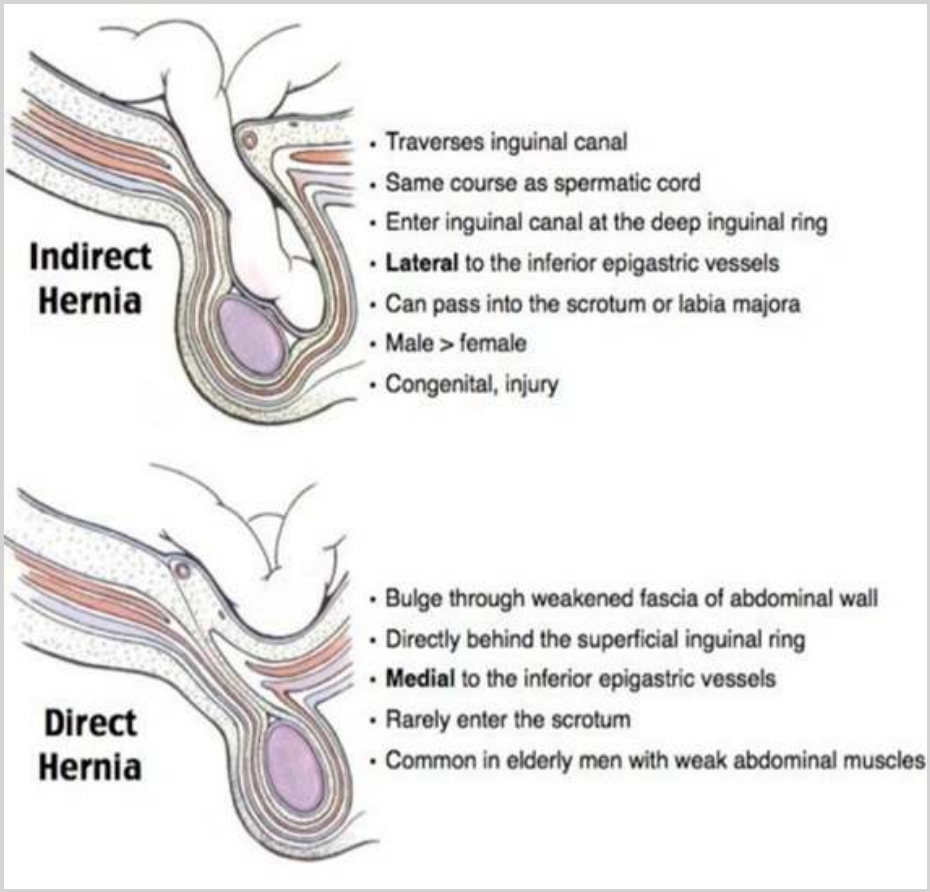
Inguinal hernia

Inguinal hernia occurs when a portion of the small intestine enters the inguinal canal

A.D.A.M.


Intestines
Weak abdominal layer
Inguinal hernia
Inguinal Hernia


Hernia Cross Section

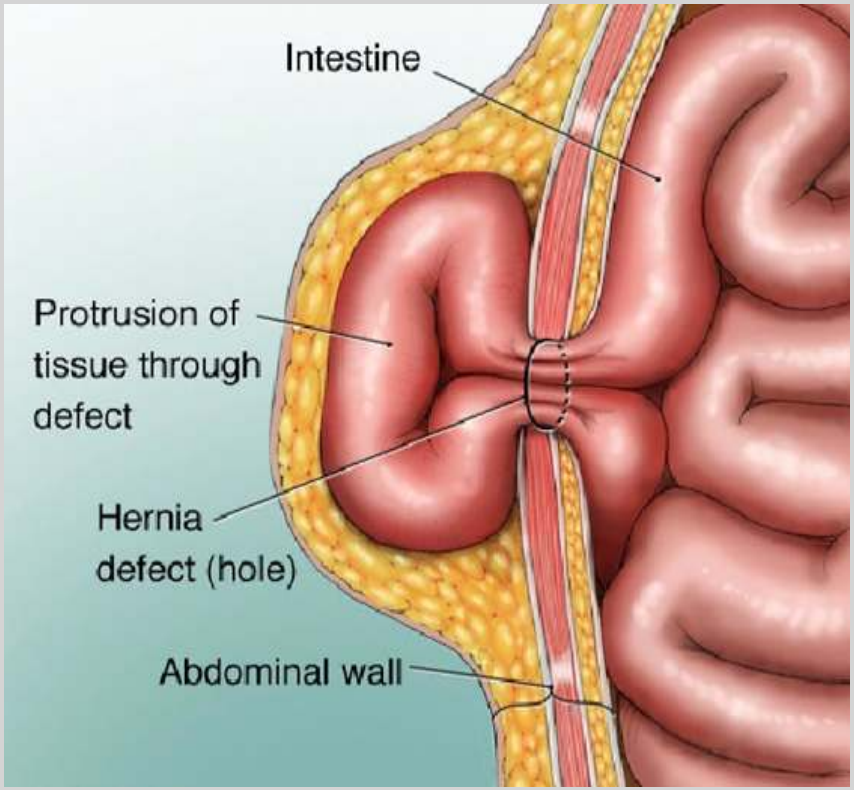

Indirect Hernia
• Traverses inguinal canal
• Same course as spermatic cord
• Enter inguinal canal at the deep inguinal ring
• **Lateral** to the inferior epigastric vessels
• Can pass into the scrotum or labia majora
• Male > female
• Congenital, injury
Direct Hernia
• Bulge through weakened fascia of abdominal wall
• Directly behind the superficial inguinal ring
• **Medial** to the inferior epigastric vessels
• Rarely enter the scrotum
• Common in elderly men with weak abdominal muscles

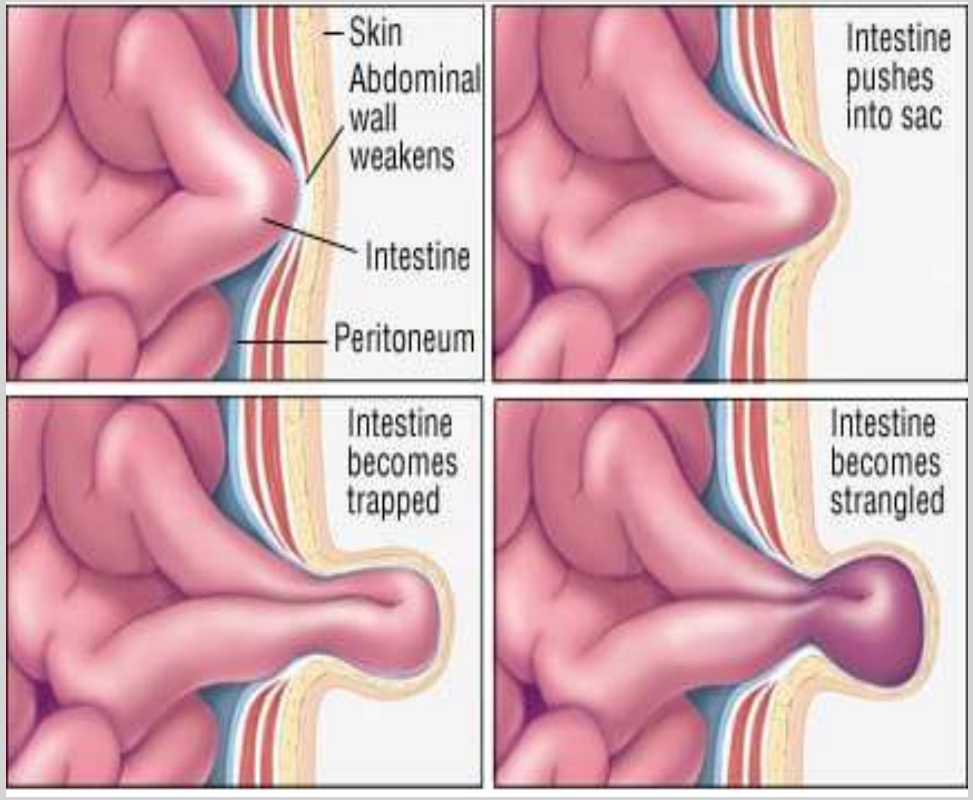

Umbilical hernia


Incisional hernia

Classification by Severity

- Reducible—the protruding mass can be placed back into abdominal cavity.
- Irreducible—the protruding mass cannot be moved back into the abdomen.
- Incarcerated—an irreducible hernia in which the intestinal flow is completely obstructed.
- Strangulated—an irreducible hernia in which the blood and intestinal flow are completely obstructed; develops when the loop of intestine in the sac becomes twisted or swollen and a constriction is


Intestine
Protrusion of tissue through defect
Hernia defect (hole)
Abdominal wall


Skin
Abdominal wall weakens
Intestine
Peritoneum
Intestine pushes into sac
Intestine becomes trapped
Intestine becomes strangled

Clinical Manifestations

- Bulging over herniated area appears when patient stands or strains, and disappears when supine.
- Hernia tends to increase in size and recurs with intra-abdominal pressure.
- Strangulated hernia presents with pain, vomiting, swelling of hernial sac, lower abdominal signs of peritoneal irritation, fever.

Diagnostic Evaluation

- Abdominal X-rays reveal abnormally high levels of gas in the bowel.
- Laboratory studies (complete blood count, electrolytes) may show hemoconcentration (increased hematocrit), dehydration (increased or decreased sodium), and elevated white blood cell (WBC) count, if incarcerated.

Management

- Mechanical (for reducible hernia only).
  - A truss: is an appliance with a pad and belt that is held snugly over a hernia to prevent abdominal contents from entering the hernial sac.
  - Conservative measures: no heavy lifting, straining at stool, or other measures that would increase intra-abdominal pressure.



- Surgery is recommended to correct hernia before strangulation occurs, which then becomes an emergency situation.
  - Herniorrhaphy removal of hernial sac; contents replaced into the abdomen; layers of muscle and fascia sutured. Laparoscopic herniorrhaphy is a possibility and often performed as outpatient procedure.
  - Hernioplasty involves reinforcement of suturing (often with mesh) for extensive hernia repair.
  - Strangulated hernia requires resection of ischemic bowel in addition to repair of hernia.

Nursing Assessment

- Ask the patient if hernia is enlarging and uncomfortable, reducible or irreducible; determine relationship to exertion and activities.
- Assess bowel sounds and determine bowel pattern.
- Determine if the patient is exhibiting signs and symptoms of strangulation, such as distention, fever, nausea, and vomiting.

Nursing Diagnoses

- Chronic Pain related to bulging hernia (mechanical).
- Acute Pain related to surgical procedure.
- Risk for Infection related to emergency procedure for strangulated or incarcerated hernia.

Nursing Interventions

Achieving Comfort

- Fit patient with truss or belt when hernia is reduced, if ordered.
- Give stool softeners as directed.

Relieving Pain Postoperatively

- Have the patient splint the incision site with hand or pillow when coughing to lessen pain and protect site from increased intra-abdominal pressure.
- Administer analgesics, as ordered.
- Teach about bed rest, intermittent ice packs, and scrotal elevation as measures used to reduce scrotal edema or swelling after repair of an inguinal hernia

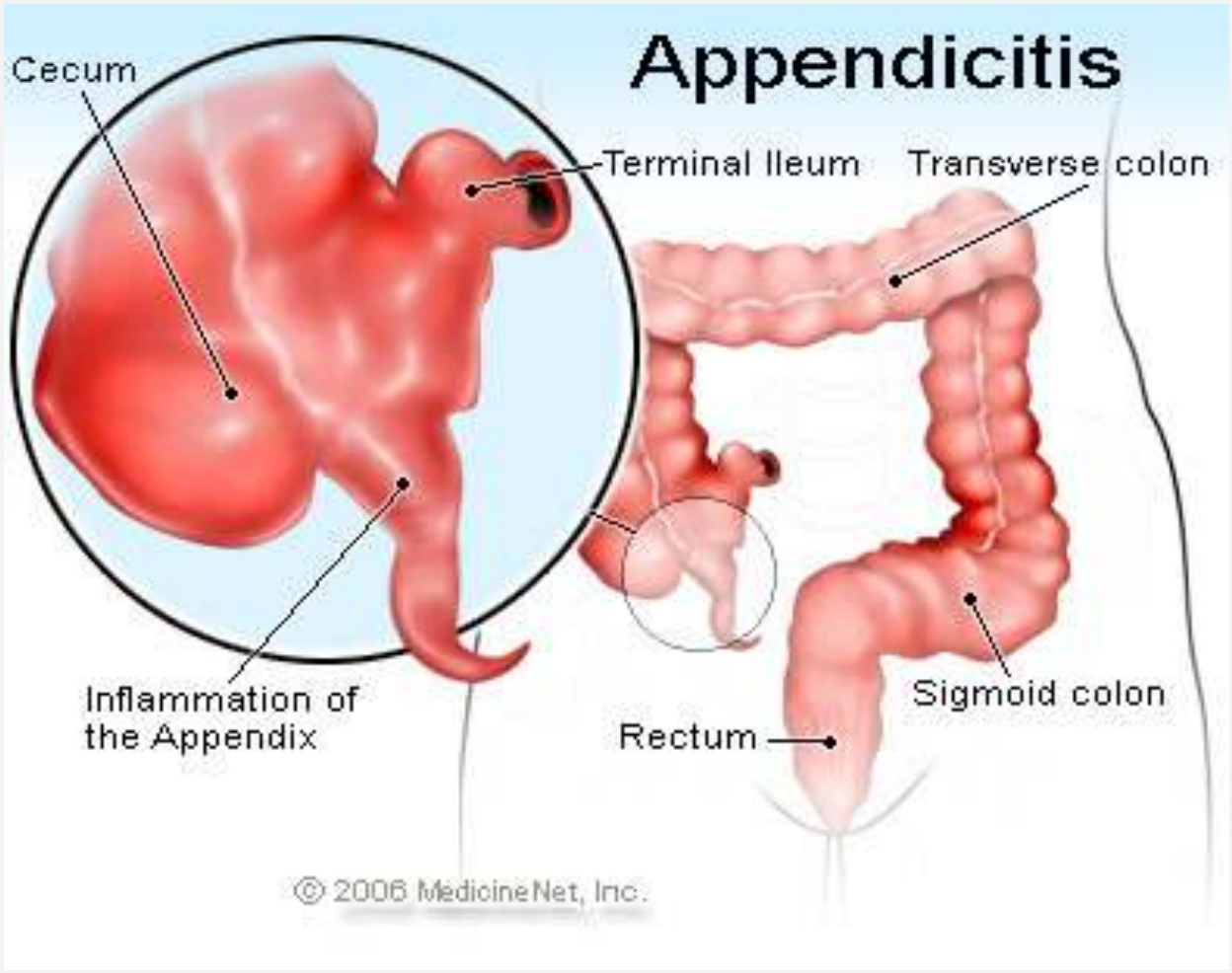
Nursing Interventions

Preventing Infection

- Check dressing for drainage and incision for redness and swelling.
- Monitor for other signs and symptoms of infection: fever, chills, malaise, diaphoresis.
- Administer antibiotics, if appropriate.

Appendicitis

- Appendicitis: is inflammation of the vermiform appendix caused by an obstruction of the intestinal lumen from infection, stricture, fecal mass, foreign body, or tumor.
- Appendicitis can affect any age-group; most common in adolescents/young adults, especially males.


Appendicitis
Cecum
Terminal Ileum
Transverse colon
Inflammation of
the Appendix
Sigmoid colon
Rectum
© 2006 MedicineNet, Inc.

Clinical Manifestations

- Generalized or localized abdominal pain in the epigastric or periumbilical areas. Within 2 to 12 hours, the pain localizes in the right lower quadrant and intensity increases.
- Anorexia, moderate malaise, mild fever, nausea and vomiting.
- Usually constipation occurs; occasionally diarrhea.
- Rebound tenderness, involuntary guarding (spasm of muscle), generalized abdominal rigidity.

Diagnostic Evaluation

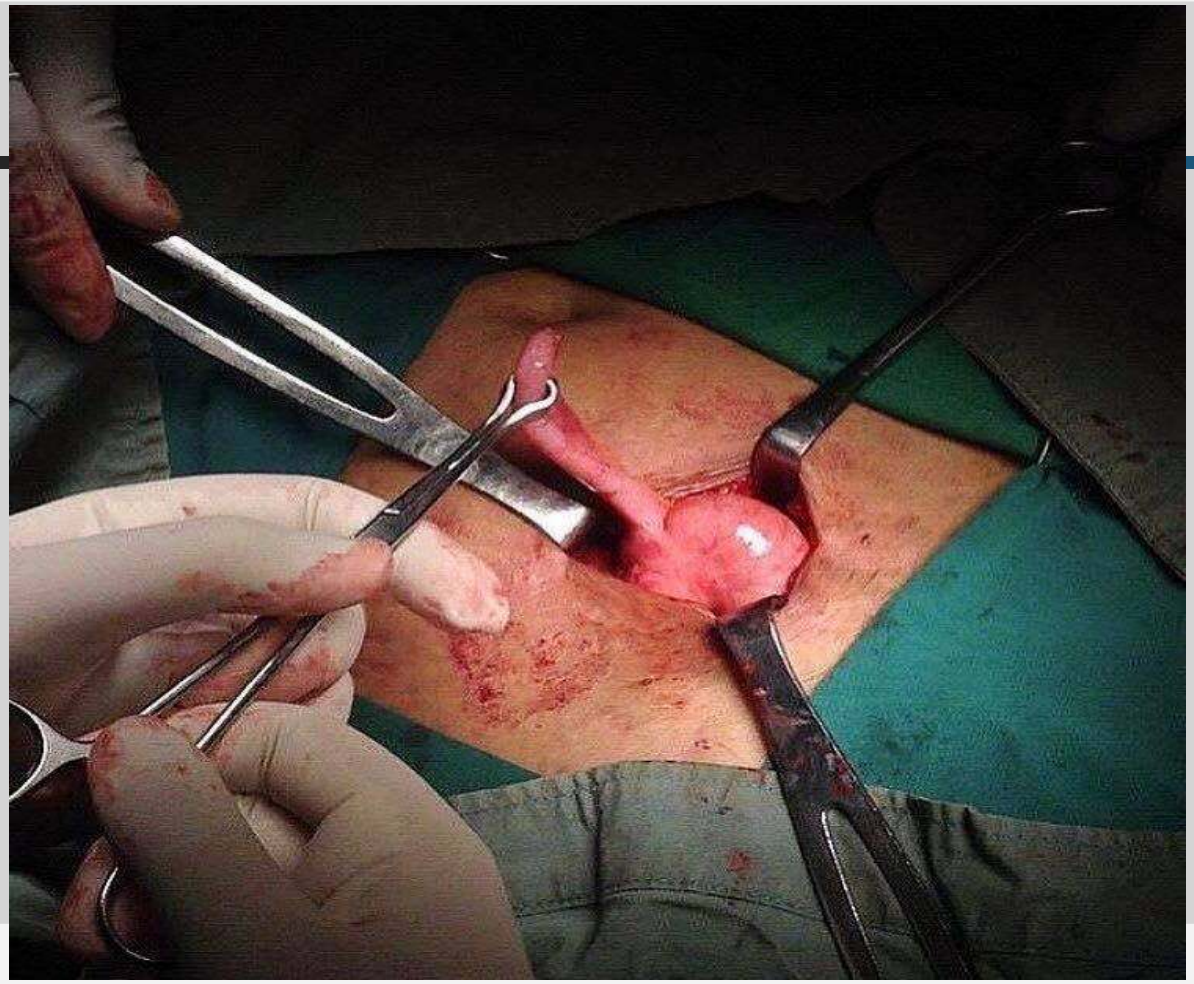
- WBC count reveals moderate leukocytosis (10,000 to 16,000/mm3).
- Urinalysis to rule out urinary disorders.
- Abdominal X-ray may visualize shadow consistent with fecalith in appendix; perforation will reveal free air.
- Abdominal ultrasound or CT scan can visualize appendix.

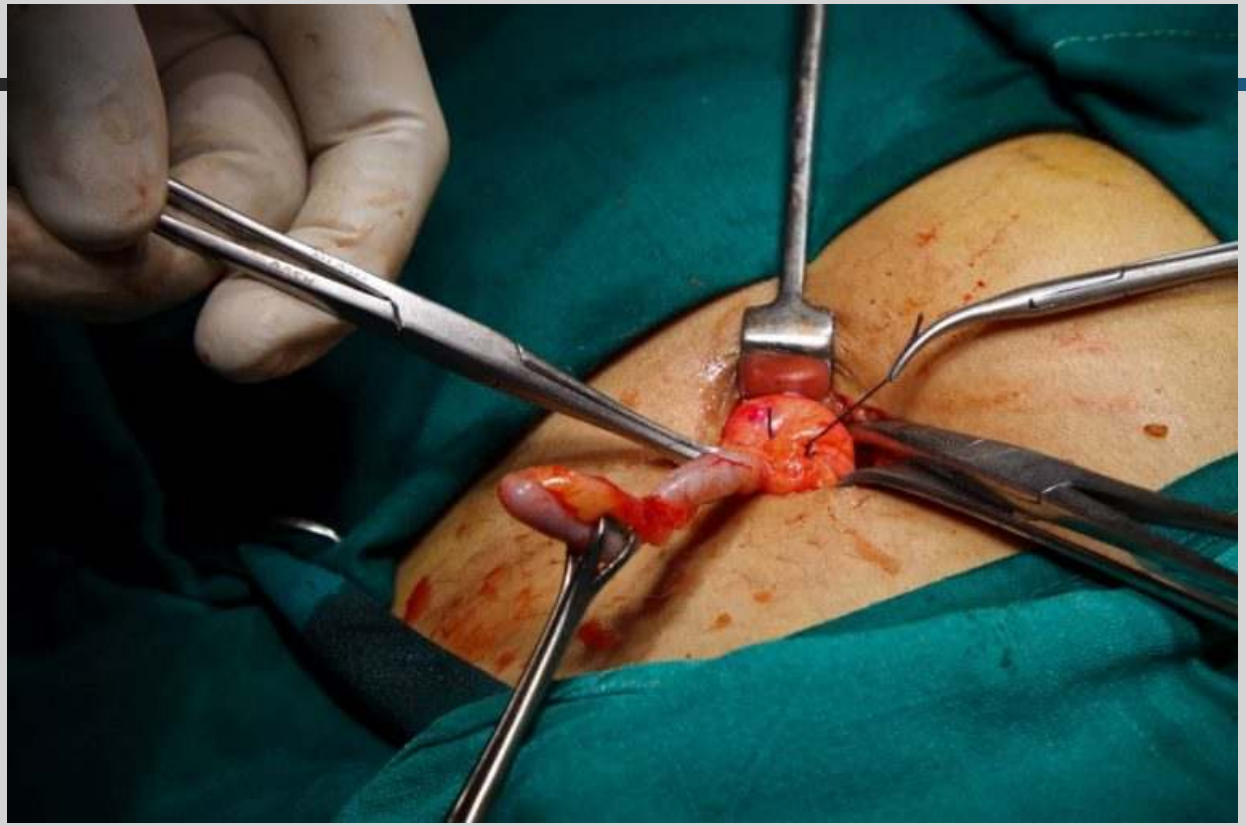
Management

Surgery (appendectomy) is indicated.

Complications

Perforation, Abscess, Peritonitis





Nursing Diagnoses

- Acute Pain related to inflamed appendix
- Risk for Infection related to perforation

Nursing Interventions

Relieving Pain

- Monitor pain level, including location, intensity, pattern.
- Assist patient to comfortable positions, such as semi-Fowler's and knees up.
- Restrict activity that may aggravate pain, such as coughing and ambulation.
- Apply ice bag to abdomen for comfort.
- Give antiemetic ordered and evaluate response.

DRUG ALERT

- Do not give analgesics to mask pain, and do not administer cathartics because they may cause rupture.

Preventing Infection

- Monitor frequently for signs and symptoms of worsening condition indicating perforation, abscess, or peritonitis (increasing severity of pain, tenderness, rigidity, distention, ileus, fever, malaise, tachycardia).
- Administer antibiotics as ordered.
- Promptly prepare patient for surgery.

Client with Intestinal Obstruction

Definition

A. Failure of intestinal contents to move through the bowel lumen;
B. Most common site is the small intestine.
C. May be partial or complete obstruction.
D. With obstruction, gas and fluid accumulate proximal to and within obstructed segment causing bowel distention.
E. Bowel distention, vomiting leads to hypovolemia, hypokalemia, renal insufficiency, shock.

Client with Intestinal Obstruction

Pathophysiology

A. Mechanical

1. Problems outside intestines: adhesions (bands of scar tissue), hernias.
2. Problems within intestines: tumors, IBD.
3. Obstruction of intestinal lumen (partial or complete).

a. Intussusception: telescoping bowel.

b. Volvulus: twisted bowel.

c. Foreign bodies.

d. Strictures.

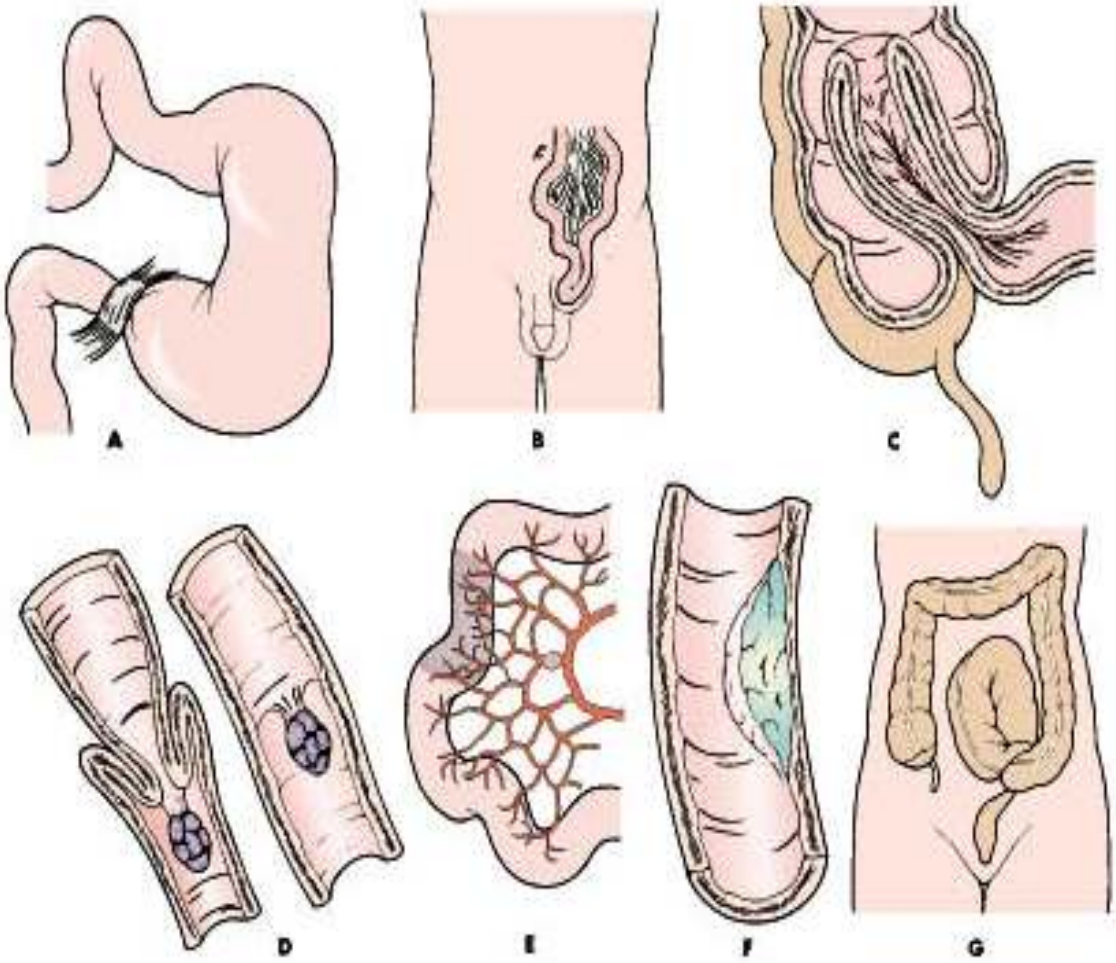


Figure 40-6 Bowel obstructions. **A,** Adhesions. **B,** Strangulated inguinal hernia. **C,** Ileocecal intussusception. **D,** Intussusception from polyps. **E,** Mesenteric occlusion. **F,** Neoplasm. **G,** Vovulus of the sigmoid colon.

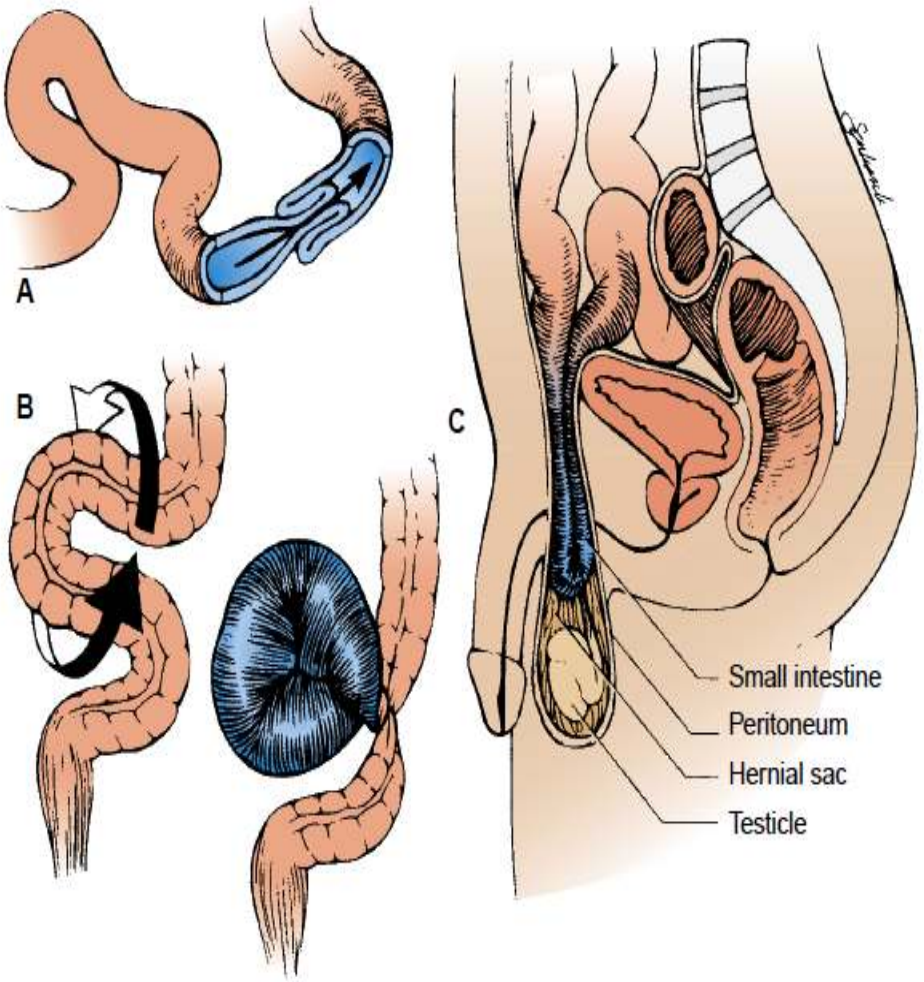
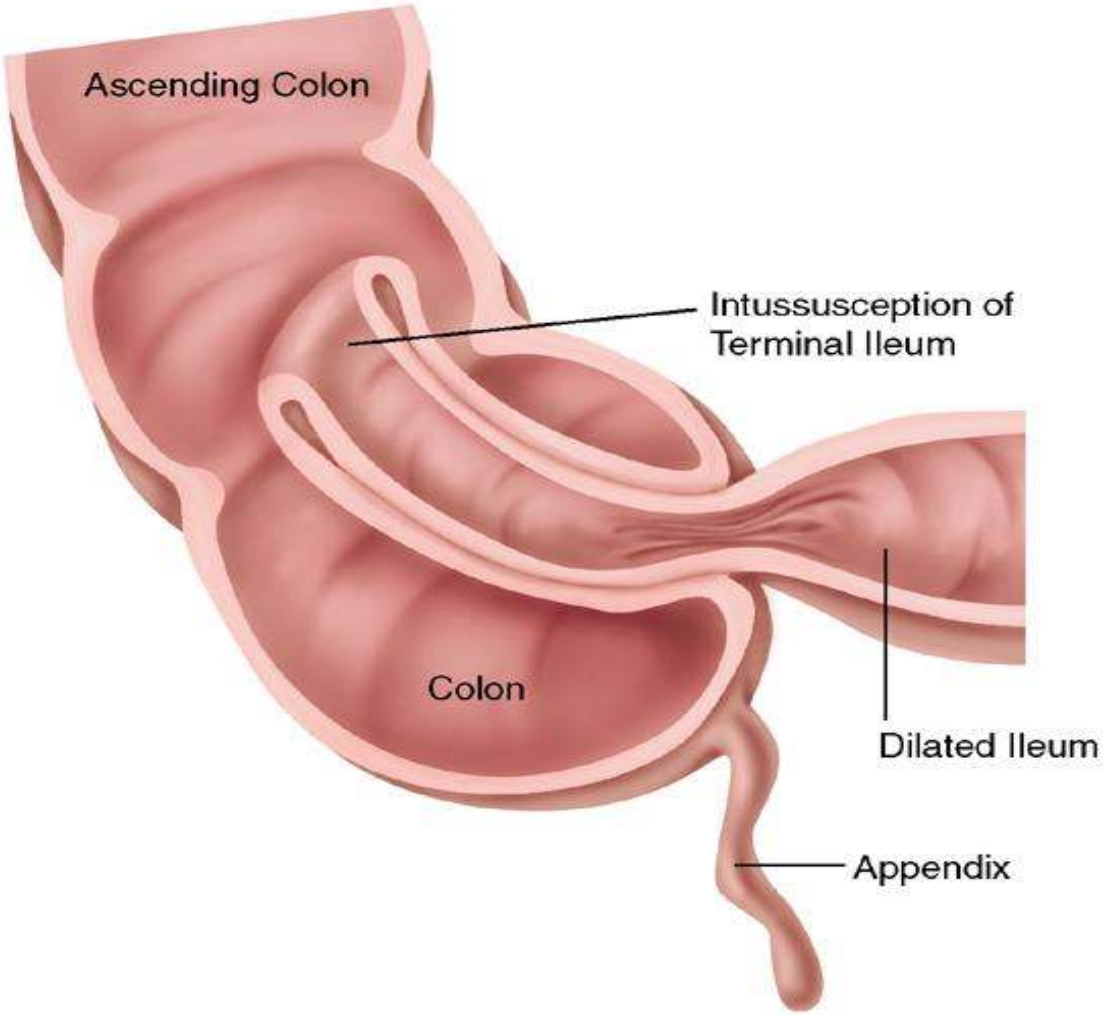


FIGURE 38-6 Three causes of intestinal obstruction. (**A**) Intussusception invagination or shortening of the colon caused by the movement of one segment of bowel into another. (**B**) Volvulus of the sigmoid colon; the twist is counterclockwise in most cases. Note the edematous bowel. (**C**) Hernia (inguinal). The sac of the hernia is a continuation of the peritoneum of the abdomen. The hernial contents are intestine, omentum, or other abdominal contents that pass through the hernial opening into the hernial sac.


Ascending Colon
Intussusception of
Terminal Ileum
Colon
Dilated Ileum
Appendix

Client with Intestinal Obstruction

B. Functional

1. Failure of peristalsis to move intestinal contents: adynamic ileus (paralytic ileus, ileus) due to neurologic or muscular impairment.
2. Accounts for most bowel obstructions.
3. Causes include:

A. Post gastrointestinal surgery.
B. Tissue anoxia or peritoneal irritation from hemorrhage, peritonitis, or perforation.
C. Hypokalemia.
D. Medications: narcotics, anticholinergic drugs, antidiarrheal medications.
E. Renal colic, spinal cord injuries, uremia.

Client with Intestinal Obstruction

Manifestations of Small Bowel Obstruction

a. Vary depend on level of obstruction and speed of development.

b. Cramping or colicky abdominal pain, intermittent, intensifying.

c. Vomiting; vomiting may become feculent.

d. Bowel sounds:

1. Early in course of mechanical obstruction: borborygmi and high-pitched tinkling, may have visible peristaltic waves.
2. Later: with paralytic ileus; diminished or absent bowel sounds throughout.

e. Signs of dehydration.



Intestinal gangrene

Client with Intestinal Obstruction

Manifestations of Large Bowel Obstruction

A. Large BO only accounts for 15% of obstructions.

B. Causes include cancer of bowel, volvulus, diverticular disease, inflammatory disorders, fecal impaction.

C. **Manifestations include:**

- Deep, cramping pain; severe, continuous pain signals bowel ischemia and possible perforation.
- Localized tenderness or palpable mass may be noted.

Client with Intestinal Obstruction

Complications

A. Hypovolemia and hypovolemic shock can result in multiple organ dysfunction (acute renal failure, impaired ventilation, death).

B. Strangulated bowel can result in gangrene, perforation, peritonitis, possible septic shock.

C. Delay in surgical intervention leads to higher mortality rate.

Client with Intestinal Obstruction

Diagnostic Tests

A. Abdominal X-rays and CT scans with contrast media; show distended loops of intestine with fluid and /or gas in small intestine, confirm mechanical obstruction.

2. If CT with contrast media meglumine diatrizoate (Gastrografin), check for allergy to iodine, need BUN and Creatinine to determine renal function.

C. Laboratory testing to evaluate for presence of infection and electrolyte imbalance: WBC, serum amylase, osmolality, electrolytes, ABG.

D. Barium enema or colonoscopy/sigmoidoscopy to identify large bowel obstruction.

Client with Intestinal Obstruction

Diagnostic Tests

A. Abdominal X-rays and CT scans with contrast media; show distended loops of intestine with fluid and /or gas in small intestine, confirm mechanical obstruction.

2. If CT with contrast media meglumine diatrizoate (Gastrografin), check for allergy to iodine, need BUN and Creatinine to determine renal function.

C. Laboratory testing to evaluate for presence of infection and electrolyte imbalance: WBC, serum amylase, osmolality, electrolytes, ABG.

D. Barium enema or colonoscopy/sigmoidoscopy to identify large bowel obstruction.

Client with Intestinal Obstruction

Management

1. Gastrointestinal Decompression

A. Treatment with nasogastric or long intestinal tube provides bowel rest and removal of air and fluid.

B. Successfully relieves many partial small bowel obstructions.

Client with Intestinal Obstruction

2. Surgery

A. Treatment for complete mechanical obstructions, strangulated or incarcerated obstructions of small bowel, persistent incomplete mechanical obstructions.

B. Preoperative care:

1. Insertion of nasogastric tube to relieve vomiting, abdominal distention, and to prevent aspiration of intestinal contents.

2. Restore fluid and electrolyte balance; correct acid and base imbalances.

3. Laparotomy: inspection of intestine and removal of infarcted or gangrenous tissue.

4. Removal of cause of obstruction: adhesions, tumors, foreign bodies, gangrenous portion of intestines and anastomosis or creation of colostomy depending on individual case.

Client with Intestinal Obstruction

Nursing Assessment

- Assess the nature and location of the patient's pain, the presence or absence of distention, flatus, defecation, emesis, obstipation.
- Assess vital signs.
- Conduct frequent checks of the patient's level of responsiveness; decreasing responsiveness may offer a clue to an electrolyte imbalance or impending shock.

Client with Intestinal Obstruction

Nursing Diagnoses

- Acute Pain related to obstruction, distention, and strangulation
- Risk for Deficient Fluid Volume related to impaired fluid intake, vomiting, and diarrhea from intestinal obstruction
- Altered bowel elimination related to obstruction
- Ineffective Breathing Pattern related to abdominal distention, interfering with normal lung expansion
- Risk for Injury related to complications and severity of illness
- Fear related to life-threatening symptoms of intestinal obstruction

Client with Intestinal Obstruction

Nursing Interventions

- **Achieving Pain Relief:**

  Analgesics. Maintain NG tube as prescribed to decompress bowel. A rectal tube may be indicated.

- **Maintaining Electrolyte and Fluid Balance:**

  Measure and record all intake and output. Administer I.V. fluids and parenteral nutrition as prescribed. Monitor urine output to assess renal function. Monitor vital signs.

- **Maintaining Normal Bowel Elimination:**

  Collect stool samples to test for occult blood if ordered. Record amount and consistency of stools. Maintain NG tube as prescribed to decompress bowel.

Client with Intestinal Obstruction

Nursing Interventions

- **Maintaining Proper Lung Ventilation**
  - Keep the patient in Fowler's position.
  - Monitor ABG levels for oxygenation levels if ordered.
- **Relieving Fears**
  - Recognize the patient's concerns, and initiate measures to provide emotional support.
  - Encourage presence of support person.

Client with Colorectal Cancer

Definition

A. Third most common cancer diagnosed
B. Affects males and females equally
C. Five-year survival rate is 90%, with early diagnosis and treatment

Risk Factors

D. Family history
E. Inflammatory bowel disease
F. Diet high in fat, calories, protein

Client with Colorectal Cancer

Pathophysiology

A. Most malignancies begin as adenomatous polyps and arise in rectum and sigmoid.

B. Spread by direct extension to involve entire bowel circumference and adjacent organs.

C. Metastasize to regional lymph nodes via lymphatic and circulatory systems to liver, lungs, brain, bones, and kidneys.

Manifestations

D. Often produces no symptoms until it is advanced.

E. Presenting manifestation is bleeding; also change in bowel habits (diarrhea or constipation); pain, anorexia, weight loss, palpable abdominal or rectal mass; anemia.

Client with Colorectal Cancer

Complications

A. Obstruction

B. Hemorrhage

C. Anemia

D. Metastasis

Client with Colorectal Cancer

Screening

Beginning at age 50, men and women should follow one of the following American Cancer Society guidelines for early detection of colon cancer.

A. FOBT (Fecal Occult Blood Test) every year and a flexible sigmoidoscopy every 5 years.

B. Colonoscopy every 10 years.

C. Double-contrast barium enema every 5 to 10 years.

Client with Colorectal Cancer

Diagnostic Tests

A. CBC: anemia from blood loss, tumor growth.

B. Fecal occult blood (guiac or Hemoccult testing): all colorectal cancers bleed intermittently.

C. Carcinoembryonic antigen (CEA): not used as screening test, but is a tumor marker and used to estimate prognosis, monitor treatment, detect reoccurrence.

D. Colonoscopy or sigmoidoscopy: tissue biopsy of suspicious lesions, polyps.

E. Chest x-ray, CT scans, MRI, ultrasounds: to determine tumor depth, organ involvement, metastasis.

Client with Colorectal Cancer

Surgery

A. Surgical resection of tumor, adjacent colon, and regional lymph nodes is treatment of choice.

B. Whenever possible anal sphincter is preserved and colostomy avoided; anastomosis of remaining bowel is performed.

C. Tumors of rectum are treated with abdominoperineal resection (A-P resection) in which sigmoid colon, rectum, and anus are removed through abdominal and perineal incisions and permanent colostomy created.

Client with Colorectal Cancer

Colostomy

1. Ostomy made in colon if obstruction from tumor.

a. Temporary measure to promote healing of anastomoses.

b. Permanent means for fecal evacuation if distal colon and rectum removed.

2. Named for area of colon is which formed:

a. Sigmoid colostomy: used with A-P resection formed on LLQ.

b. Double-barrel colostomy: 2 stomas; proximal for feces diversion, distal is mucous fistula.

c. Transverse loop colostomy: emergency procedure; loop suspended over a bridge; temporary.

d. Hartman procedure: Distal portion is left in place and oversewn; only proximal colostomy is brought to abdomen as stoma; temporary; colon reconnected at later time when client ready for surgical repair.

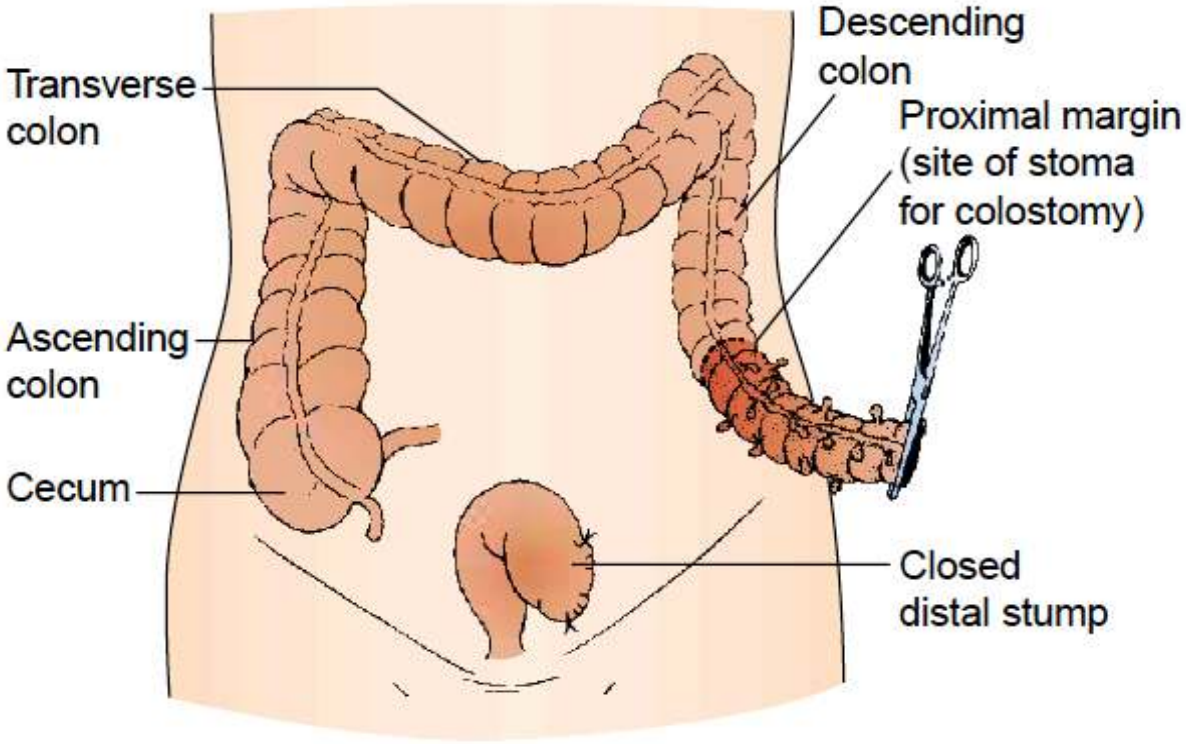


FIGURE 38-3 The Hartmann procedure for diverticulitis: primary resection for diverticulitis of the colon. The affected segment (*clamp attached*) has been divided at its distal end. In a primary anastomosis, the proximal margin (*dotted line*) is transected and the bowel attached end-to-end. In a two-stage procedure, a colostomy is constructed at the proximal margin with the distal stump oversewn (Hartmann procedure, as shown) or brought to the outer surface as a mucous fistula. The second stage consists of colostomy takedown and anastomosis.

Client with Colorectal Cancer

Radiation Therapy

A. May be used preoperatively to improve resectability of the tumor.

B. May be used postoperatively as adjuvant therapy to treat residual disease.

Chemotherapy

C. May be used as adjuvant therapy to improve survival time.

D. May be used for residual disease, recurrence of disease, unresectable tumors, and metastatic disease.

Client with Colorectal Cancer

Prevention of Colorectal Cancer

A. Prevention is a primary issue.
B. Client teaching:
3. Diet: decrease amount of fat, refined sugar, red meat; increase amount of fiber; diet high in fruits and vegetables, whole grains, legumes.
4. Screening recommendations.
5. Seek medical attention for bleeding and warning signs of cancer.
6. Risk may be lowered by aspirin or NSAID use.

Client with Colorectal Cancer

Prevention of Colorectal Cancer

- People with positive FOBTs (Fecal Occult Blood Test) usually undergo colonoscopy with removal of polyps, if present.

- Genetic testing can confirm a hereditary diagnosis such as FAP (Familial adenomatous polyposis) or HNPCC (Hereditary Nonpolyposis Colorectal Cancer).

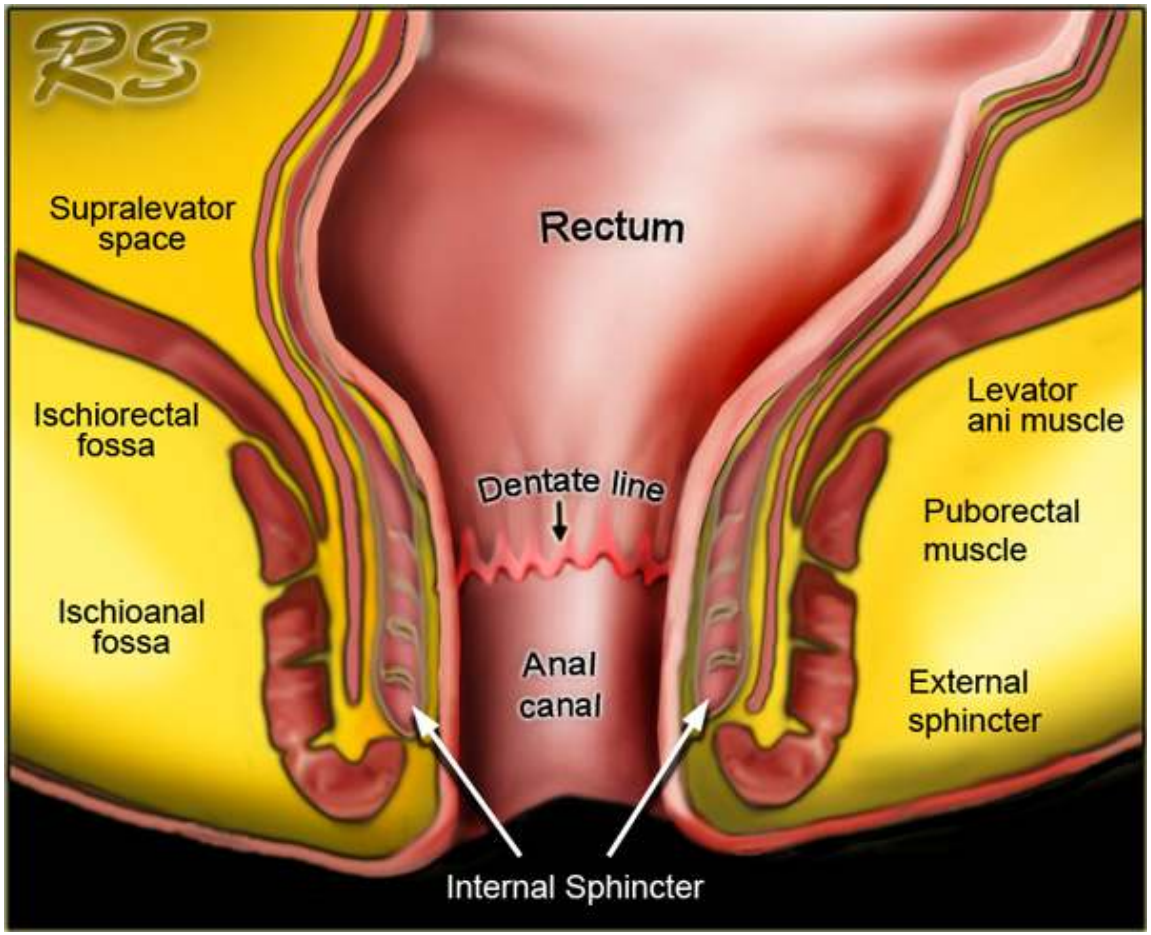
HEMORRHOIDS

HEMORRHOIDS

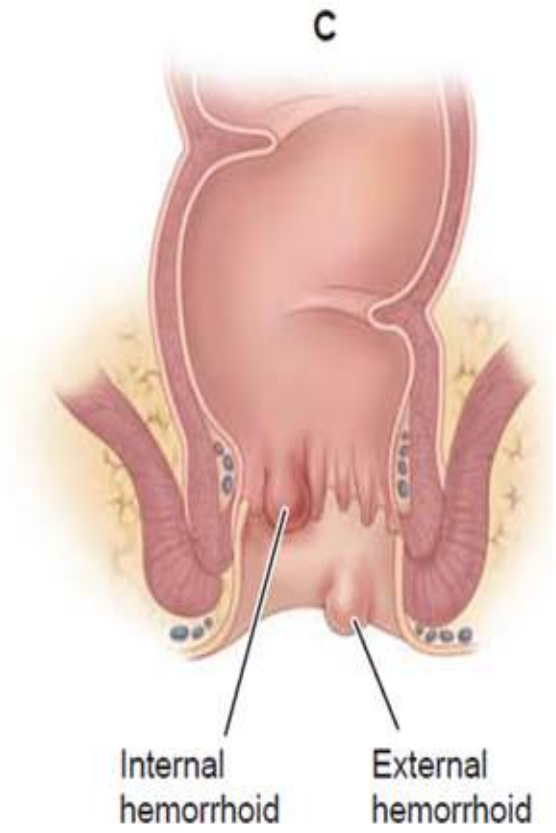
- Hemorrhoids basically means "blood flow" [Greek 'haima' meaning "blood" + 'rhoia' meaning "flow"].
- Hemorrhoids are defined as a vascular masses in the lower rectum or anus.
- The most common symptom of hemorrhoids is rectal bleeding associated with bowel movement.

TYPES OF HEMORRHOIDS

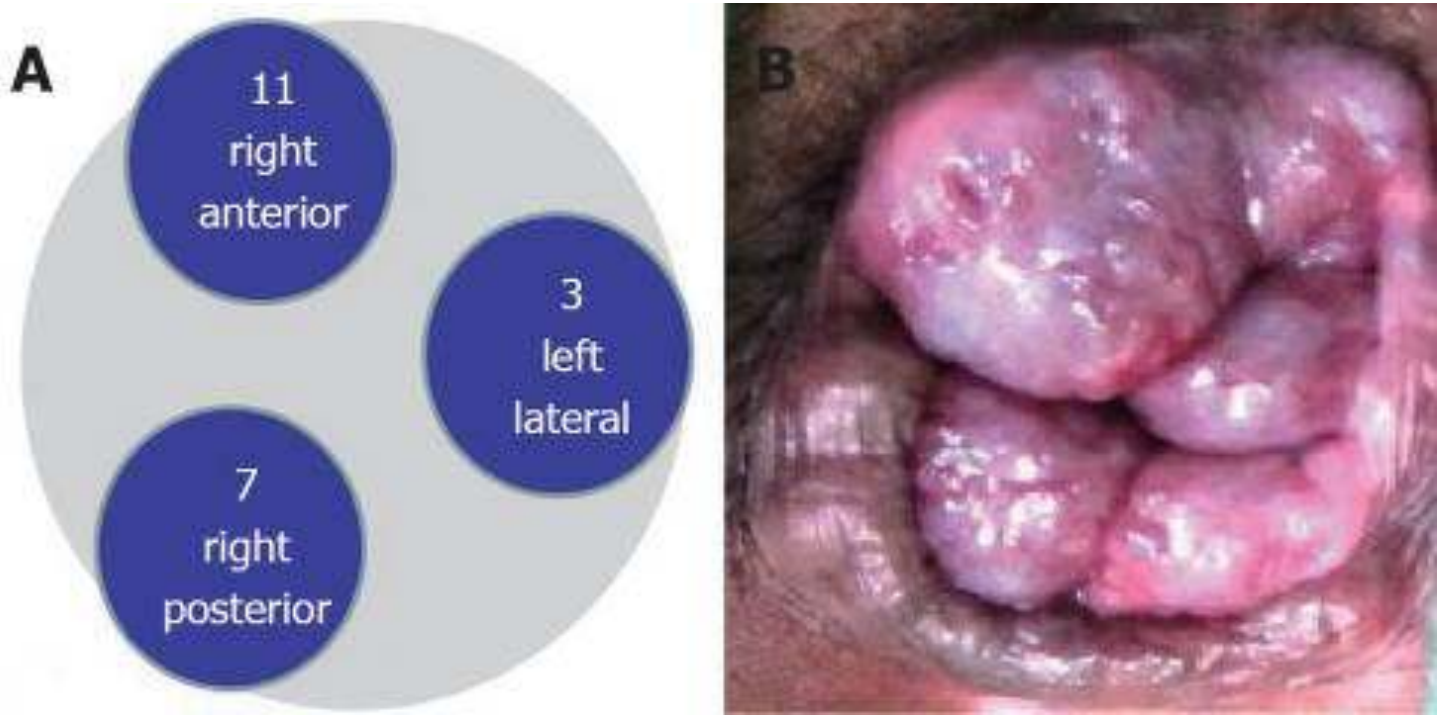
- External hemorrhoids appear outside the external sphincter.
- Internal hemorrhoids appear above the internal sphincter.
- When blood within the hemorrhoids becomes clotted due to obstruction, the hemorrhoids are referred to as thrombosed.


RS
Supralevator space
Rectum
Ischiorectal fossa
Levator ani muscle
Dentate line
Puborectal muscle
Ischioanal fossa
Anal canal
External sphincter
Internal Sphincter

TYPES OF HEMORRHOID



ARRANGEMENT OF THE PILES



PATHOPHYSIOLOGY AND ETIOLOGY

- The exact pathogenesis remains controversial.
- Theories include:

  Abnormal dilatation of veins of internal hemorrhoidal venous plexus.
- Predisposing factors include:

  - Pregnancy, prolonged sitting/standing.
  - Straining at stool, chronic constipation/diarrhea.
  - Anal infection, rectal surgery, or episiotomy.
  - Hereditary factor.

CLINICAL PICTURE

COMPLICATIONS

- Bleeding
- Strangulation
- Thrombosis
- Ulceration
- Gangrene
- Fibrosis
- Suppuration

8.Incontinence

DIAGNOSIS

TREATMENT

MEDICAL THERAPY

- Correct Bowel habits.
- Analgesics as needed.
- Topical creams, lotions, and suppositories to provide comfort (Anusol).

MINIMALLY INVASIVE TECHNIQUES

Rubber band ligation

Cryosurgery

Bipolar, infrared, and laser coagulation

Laser hemorrhoidectomy

Sclerotherapy

Doppler-guided hemorrhoidal artery ligation

INDICATIONS OF SURGERY

- Failure of medical and non-operative therapy
- Prolonged bleeding
- Disabling pain
- Prolapse
- Symptomatic third-degree, fourth-degree
- Fibrosed hemorrhoids
- External hemorrhoids

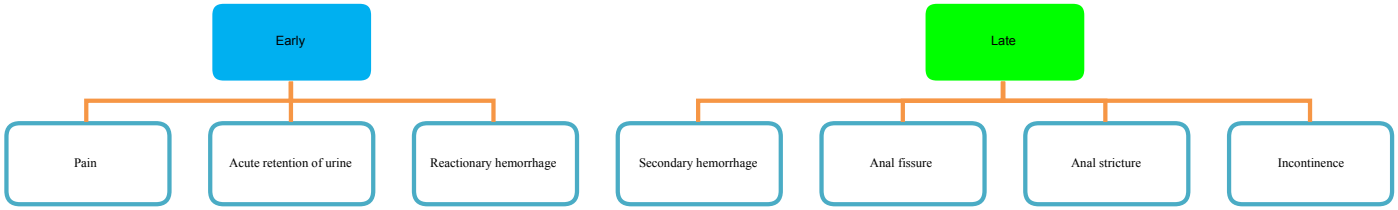
SURGICAL THERAPY

- Dilatation of the anal canal and lower rectum under general anesthesia.
- Incision and removal of clot from acutely thrombosed hemorrhoid.
- Hemorrhoidectomy excision of internal/external hemorrhoids.

POST OP CARE

- Sitz bath
- Analgesics
- Antibiotics
- Laxative
- Dressing
- P/R after 3 weeks

COMPLICATIONS OF SURGERY



HEPATIC DISORDERS

Hepatic Disorders

Hepatitis

- Hepatitis is a viral infection of the liver associated with a broad spectrum of clinical manifestations from non-symptom-producing infection through icteric hepatitis to hepatic necrosis.
- Five types of hepatitis virus have been identified.

Pathophysiology and Etiology

Type A Hepatitis

- Hepatitis A (HAV) is caused by a ribonucleic acid (RNA) virus of the enterovirus family.
- Mode of transmission is primarily fecal-oral
  - Prevalent in underdeveloped countries or in instances of overcrowding and poor sanitation.
  - Infected food handler can spread the disease, and people can contract it by consuming water or shellfish from contaminated waters.
  - Commonly spread by person to person contact and rarely by blood transfusion.

Pathophysiology and Etiology

Type A Hepatitis

- Incubation period is 3 to 5 weeks, with the average being 4 weeks.
- Occurrence is worldwide, usually among children and young adults.
- Mortality is 0% to 1%, with recovery as the rule.

Clinical Manifestations

Type A Hepatitis

- May have no symptoms.
- Prodromal symptoms: fatigue, anorexia, malaise, headache, low-grade fever, nausea and vomiting.
- Highly contagious during this period, usually 2 weeks before the onset of jaundice.
- Icteric phase: jaundice, tea-colored urine, clay-colored stool, and right upper quadrant tenderness.
- Symptoms may be mild in children; adults are more likely to have severe symptoms and a prolonged course of disease.

Management

All Types of Hepatitis

- Rest according to patient's level of fatigue.
- Hospitalization for protracted nausea and vomiting or life-threatening complications.
- Small, frequent feedings of a high-caloric, low-fat diet; proteins are restricted when the liver cannot metabolize protein by-products, as demonstrated by symptoms.
- Vitamin K injected subcutaneously if PT is prolonged.
- I.V. fluid and electrolyte replacement as indicated.
- Administration of antiemetic for nausea and vomiting

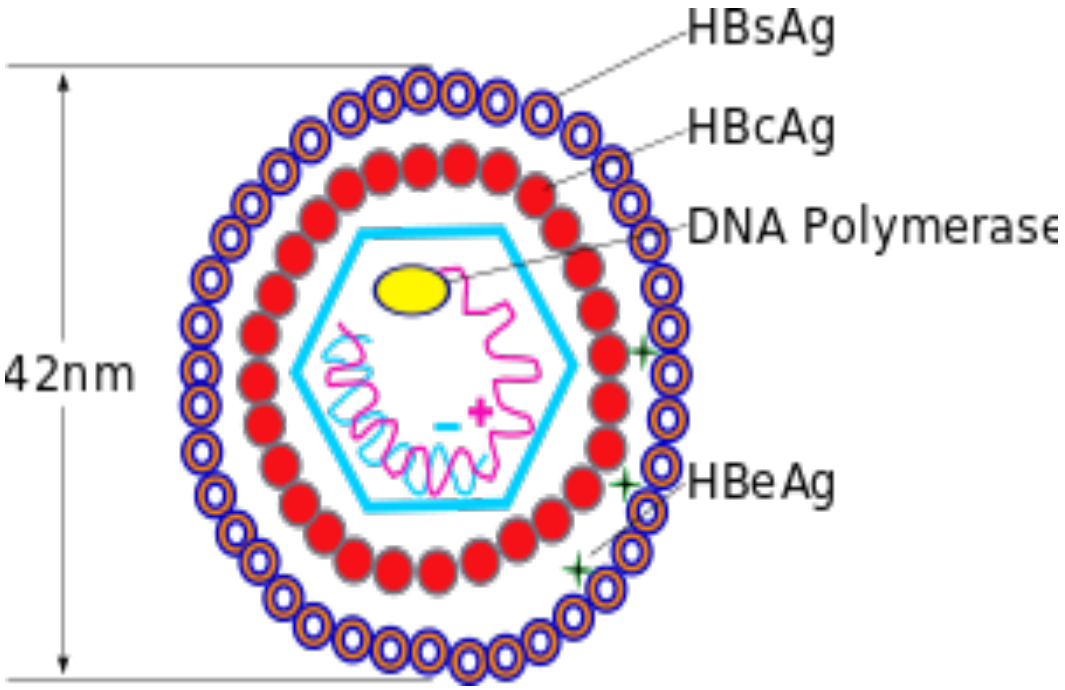
Management

Type A Hepatitis

- Bed rest during the acute stage and instructions for diet.
- During the period of anorexia, the patient should receive frequent small feedings, supplemented, if necessary, by IV fluids with glucose.
- Because this patient often has an aversion to food, gentle persistence and creativity may be required to stimulate the appetite.
- Optimal food and fluid levels are necessary to counteract weight loss and slow recovery.

Type B Hepatitis

- Hepatitis B (HBV) is a double-shelled particle containing DNA. This particle is composed of the following:
  - HBcAg hepatitis B core antigen (antigenic material in an inner core).
  - HBsAg hepatitis B surface antigen (antigenic material in an outer coat).
  - HBeAg an independent protein circulating in the blood.


HBsAg
HBcAg
DNA Polymerase
42nm
HBeAg

Type B Hepatitis

- Mode of transmission is primarily through blood.
- Oral route through saliva or through breast-feeding.
- Sexual activity through blood, semen, saliva, or vaginal secretions.
- Incubation period is 2 to 5 months.
- Occurrence is for all ages, but mostly affects young adults worldwide.
- Mortality can be as high as 10%, with another 10% of patients progressing to carrier status or developing chronic hepatitis. It is the main cause of cirrhosis and hepatocellular carcinoma worldwide.

Clinical Manifestations

Type B Hepatitis

- Symptoms onset usually more insidious and prolonged compared with HAV.
- May be asymptomatic.
- One week to 2 months of prodromal symptoms: fatigue, anorexia, transient fever, abdominal discomfort, nausea and vomiting, headache.
- Extrahepatic manifestations may include myalgia, photophobia, arthritis, angioedema, urticaria, maculopapular eruptions, skin rashes, vasculitis.

Clinical Manifestations

Type B Hepatitis

- Jaundice in icteric phase.
- In rare cases, it may progress to hepatic failure.
- May become chronic active or chronic persistent (asymptomatic) hepatitis.

Diagnostic Findings for Type B Hepatitis

- anti-HBc—antibody to core antigen or HBV; persists during the **acute phase of illness**; may indicate **continuing HB** in the liver
- anti-HBs—antibody to surface determinants on HBV; detected during **late convalescence**; usually indicates **recovery and development of immunity**
- anti-HBe—antibody to hepatitis B e-antigen; usually **signifies reduced infectivity**
- anti-HBxAg—antibody to the hepatitis B x-antigen; may indicate **ongoing replication of HBV**

Management

For HBV patients

- Alpha interferon as the single modality of therapy offers the most promise.
- This regimen of 5 million units daily or 10 million units three times weekly for 4 to 6 months results in remission of disease in approximately one third of patients.
- Two antiviral agents (lamivudine [Epvir] and adefovir [Hepsera]).

Type C Hepatitis

- Hepatitis C (HCV) was formerly called non-A, non-B hepatitis; an RNA virus.
- Found among I.V. drug users and renal dialysis patients.
- Can be transmitted through sexual intercourse.
- Incubation period varies from 1 week to several months.
- Occurs in all age-groups.

Type C Hepatitis

- Similar to those associated with HBV but usually less severe.
- Symptoms usually occur 6 to 7 weeks after transfusion but may be attributed to another viral infection and not diagnosed as hepatitis.
- Approximately 50% develop chronic liver disease and at least 20% progress to cirrhosis.

Management

For HCV patients

- Treatment of the virus with long-acting injectable interferons in combination with the oral antiviral ribavirin (Virazole) may induce a sustained response of undetectable viral levels in about 50% of people.
- Close monitoring during long treatment period is imperative.

Complications of Hepatitis

- Dehydration, hypokalemia.
- Chronic carrier hepatitis or chronic active hepatitis.
- Cholestatic hepatitis.
- Fulminant hepatitis (liver transplantation may be necessary).
- HBV and HCV carriers have a higher risk of developing hepatocellular carcinoma.

Nursing Diagnoses

All Types of Hepatitis

- Imbalanced Nutrition: Less Than Body Requirements related to effects of liver dysfunction
- Deficient Fluid Volume related to nausea and vomiting
- Activity Intolerance related to anorexia and liver dysfunction
- Deficient Knowledge related to transmission

Nursing Interventions

- **Maintaining Adequate Nutrition:** small feedings of high-calorie, low-fat diet. Avoid large quantities of protein during acute phase of illness.
- **Maintaining Adequate Fluid Intake:** Provide frequent oral fluids as tolerated. Administer I.V. fluids for patients with inability to maintain oral fluids. Monitor intake and output.
- **Maintaining Adequate Rest and Activity**
- **Ensuring Prevention of Disease Transmission**
- **Preventing and Controlling Bleeding**

Hepatic Cirrhosis

- Cirrhosis of the liver is characterized by scarring. It is a chronic disease in which there has been diffuse destruction and fibrotic regeneration of hepatic cells.

- As necrotic tissue is replaced by fibrotic tissue, normal liver structure and vasculature is altered, impairing blood and lymph flow, resulting in hepatic insufficiency and portal hypertension.

Pathophysiology and Etiology

- Alcoholic cirrhosis.
  - Fibrosis mainly around central veins and portal areas.

- Biliary cirrhosis
  - Scarring around bile ducts and lobes of the liver.
  - Results from chronic biliary obstruction and infection (cholangitis).

Clinical Manifestations

- Onset is insidious; may take years to develop.
- Early complaints include fatigue, anorexia, ankle edema in the evening, epistaxis and bleeding gums, and weight loss.
- Bleeding tendencies.

Clinical Manifestations Cont. ……

- Later complaints because of chronic failure of the liver and obstruction of portal circulation.
  - Chronic dyspepsia, constipation, or diarrhea.
  - Esophageal varices, internal hemorrhoids, ascites, splenomegaly.
  - Plasma albumin is reduced, leading to edema and contributing to ascites.
  - Anemia and poor nutrition lead to fatigue and weakness, wasting, and depression.
  - Deterioration of mental function from lethargy to delirium to coma and eventual death.

Diagnostic Evaluation

- Liver biopsy.
- Liver scan shows abnormal thickening and a liver mass.
- CT scan determines the size of the liver and its irregular nodular surface.
- Esophagoscopy to determine esophageal varices.
- Paracentesis to examine ascitic fluid for cells, protein, and bacterial counts.
- Serum liver function test results are elevated.

Management

- Minimize further deterioration of liver function through the withdrawal of toxic substances, alcohol, and drugs.

- Correction of nutritional deficiencies with vitamins and nutritional supplements and a high-calorie and moderate- to high-protein diet.

Management Cont. …

Treatment of ascites, fluid and electrolyte imbalances.

Restrict sodium and water intake, depending on amount of fluid retention.

Bed rest to aid in diuresis.

Diuretic therapy, frequently with spironolactone (Aldactone), a potassium-sparing diuretic that inhibits the action of aldosterone on the kidneys. Furosemide.

Abdominal paracentesis to remove fluid and relieve symptoms; ascitic fluid may be ultrafiltrated and reinfused through a central venous access.

Management Cont. ….

- Symptomatic relief measures, such as pain medication and antiemetics.

- Orthotopic liver transplantation may be necessary.

Nursing Diagnoses

- Activity Intolerance related to fatigue, general debility, and discomfort
- Imbalanced Nutrition: Less Than Body Requirements related to anorexia and GI disturbances
- Impaired Skin Integrity related to edema, jaundice, and compromised immunologic status
- Risk for Injury related to altered clotting mechanisms
- Disturbed Thought Processes related to deterioration of liver function and increased serum ammonia level

Nursing Interventions

- **Promoting Activity Tolerance**
- **Improving Nutritional Status**
- **Protecting Skin Integrity**
- **Preventing Injury Through Bleeding**
- **Promoting Improved Thought Processes**

DRUG ALERT

- Opioids, sedatives, and barbiturates are used cautiously in the restless patient with cirrhosis to prevent precipitation of hepatic coma.

Biliary Disorders

- **Cholelithiasis:** is stones in the gallbladder.
- **Cholecystitis:** inflammation of the gallbladder, may be acute or chronic.
- **Choledocholithiasis:** is stones in the common bile duct.
- **Cholecystitis**
- Acute cholecystitis, an acute inflammation of the gallbladder, is most commonly caused by gallstone obstruction.
- Secondary bacterial infection may occur and progress to empyema (purulent effusion of the gallbladder).

- Acalculous cholecystitis is acute gallbladder inflammation without obstruction by gallstones.
- Occurs after major surgical procedures, severe trauma, or burns.
- Chronic cholecystitis occurs when the gallbladder becomes thickened, rigid, and fibrotic and functions poorly. Results from repeated attacks of cholecystitis, calculi, or chronic irritation.

CHOLECYSTECTOMY

- Cholecystectomy is surgical removal of the gallbladder for acute and chronic cholecystitis.

  It is one of the most frequent surgical procedures.

Procedure

- Open laparotomy gallbladder removed after making an abdominal incision.
- Laparoscopy gallbladder removed from a small opening just above the umbilicus by the use of a laparoscope (also called an endoscope) for viewing.

Preoperative Management

- I.V. fluids are given before surgery to improve hydration status if the patient has been vomiting.
- Antibiotics are ordered for acute cholecystitis.
- Educate the patient about the procedure and what to expect postoperatively.
- Patient must remain NPO from midnight the night before surgery and must void before surgery.

Postoperative Management

- Postoperatively, the patient is evaluated for:
  - Vital signs, LOC.
  - Level of pain.
  - Wound or puncture sites appearance: wound drain or T-tube patency, security and drainage (if present).
  - Intake and output.
- Early ambulation is encouraged to prevent thromboembolus, to facilitate voiding, and to stimulate peristalsis.
- Complications to assess for include: incisional infection, hemorrhage, and bile duct injury (persistent pain, fever, abdominal distention, nausea, anorexia, or jaundice).

Potential Complications

- Deep vein thrombosis or pulmonary embolism
- Pneumonia or atelectasis
- Infection, hemorrhage, or bile duct injury

Nursing Diagnoses

- Acute Pain related to surgical procedure
- Risk for Infection related to surgical procedure
- Impaired Skin Integrity related to surgical procedure
- Imbalanced Nutrition: Less Than Body Requirements related to surgical procedure, wound pain, or tube drainage

Nursing Interventions

Relieving Pain

- Assess pain location, level, and characteristics.
- Administer prescribed pain medications analgesia.
- Encourage splinting of incision when moving.
- Encourage ambulation as soon as prescribed to decrease flatus and abdominal distention and to promote bowel motility.
- Instruct patient that usual activities can normally be resumed within 5 to 7 days after laparoscopic cholecystectomy or within 4 to 6 weeks of open cholecystectomy.

Nursing Interventions

Preventing Infection

- Assess wound dressings for any increased or purulent drainage.
- Assess wound drain or T-tube site for drainage, and note amount, color, and odor.
- Assess bile drainage from T-tube into bile bag:
- Administer antibiotics as prescribed.

Pancreatic Disorders

Pancreatic Disorders

- The pancreas secretes pancreatic enzymes, including amylase and lipase, through the pancreatic duct when stimulated by cholecystokinin and secretin to aid in digestion of carbohydrates and fat in the small intestine.

- The pancreas also secretes hormones, such as insulin and glucagon, which help to regulate and maintain normal serum glucose.

Acute Pancreatitis

- Acute pancreatitis: is an inflammation of the pancreas, ranging from mild edema to extensive hemorrhage, resulting from various insults to the pancreas.

- It is defined by a discrete episode of abdominal pain and serum enzymes elevations.

  The structure and function of the pancreas usually return to normal after an acute attack.

Pathophysiology and Etiology

- Excessive alcohol consumption.
- Cholelithiasis, acute and chronic cholecystitis.
- Bacterial or viral infection, blunt abdominal trauma, peptic ulcer disease, ischemic vascular disease, hyperlipidemia, hypercalcemia; the use of corticosteroids, thiazide diuretics, and oral contraceptives; surgery on or near the pancreas or after instrumentation of the pancreatic duct by ERCP; tumors of the pancreas or ampulla; and a low incidence of hereditary pancreatitis.

Pathophysiology and Etiology

- Mortality is high (10%) because of shock, anoxia, hypotension, or multiple organ dysfunction.
- Attacks may resolve in complete recovery, may recur without permanent damage, or may progress to chronic pancreatitis.
- Autodigestion of all or part of the pancreas is involved, but the exact mechanism is not completely understood.

Clinical Manifestations

- Abdominal pain, usually constant, midepigastric or periumbilical, radiating to the back or flank.
- Nausea and vomiting.
- Fever.
- Dry mucous membranes; hypotension; cold, clammy skin; cyanosis; and tachycardia.
- Shock.
- Purplish discoloration of the flanks (Turner's sign) or of the periumbilical area (Cullen's sign) occurs in extensive hemorrhagic necrosis of the pancreas.

Diagnostic Evaluation

- Serum amylase, lipase, glucose, bilirubin, alkaline phosphatase, lactate dehydrogenase, (aspartate aminotransferase)AST, (Alanine transaminase) ALT, and cholesterol may be elevated.
- Serum albumin, calcium, sodium, magnesium and, possibly, potassium may be low due of dehydration, vomiting.
- Abdominal X-ray to detect an ileus or isolated loop of small bowel overlying pancreas. Pancreatic calcifications or gallstones may suggest an alcohol or biliary etiology.

Diagnostic Evaluation

- CT scan is the most definitive study for determining pancreatic changes.

- Chest X-ray for detection of pulmonary complications. Pleural effusions are common, especially on the left, but may be bilateral.

Management

- Restoration of circulating blood volume with I.V. crystalloid or colloid solutions or blood products.
- Maintenance of adequate oxygenation (reduced by pain, anxiety, acidosis, abdominal pressure, or pleural effusions).
- Pain control to alleviate pain and anxiety, which increases pancreatic secretions.
- Rest of the GI tract.
  - Withhold oral feedings to decrease pancreatic secretions.
  - NG intubation and suction to relieve gastric

Management Cont. …..

- Maintenance of alkaline gastric pH with H2-receptor antagonists and antacids to suppress acid drive of pancreatic secretions and to prevent stress ulcer complications of acute illness.
- parenteral feedings as needed.
- Pharmacotherapy.
- Electrolyte replacements as needed.
- Sodium bicarbonate to reverse

Management Cont. …..

- **Surgical intervention if complications occur**

- Incision and drainage of infection and pseudocysts.
- Debridement or pancreatectomy to remove necrotic pancreatic tissue.
- Cholecystectomy for gallstone pancreatitis.

Complications

- Pancreatic ascites, abscess, or pseudocyst.
- Pulmonary infiltrates, pleural effusion, acute respiratory distress syndrome.
- Hemorrhage with hypovolemic shock.
- Acute renal failure.
- Sepsis and multiple-organ dysfunction syndrome.

Nursing Diagnoses

- Acute Pain related to disease process
- Deficient Fluid Volume related to vomiting, self-restricted intake, fever, and fluid shifts
- Ineffective Breathing Pattern related to severe pain and pulmonary complications

Nursing Interventions

Controlling Pain

- Administer opioid analgesics as ordered to control pain.
- Maintain NPO status to decrease pancreatic enzyme secretion.
- Provide frequent oral hygiene and care.
- Administer antacids or H2-receptor antagonists as prescribed.

Nursing Interventions Cont. …

Restoring Adequate Fluid Balance

- Monitor vital signs.
- Monitor intake and output.
- Observe and measure abdominal girth if pancreatic ascites is suspected.

Nursing Interventions Cont. …

Improving Respiratory Function

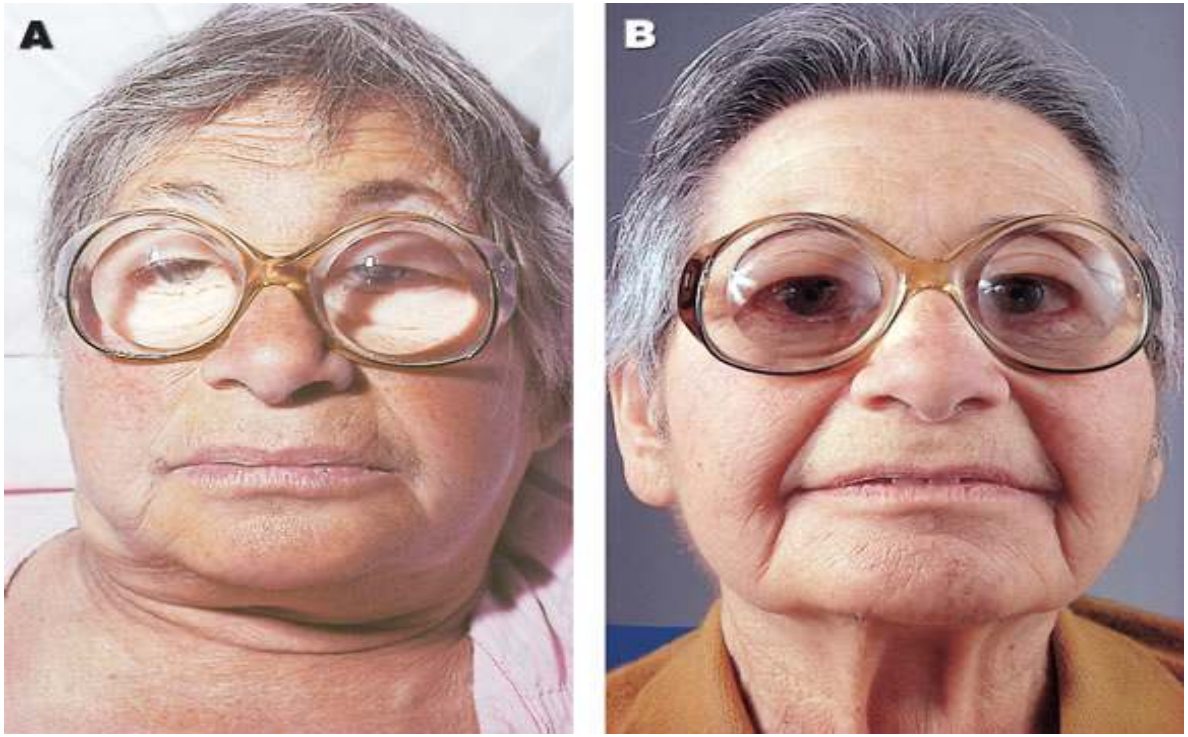
- Assess respiratory rate and rhythm, effort, oxygen saturation, and breath sounds frequently.
- Position in upright or semi-Fowler's position.
- Administer oxygen supplementation.
- Report signs of respiratory distress immediately.

Disorders of the Thyroid Gland

Thyroid Gland

- Found at the base of the throat
- Consists of two lobes and a connecting isthmus
- Major metabolic hormone
- Composed of two active iodine-containing hormones
  - Thyroxine (T4) – secreted by thyroid follicles
  - Triiodothyronine (T3) – conversion of T4 at target tissues

Hypothyroidism



Before treatment

After treatment

Diagnostic Evaluation

- Low T3 and T4 levels.
- Elevated TSH levels in primary hypothyroidism.
- Elevation of serum cholesterol.
- Electrocardiogram (ECG); sinus bradycardia, low voltage of QRS complexes, and flat or inverted T waves.

Management

- Depends on severity of symptoms; may necessitate **replacement therapy** in mild cases.
- As thyroid hormone levels gradually return to normal, the patient is monitored closely to prevent complications resulting from sudden increases in metabolic rate and oxygen requirements.

Complications

Myxedema coma:

- Hypotension, unresponsiveness, brady-cardia, hypoventilation, hyponatremia, (possibly) convulsions, hypothermia, cerebral hypoxia.
- High mortality in myxedema coma.

Nursing diagnosis

- Potential alteration in cardiac output related to decreased metabolic rate, decreased cardiac conduction, elevated cholesterol level, atherosclerosis and coronary artery disease.
- Activity intolerance related to lethargy, fatigue, and depressed neuromuscular status.
- Alteration in fluid and nutritional status related to decreased metabolic rate, poor appetite and decreased gastrointestinal function.

Nursing Interventions

Increasing Cardiac Output

- Monitor vital signs frequently.
- Monitor ECG to detect arrhythmias and deterioration of CV status.
- Prevent chilling to avoid increasing metabolic rate, Provide bed socks, bed jacket, warm environment.
- Avoid rapid rewarming because the resulting increased oxygen requirements and peripheral vasodilatation may worsen cardiac failure.
- Administer fluids cautiously, even though hyponatremia is present.
- Administer all prescribed drugs with caution before and after thyroid replacement begins.
- Report occurrence of angina.
- Monitor ABGs levels to assess cardiopulmonary function.

Hyperthyroidism

- This hypermetabolic condition is characterized by excessive amounts of thyroid hormone in the bloodstream.

Causes:

- Graves‘ disease: diffuse hyperfunction of the thyroid gland with autoimmune etiology.
- Exogenous hyperthyroidism; (iatrogenic, iodine induced)
- Thyroiditis.
- Toxic nodular goiter.
- Thyroid cancer.

Nontoxic goiter


Copyright © 2007 Lippincott Williams & Wilkins

Clinical Manifestations

- Nervousness, irritability, apprehension.
- Rapid pulse at rest and on exertion (ranges between 90 and 160); palpitations.
- Heat intolerance; profuse perspiration; flushed skin (e.g. hands may be warm, soft, moist).
- Fine tremor of hands; change in bowel habits constipation or diarrhea.
- Increased appetite and progressive weight loss; frequent stools.
- Muscle fatigability and weakness; amenorrhea.
- Atrial fibrillation possible.

Clinical Manifestations Cont. …

- Bulging eyes (exophthalmos) seen only in Graves' disease.
- Thyroid gland may be palpable.
- **Thyroid storm or crisis:**

- An extreme form of hyperthyroidism, is characterized by hyperpyrexia, diarrhea, dehydration, tachycardia, arrhythmias, extreme irritation, delirium, coma, shock, and death if not adequately treated.

- Thyroid storm may be precipitated by stress (surgery, infection) or inadequate preparation for surgery in a patient with known hyperthyroidism.

Management

- Treatment depends on causes, age of patient, severity of disease, and complications.
- Antithyroid drugs, radiation, or surgery may be used for treatment.
- Nodular toxic goiter: surgery or use of radioiodine is preferred.
- Thyroid carcinoma: surgery or radiation is used.

Management Cont. …

Pharmacotherapy

- Drugs that inhibit hormone formation (PTU; Propylthiouracil).
- Drugs to control peripheral manifestations of hyperthyroidism.
  - Propranolol (Inderal).
  - Glucocorticoids decrease the peripheral conversion of T4 to T3, a more potent thyroid hormone.

Radioactive Iodine

- Action: limits secretion of thyroid hormone by destroying thyroid tissue.
- Dosage is controlled so that hypothyroidism does not occur.

Management Cont. …

DRUG ALERT

➢ Observe the patient for evidence of iodine toxicity: swelling of buccal mucosa, excessive salivation, coryza, skin eruptions. If these occur, iodides are discontinued.

Nursing diagnosis

1. Alteration in fluid, electrolyte and nutritional balance related to hypermetabolic state.
2. Potential impaired skin integrity related to extreme diaphoresis.
3. Apprehension and anxiety related to concern about upcoming surgery.

Nursing Interventions

- **Providing Adequate Nutrition**
- **Maintaining Skin Integrity**
- **Relieving Anxiety**

Care of the Patient Undergoing Thyroidectomy

- Thyroidectomy: involves the partial or complete removal of the thyroid gland to treat thyroid tumors, hyperthyroidism, or hyperparathyroidism.

Types of Procedures

- Thyroidectomy can be total (removal of the entire thyroid gland); subtotal (95% of gland removed) to prevent damage to the parathyroid glands; and partial (one lobe or isthmus removed) to treat nodular disease.
- The parathyroid glands are usually spared to prevent hypocalcaemia.
- Indications for thyroidectomy include Graves' disease, large goiters, adenoma (thyroid cancer), and some nodules.

Preoperative Management

- The patient must be euthyroid at time of surgery.
- Iodide is given to increase firmness of thyroid gland and to reduce its vascularity after blood loss.
- Maintaining a restful and therapeutic environment.
- The patient is prepared for surgery physically and emotionally in the following ways:
  - Make a special effort to ensure that patient has a good night's rest preceding surgery.
  - Explain to the patient that speaking is to be minimized immediately postoperatively and that oxygen may be administered to facilitate breathing.
  - Explain that postoperatively, fluids may be given I.V. to maintain fluid, electrolyte, and nutritional needs.

Postoperative Management

- The patient is monitored for bleeding and respiratory distress that indicates laryngeal edema, secondary to swelling in the area of surgery.
- Signs of hypocalcaemia are watched for irritability, twitching, spasms of hands and feet.
  - I.V. calcium is used cautiously in patients who have renal disease or who are taking digoxin.

Complications

- Hemorrhage, edema of the glottis, damage to laryngeal nerve.
- Hypothyroidism occurs in 5% of patients in first postoperative year.
- Hypoparathyroidism occurs in about 4% of patients.

Nursing Interventions

- **Observing for Hemorrhage and Airway Edema:**

– Humidified oxygen, semi-Fowler's position, vital signs.

– Be alert for voice changes, Keep a tracheostomy set in the patient's room.

- **Preventing Tetany:** (hypocalcemic tetany)

– Position the patient for optimal ventilation; pillow removed to prevent head from bending forward and compressing trachea.

– Keep side rails padded and elevated and position the patient to prevent injury if a seizure occurs.

– Administer I.V. calcium as directed.

Disorders of the Parathyroid Glands

Parathyroid Glands

- The parathyroid glands are small, bean-sized structures embedded in the posterior section of the thyroid gland. Functions include the production, storage, and release of parathyroid hormone (PTH) in response to the serum level of ionized calcium. PTH increases serum calcium by decreasing elimination of calcium ions in the urine by the kidney, increasing absorption of calcium ions from the gut, and increasing bone contribution of calcium ions to the plasma.

Hyperparathyroidism

Hyperparathyroidism is hypersecretion of PTH.

Pathophysiology and Etiology

- Disorder is most common among women older than age 50.
- Primary hyperparathyroidism.
  - Single parathyroid adenoma is the most common cause (approximately 80% of cases).
  - Parathyroid hyperplasia accounts for approximately 20% of cases.
  - Parathyroid carcinoma accounts for less than 1% of cases.
- Secondary hyperparathyroidism.
  - Primarily the result of renal failure.
- Excess secretion of PTH results in increased serum calcium levels.

Clinical Manifestations

- Decalcification of bones.
  - Skeletal pain, backache, pain on weight-bearing, pathologic fractures, deformities, formation of bony cysts.
  - Formation of bone tumors overgrowth of osteoclasts.
  - Formation of calcium-containing kidney stones.
- Depression of neuromuscular function.
  - The patient may trip, drop objects, show general fatigue, lose memory for recent events, experience emotional instability, have changes in level of consciousness, with stupor and coma.
  - Cardiac arrhythmias, hypertension, cardiac standstill.

Diagnostic Evaluation

- Persistently elevated serum calcium (11 mg/100 mL).
- Exclusion of other causes of hypercalcemia.
- PTH levels are increased.
- Serum calcium and alkaline phosphatase levels are elevated and serum phosphorus levels are decreased.
- Skeletal changes are revealed by X-ray.
- Cine computed tomography (CT).

Treatment of Hypercalcemia: Management

- (Lasix) to increase urinary excretion of calcium in patients not in renal failure.
- Oral phosphate may be used as an antihypercalcemic agent.
- Dietary calcium is restricted, and all drugs that might cause hypercalcemia (thiazides, vitamin D) are discontinued.
- Dialysis may be necessary in patients with resistant hypercalcemia or those with renal failure.

Management Cont. …

- Digoxin is reduced because patient with hypercalcemia is more sensitive to toxic effects of this drug.
- Monitoring of daily serum calcium, blood urea nitrogen (BUN), potassium, and magnesium levels.

Treatment of Primary Hyperparathyroidism

- Surgery for removal of abnormal parathyroid tissue.

Complications

- Formation of renal stones, calcification of kidney parenchyma, renal shutdown.
- Ulceration of upper GI tract leading to hemorrhage and perforation.
- Demineralization of bones, cysts, and fibrosis of marrow leads to fractures, especially of vertebral bodies and ribs.
- Hypoparathyroidism after surgery.

Nursing Interventions

- Achieving Fluid and Electrolyte Balance
- Promoting Urinary Elimination
- Increasing Physical Mobility
- Relieving Anxiety
- Monitoring for Hypocalcemia Postoperatively

Hypoparathyroidism

- Hypoparathyroidism results from a deficiency of PTH and is characterized by hypocalcemia and neuromuscular hyperexcitability.

Pathophysiology and Etiology

- The most common cause is **accidental removal or destruction of parathyroid tissue** or its blood supply during thyroidectomy or radical neck dissection for malignancy.
- Decrease in gland function (**idiopathic hypoparathyroidism**); may be autoimmune or familial in origin.
- **Malignancy** or metastasis from a cancer to the parathyroid glands.

Clinical Manifestations

- **Tetany general muscular hyperactivity**: attempts at voluntary movement result in tremors and spasmodic or uncoordinated movements; fingers assume classic tetanic position.
  - **Chvostek's sign**: a spasm of facial muscles that occurs when muscles or branches of facial nerve are tapped.
  - **Trousseau's sign**: carpopedal spasm within 3 minutes after a BP cuff is inflated 20 mm Hg above the patient's systolic pressure.
  - **Laryngeal spasm.**
- **Severe anxiety and apprehension**.
- **Renal colic** is usually present if the patient has a history of **stones**; preexisting stones loosen and migrate into the ureter

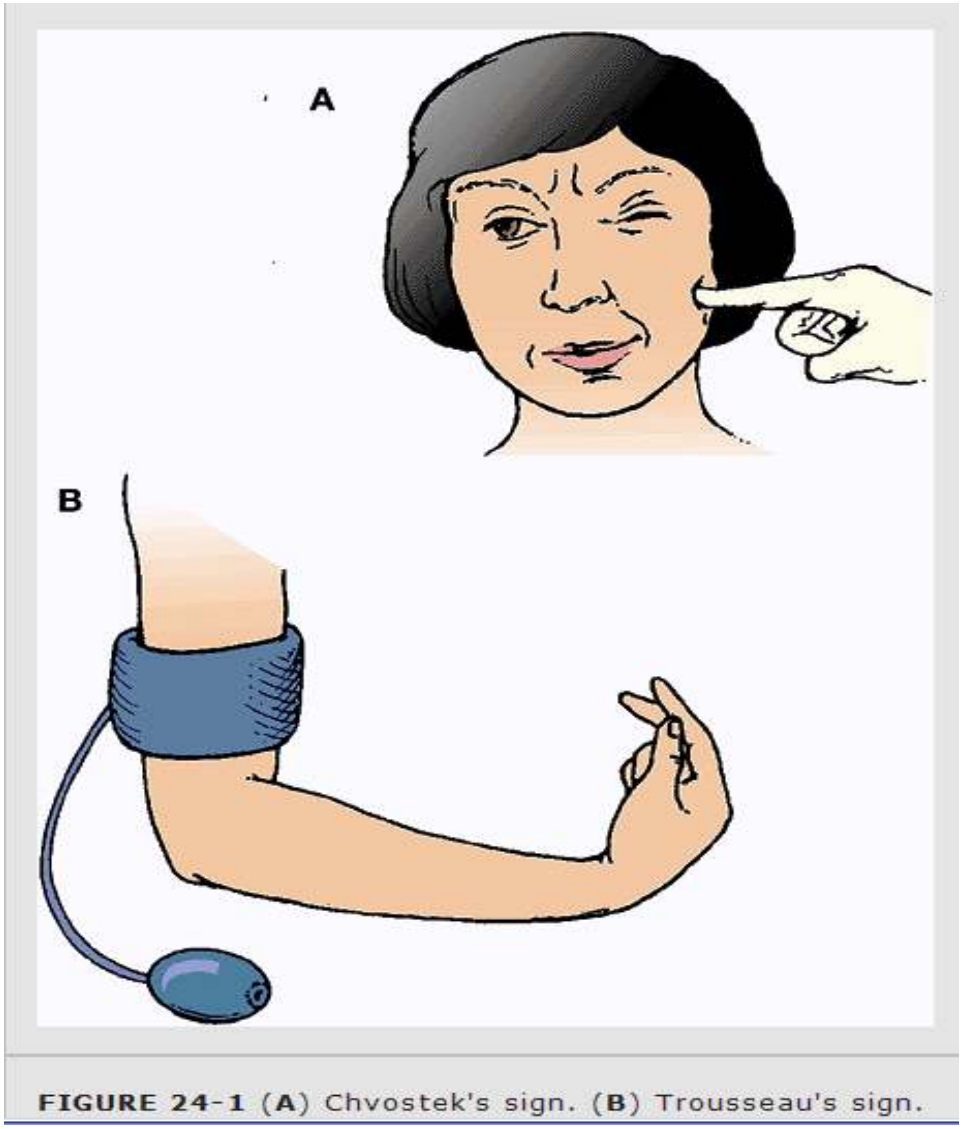


FIGURE 24-1 (**A**) Chvostek's sign. (**B**) Trousseau's sign.

Diagnostic Evaluation

- **Phosphorus level** in blood is elevated.
- Decrease in **serum calcium level** to a low level (7.5 mg/100 mL or less).
- **PTH levels are low in most cases**; may be normal or elevated in pseudohypoparathyroidism.

Management

- **I.V. Calcium Administration**
  - **All I.V. calcium preparations are administered slowly. It is highly irritating, stings, and causes thrombosis; patient experiences unpleasant burning flush of skin and tongue.**
- **Later, vitamin D is added to calcium intake** increases absorption of calcium, **Thiazide diuretics** may also be added because of their calcium-retaining effect on the kidney; doses of calcium and vitamin D may be lowered.
- **Administration of I.V. calcium seems to cause rapid relief of anxiety.**

Complications

- Acute complications related to hypocalcemia include **seizures, tetany, and mental disorders**, all of which can be reversed with calcium therapy.
- If onset of hypocalcemia is acute, the major concerns are **laryngeal spasm, acute airway obstruction, and cardiovascular failure.**
- Long-term complications include subcapsular cataracts, calcification of the basal ganglia, and papilledema.

Nursing Interventions

Maintaining Normal Serum Calcium Levels

- Assess **neuromuscular status** frequently in patients with hypoparathyroidism.
- Assess **respiratory status** frequently in acute hypocalcemia and postoperatively.
- Monitor **serum calcium** and phosphorus levels.
- Promote **high-calcium diet** if prescribed dairy products, green, leafy vegetables.

Nursing Interventions Cont. …

- Use caution in administering other drugs to the patient with hypocalcemia.
- The hypocalcemic patient is **sensitive to digoxin** (Lanoxin); as hypocalcemia is reversed, the patient may rapidly develop digitalis toxicity.
- **Cimetidine** (Tagamet) interferes with normal parathyroid function, especially with renal failure, which increases the risk of hypocalcemia.

Disorders of the Adrenal Glands

- **The adrenal medulla, or inner portion of the gland**
- Is not necessary to maintain life, but enables a person to cope with stress. **It secretes two hormones:**
  - **Epinephrine (adrenalin):**

1. acts on alpha and beta receptors to increase contractility and excitability of heart muscle, leading to increased cardiac output; facilitates blood flow to muscles, brain, and viscera;
2. enhances blood sugar by stimulating conversion of glycogen to glucose in liver; and inhibits smooth muscle contraction.

  - **Norepinephrine (noradrenaline):**

- acts primarily on alpha receptors to increase peripheral vascular resistance, leading to increases in diastolic and

- **The adrenal cortex, or outer portion of the gland:**

- Is essential to life. It secretes adrenocortical hormones synthesized from cholesterol.

1. **Glucocorticoids (cortisone and hydrocortisone):** enhance protein catabolism and inhibit protein synthesis; antagonize action of insulin and increase blood sugar; increase synthesis of glucose by liver; influence defense mechanism of body and its reaction to stress; and influence emotional reaction.
2. **Mineralocorticoids (aldosterone and desoxycorticosterone):** regulate reabsorption of sodium; regulate excretion of potassium by renal tubules.
3. **Adrenosterones (adrenal androgens)** exert minimal effect on sex characteristics and function.

Cushing's Syndrome

- Cushing's syndrome: is a condition in which the plasma cortisol levels are elevated, causing signs and symptoms of hypercortisolism.
- **Clinical Manifestations**

Manifestations Caused by Excess Glucocorticoids

- Weight gain or obesity, Heavy trunk; thin extremities.
- Buffalo hump (fat pad) in neck and supraclavicular area.
- Moon face.
- Fragile and thin skin.
- Muscles wasted because of excessive catabolism.
- Osteoporosis characteristic kyphosis, backache.
- Mood changes, psychosis.
- Increased susceptibility to infections.

Manifestations Caused by Excess Mineralocorticoids

- Hypertension.
- Hypernatremia, hypokalemia.
- Weight gain.
- Expanded blood volume.
- Edema.

Manifestations Caused by Excess Androgens

- Women experience virilism (masculinization):
  - Hirsutism excessive growth of hair on the face and midline of trunk.
  - Breasts atrophy Voice masculine Loss of libido
- If exposed in utero possible hermaphrodite.
- Males loss of libido

Management

Surgical and Radiation

- Tumor (adrenal or pituitary) is removed or treated with irradiation.

Replacement Therapy Postoperatively

- Adrenalectomy patients require a lifelong replacement therapy with the following:
  - A glucocorticoid cortisone (Cortef).
  - A mineralocorticoid fludrocortisone (Florinef).

Medical Treatment

- If patients cannot undergo surgery, cortisol synthesis-inhibiting medications may be used.

Nursing Interventions

- Maintaining Skin Integrity
- Encouraging Active Participation in Self-Care
- Strengthening Body Image
- Reducing Anxiety



Adrenal Insufficiency

- Adrenal cortex function is inadequate to meet the needs for cortical hormones
- Primary: Addison's Disease
- Secondary: May be the result of adrenal suppression by exogenous steroid use

Addison's Disease

- Adrenocortical insufficiency
- Autoimmune or idiopathic atrophy
- Can be caused by inadequate ACTH from pituitary
- Therapeutic use of steroids

Manifestations

- Muscle weakness
- Anorexia
- Dark pigmentation
- Hypotension
- Hypoglycemia
- Low sodium levels
- High potassium levels
- Can result in *Addisonian crisis*

Addisonian crisis

- Circulatory shock
- Pallor, apprehension, weak & rapid pulse, rapid respirations and low blood pressure
- Headache, nausea, abdominal pain and diarrhea
- Can be brought on by overexertion, exposure to cold, acute infection, decrease in salt intake

Assessment and Diagnostic Findings

- Early morning serum cortisol and plasma ACTH are performed. Will distinguish between primary and secondary adrenal insufficiency. In primary, will have elevated ACTH levels and below normal cortisol levels.
- If the adrenal cortex is not stimulated by the pituitary, a normal response to doses of exogenous ACTH (see text)
- Blood sugar levels and electrolyte values

Management

- Restore circulatory status—fluids, steroids
- May need antibiotics if infection precipitated crisis
- May need lifelong steroid therapy and mineralocorticoid therapy
- May need additional salt intake
- Check orthostatics
- Daily weights
- Aware that stressors can precipitate crises
- Medic alert bracelet or similar identification of history

Nursing Interventions for a Patient Receiving Steroid Therapy

Preventing Infection

- Steroids may affect the circulating blood, resulting in decreased eosinophils and lymphocytes, increased red cells, and increased incidence of thrombophlebitis and infection.
- Encourage the patient to avoid crowds and the possibility of exposure to infection.
- Encourage exercise to prevent venous stasis.
- Be aware that signs of infection/inflammation may be masked fever, redness, swelling.
- Practice and encourage good hand-washing technique and asepsis.

Preventing Nutritional and Metabolism Complications

- Determine whether the patient needs assistance in dietary control. Steroids may cause weight gain and an increase in appetite.
- Encourage a high-protein, high-carbohydrate diet. Because steroids affect protein metabolism, there may be negative nitrogen balance.
- Encourage the patient to take steroids with milk or with food. Because steroids increase secretion of gastric acid and have an inhibiting effect on secretion of mucus in the stomach, they may cause peptic ulcer.
- Check fasting blood glucose levels.
  - Steroids precipitate gluconeogenesis and insulin antagonism, which results in hyperglycemia, glucosuria, decreased carbohydrate tolerance.
  - Temporary insulin injections may be necessary.

Observing for Bone Complications

- Possibility of pathologic fractures.
  - Steroids affect the musculoskeletal system, causing potassium depletion and muscular weakness.
  - Steroids cause increased output of calcium and phosphorus, which may lead to osteoporosis.

- Administer a diet high in calcium and protein.

- Recommend a program of activities of daily living (ADLs) and weight-bearing; normal range of motion and safe repositioning for the bedridden.

Avoiding Electrolyte Disturbance

- Restrict sodium intake and increase potassium intake.
  - Mineralocorticoids differ from other steroids, resulting in sodium retention and potassium depletion: edema, weight gain.
  - Lemon juice is high in potassium and low in sodium.
  - Avoid saline as a diluents in preparing injectable medications.
- Check BP frequently and weigh the patient daily.
- Observe for edema.

Monitoring Behavioral Reactions

- Watch for convulsive seizures (especially in children).
- Recognize and report any mood that deviates from the usual behavior patterns.
- Report unusual behavior, haunting dreams, withdrawal, or suicidal tendencies.

Preventing Stress Reactions

- Recommend that the patient carry an identification card that indicates steroid therapy and name of health care provider.
- Be aware that steroids affect the hypothalamic-pituitary-adrenal system; this affects the patient's ability to respond to stress.
- Advise the patient to avoid extremes of temperature, infections, and upsetting situations.

Preventing Injury and Promoting Healing

- Instruct the patient to avoid injury; stress safety precautions. Because steroids interfere with fibroblasts and granulation tissue, altered response to injury results in impaired growth and delayed healing.
- Observe daily the healing process of wounds, particularly surgical wounds, to recognize the potential for wound dehiscence.

Disorders of the Pituitary Gland

- The pituitary gland (hypophysis) exerts prime control over the body's hormonal functions. Its function is regulated by the hypothalamus. The pituitary consists of two parts that are structurally and functionally separate, the anterior pituitary and the posterior pituitary.

  Hypothalamic control of the anterior pituitary is mediated by releasing factors secreted by the hypothalamus; the posterior pituitary is regulated through direct neural stimulation.

Hormones of the Pituitary Gland

HORMONE	TARGET TISSUE
Anterior Pituitary	
• Growth hormone	Multiple sites
• Thyroid-stimulating hormone	Thyroid gland
• Adrenocorticotropic hormone	Adrenal glands
• Prolactin	Breasts
• Luteinizing hormone	Ovaries, testes
• Follicle-stimulating hormone	Ovaries, testes
• Melanocyte-stimulating hormone (skin)	Melanocytes
Posterior Pituitary	
• Oxytocin	Uterus, breasts
• Antidiuretic hormone	Kidneys

Diabetes Insipidus

- DI: is a disorder of water metabolism caused by deficiency of ADH, also called vasopressin, secreted by the posterior pituitary or by inability of the kidneys to respond to ADH (nephrogenic DI).

Etiology

- Primary: idiopathic.
- Secondary: head trauma, neurosurgery, tumors, vascular disease (aneurysms, infarct), infection (meningitis, encephalitis).
- Nephrogenic DI—relates to failure of the renal tubules to respond to ADH.
- DI may be transient or permanent.
- Deficiency of ADH results in decreased renal water reabsorption; it may be partial or complete.

Manifestations

- Marked polyuria: daily output of 5 to 20 L of dilute urine; appearance of urine like that of water, with a specific gravity of 1.000 to 1.005.
- Polydipsia (intense thirst): drinks 4 to 40 L of fluid daily; has craving for cold water.
- High serum osmolality (above 295 mOsm), and high serum sodium level (greater than 145 mEq/L).

Diagnostic Findings

- **Fluid deprivation test**
  - Withhold fluids for 8-12 hours.
  - Weigh patient frequently.
  - Inability to slow down the urinary output and fail to concentrate urine are diagnostic.
  - Stop test if patient is tachycardic or hypotensive.
- Serum osmolality: high
- Urine osmolality: low
- Measurements of serum and urine ADH: decreased to absent.

Management

- Desmopressin acetate (DDAVP)—vasopressin derivative administered into the nose through a soft, flexible nasal tube or by nasal spray.
- Duration of action 12 to 24 hours.
- Can also use Diabenese and thiazide diuretics in mild disease as they potentiate the action of ADH.
- If renal in origin—thiazide diuretics, NSAIDs (prostaglandin inhibition) and salt depletion may help.
- Educate patient about actions of medications,

Clinical Manifestations Associated with Hormone Effects of Pituitary Tumors

HORMONE	HYPERPITUITARISM (INCREASED SECRETION)	HYPOPITUITARISM (DIMINISHED SECRETION)
Growth hormone	o Gigantism (child) o Acromegaly (adult)	o Shortness of stature (child) o Silent (adult)
Prolactin	o Infertility and galactorrhea (female)	o Postpartum lactation failure
Adrenocorticotropic hormone	o Cushing's disease	o Adrenocortical insufficiency
Thyroid-stimulating hormone	o Hyperthyroidism	o Hypothyroidism
Luteinizing hormone and follicle-stimulating	o Gonadal dysfunction	o Hypogonadism

Diabetes Mellitus

Pathophysiology and Etiology

- Absolute deficits in insulin result from decreased production of endogenous insulin by the beta cell of the pancreas.

- Relative deficits in insulin are caused by inadequate utilization of insulin by the cell.

Classification Of Diabetes

Type 1 Diabetes Mellitus

- Type 1 diabetes mellitus was formerly known as insulin dependent diabetes mellitus and juvenile diabetes mellitus.
- Little or no endogenous insulin, requiring injections of insulin to control diabetes and prevent ketoacidosis.
- 5% to 10% of all diabetic patients have type 1.

- Etiology: autoimmunity, viral, and certain histocompatibility antigens as well as a genetic component.
- Usual presentation is rapid with classic symptoms of polydipsia, polyphagia, polyuria, and weight loss.
- Most commonly seen in patients under age 30 but can be seen in older adults.

Type 2 Diabetes Mellitus

- Type 2 diabetes mellitus was formerly known as noninsulin dependent diabetes mellitus or adult onset diabetes mellitus.
- Caused by a combination of insulin resistance and relative insulin deficiency some individuals have predominantly insulin resistance, whereas others have predominantly deficient insulin secretion, with little insulin resistance.

- Approximately 90% of diabetic patients have type 2.
- Etiology: strong hereditary component, commonly associated with obesity.
- Usual presentation is slow and typically insidious with symptoms of fatigue, weight gain, poor wound healing, and recurrent infection.

- Found primarily in adults over age 30; however, may be seen in younger adults and adolescents who are overweight.
- Patients with this type of diabetes, but who eventually may be treated with insulin, are still referred to as having type 2 diabetes.

DIAGNOSTIC TESTS

- **Blood Glucose**
  - Fasting blood sugar (FBS): drawn after at least an 8-hour fast, to evaluate circulating amounts of glucose.
    - FBS greater than or equal to 126 mg/dL on two occasions
  - Random glucose: drawn at any time, nonfasting.
    - Random blood sugar greater than or equal to 200 mg/dL and presence of

- Postprandial test: drawn usually 2 hours after a well-balanced meal, to evaluate glucose metabolism.
- Oral Glucose Tolerance Test: the oral glucose tolerance test (OGTT) evaluates insulin response to glucose loading.
  - Diagnostic for diabetes mellitus if 2-hour value is 200 mg/dL or greater.

- **Glycated Hemoglobin (Glycohemoglobin, HbA1c)**
- Measures glycemic control over a 60- to 120-day period by measuring the irreversible reaction of glucose to hemoglobin through freely permeable erythrocytes during their 120-day lifecycle.

- **Urinalysis**
- Glycosuria, Ketone bodies

Clinical Manifestations

➢ Hyperglycemia

- Weight loss, fatigue
- Polyuria, polydipsia, polyphagia
- Blurred vision

➢ Altered Tissue Response

- Poor wound healing
- Recurrent infections, particularly of the skin

Management

- Diet
- Dietary control with caloric restriction of carbohydrates and saturated fats to maintain ideal body weight.
- The goal of meal planning is to control blood glucose and lipid levels.
- Weight reduction is a primary treatment for type 2 diabetes.

Management Cont. …..

- Exercise
- Regularly scheduled, moderate exercise performed for at least 30 minutes most days of the week promotes the utilization of carbohydrates, assists with weight control, enhances the action of insulin, and improves cardiovascular fitness.

Management Cont. …..

- Pharmacologic

A. Oral Antidiabetic Agents:

- Oral antidiabetic agents for patients with type 2 diabetes who do not achieve glucose control with diet and exercise only.
- First-generation sulfonylureas such as: Daonil and diabenase.
- Second generation sulfonylureas such as: micronase.
- Biguanides such as: Metformin.

B. Insulin therapy

- Insulin therapy for patients with type 1 diabetes who require replacement.
  - May also be used for type 2 diabetes when unresponsive to diet, exercise, and oral antidiabetic therapy.
  - Commonly results in increased appetite and weight gain.
  - Short-acting (Regular); Onset:1⁄2–1 h, Peak: 2–3 h, Duration: 4–6 h.
  - Intermediate-acting (NPH) (neutral protamine Hagedorn); Onset: 2–4 h, Peak: 4–12 h, Duration: 16–20 h.

NURSING ALERT

- Regular insulin is the only insulin that may be administered I.V.;
- All other insulin formulations are suspensions for subcutaneous injection only.

Acute Complications of D.M.

1. Hypoglycemia

occurs as a result of an imbalance in food, activity, and insulin/oral antidiabetic agent.

Preventing Injury Secondary to Hypoglycemia

- Closely monitor blood glucose levels to detect hypoglycemia.
- Instruct patient in the importance of accuracy in insulin preparation and meal timing to avoid hypoglycemia.
- Assess patient for the signs and symptoms of hypoglycemia.
- Treat hypoglycemia promptly with 15 to 20 g of fast-acting carbohydrates.
- Encourage patient to carry a portable treatment for hypoglycemia at all times.

- Assess patient for cognitive or physical impairments that may interfere with ability to accurately administer insulin.
- Between-meal snacks as well as extra food taken before exercise should be encouraged to prevent hypoglycemia.
- Encourage patients to wear an identification bracelet or card that may assist in prompt treatment in a hypoglycemic emergency.

2. DIABETIC KETOACIDOSIS

DKA: is an acute complication of diabetes mellitus (usually type 1 diabetes) characterized by hyperglycemia, ketonuria, acidosis, and dehydration.

Pathophysiology and Etiology

- Insulin deficiency prevents glucose from being used for energy, forcing the body to metabolize fat for fuel.
- Ketone bodies are organic acids that cause metabolic acidosis.

- Osmotic diuresis caused by hyperglycemia creates a shift in electrolytes, with losses in potassium, sodium, phosphate, and water.
- Caused by inadequate amounts of endogenous or exogenous insulin.
  - Frequently occurs due to failure to increase the dose of insulin during periods of stress (eg, infection, surgery, pregnancy).

Clinical Manifestations

- Early
  - Polydipsia, polyuria
  - Fatigue, malaise, drowsiness
  - Anorexia, nausea, vomiting
  - Abdominal pains, muscle cramps
- Later
  - Kussmaul respiration (deep respirations)
  - Fruity, sweet breath
  - Hypotension, weak pulse
  - Stupor and coma

Diagnostic Evaluation

- Serum glucose level is usually elevated over 300 mg/dL; may be as high as 1,000 mg/dL.
- Serum and urine ketone bodies are present.
- Serum bicarbonate and pH are decreased due to metabolic acidosis, and partial pressure of carbon dioxide is decreased as a respiratory compensation mechanism.

- BUN, creatinine, hemoglobin, and hematocrit are elevated due to dehydration.
- Urine glucose is present in high concentration and specific gravity is increased, reflecting osmotic diuresis and dehydration.

Management

- I.V. fluids to replace losses from osmotic diuresis, vomiting.
- I.V. insulin drip regular insulin infused only to increase glucose utilization and decrease lipolysis.
- Electrolyte replacement sodium chloride and phosphate as required, potassium chloride and bicarbonate based on laboratory results.

3. HYPEROSMOLAR HYPERGLYCEMIC NONKETOTIC SYNDROME

HHNKS: is an acute complication of diabetes mellitus (particularly type 2 diabetes) characterized by hyperglycemia, dehydration, stupor, and hyperosmolarity, but little or no ketosis.

Chronic Complications of D.M.

- Cerebrovascular Disease
- Coronary Artery Disease (CAD)
- Peripheral Vascular Disease
- Retinopathy
- Nephropathy
- Peripheral Neuropathy
- Autonomic Neuropathy
- Gastroparesis
- Diarrhea
- Impotence/Sexual Dysfunction
- Orthostatic Hypotension

Foot and Leg Problems:

- Between 50% and 75% of lower extremity amputations are performed on people with diabetes.
- More than 50% of these are preventable by providing foot care measures and practice them on a daily basis.

Complications of DM lead to foot problems and infections:

- Neuropathy: Sensory neuropathy leads to loss of pain and pressure sensation, and autonomic neuropathy leads to increased dryness and fissuring of the skin (secondary to decreased sweating).
- Peripheral vascular disease: Poor circulation of the lower extremities contributes to poor wound healing and the development of gangrene.
- Immunocompromise: Hyperglycemia impairs the ability of specialized leukocytes to destroy bacteria.

Causes of foot injuries:

1. Thermal: Patients with an insensitive foot do not feel injuries, which may be (eg, from using heating pads, walking barefoot on hot concrete, testing bath water with the foot).
2. Chemical (eg, burning the foot while using caustic agents).
3. Traumatic (eg, injuring skin while cutting nails, walking with an undetected foreign object in the shoe, or wearing ill-fitting shoes and socks).

Nursing interventions of preventive foot care that are taught to patients and families include the following:

- Daily visual and manual inspection of the feet.
- The feet should be examined during every health care visit or at least once per year.
- Properly bathing, drying, and lubricating the feet, taking care not to allow moisture (water or lotion) to accumulate between the toes.

- Wearing closed-toed shoes that fit well.
- Trimming toenails straight across and filing sharp corners to follow the contour of the toe.
- Reducing risk factors, such as smoking and elevated blood lipids, that contribute to peripheral vascular disease.
- Avoiding home remedies, over-the-counter agents, and self medicating to treat foot problems.

BOX 38–8 Foot Care Tips

Wash and dry feet every day. Use warm (not hot) water.
Apply lotion that does not contain alcohol, avoiding areas between toes.
Inspect feet for sores or red areas daily (have a family member help if necessary).
Report any abnormalities immediately.
Wear leather shoes and cotton socks.
Never go barefoot.
Avoid garters and tight socks.
Avoid crossing legs.
Cut toenails to natural shape of nail—not into corners.
See a podiatrist for calluses or problem toenails (avoid "bathroom surgery").
Have feet checked at least once a year, preferably 3–4 times a year, for loss of sensation.

Management of Patients with Dysrhythmias and Conduction Problems

Dysrhythmias: disorders of the formation or conduction (or both) of the electrical impulses in the heart

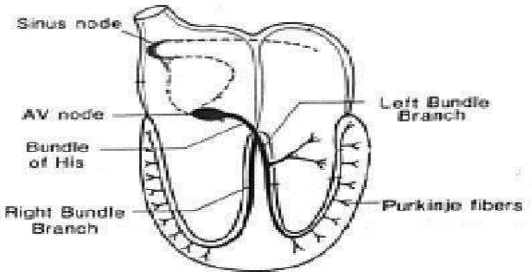
- These disorders can cause disturbances of:
  - ✓ Rate
  - ✓ Rhythm
  - ✓ Both rate and rhythm
- Potentially can alter blood flow & cause hemodynamic changes
- Diagnosed by analysis of ECG waveform

Conducting system of the heart

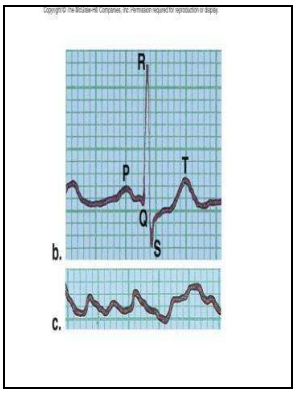
- The SA (sinoatrial) node, or pacemaker, initiates the heartbeat approximately 60-100 impulses / min.
- The AV (atrioventricular) node conveys the stimulus and initiates contraction of the ventricles, located right Atrial wall, similar to S.A node but with impulses about 40-60/ min.

The signal for the ventricles to contract travels from the AV node through the atrioventricular bundle to the smaller

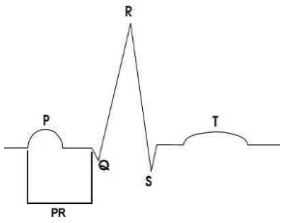
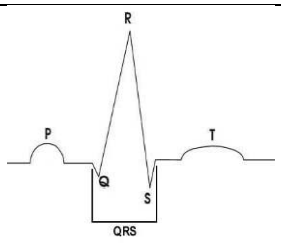
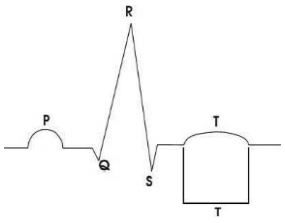
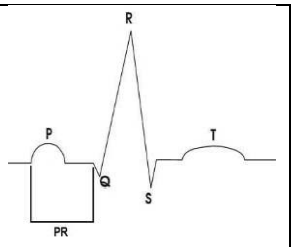
Normal Impulse Conduction

Sinoatrial node ↓ AV node ↓ Bundle of His ↓ Bundle Branches ↓ Purkinje fibers	

Electrocardiogram



- Electro- : Refers to electrical activity
- Gram-: Picture
- Cardio-: Heart
- ECG is a pictorial representation of the electrical activity occurring within the heart
- Electrical impulse stimulates the heart to contract
- A small amount of electricity spreads on the body surface
- This electricity can be measured and recorded on the ECG
- **P wave** represents depolarization (contraction) of the atrium
- **QRS** wave represent the depolarization of the ventricles
- **PR interval** represents the impulses for the time it leave the SA node through the delay at the AV node and its arrival at purkinje fibers
- P – R interval:- the time required for atrial depolarization.
- ST segment: End of the QRS complex (early ventricular repolarization)
- T wave represents the repolarization (relaxation) of the ventricles
- U wave:- repolarization of purkinje fibers sometimes seen ↓ K and approximate the same size as P wave

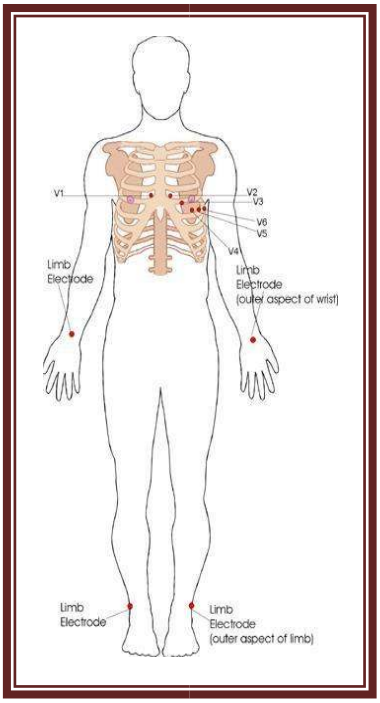
			
P wave represents depolarization of the atrium	**QRS wave** represent the depolarization of the ventricles	**T wave** represents the repolarization (relaxation) of the ventricles	**P – R** interval:- the time required for atrial depolarization.

Electrocardiogram

- Recording of hearts electrical activity.
- Allows the physician to evaluate the hearts:
  - Rate
  - Rhythm
  - Certain cardiac problems
- Quick – safe inexpensive non invasive

ECG leads

- **For standard 12** lead ECG, We have 6 Limb leads and 6 Chest leads, 10 electrodes (six on the chest and four on the limbs) are placed on the body.
- The 12-Lead ECG sees the heart from 12 different views.
- Therefore, the 12-Lead ECG helps you see what is happening in different portions of the heart.
- The rhythm strip is only 1 of these 12 views.



- ❖ Four Limbs: RA, LA, RL and LL.
- ❖ **Chest leads**: from V1 to V6

V1 4th intercostal space just to the right of the sternum

V2 4th intercostal space just to the left of the sternum

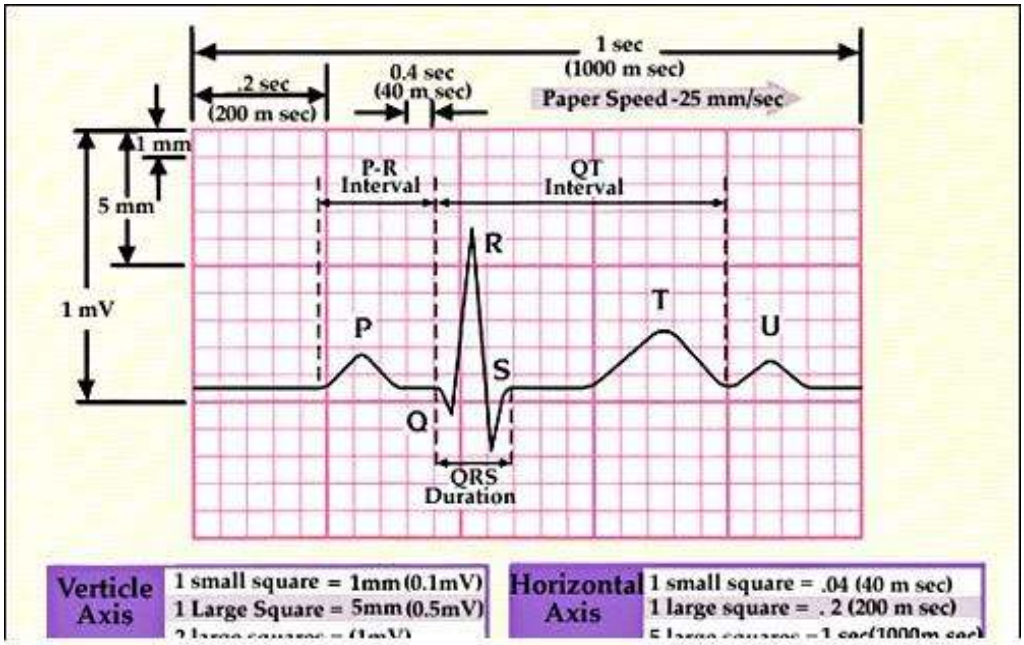
V3 Halfway between V2 and V4

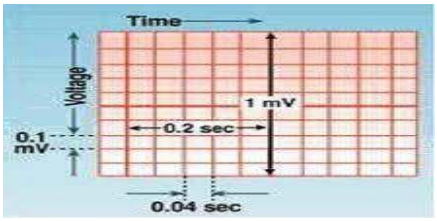
V4 5th intercostal space in the left mid-clavicular line

V5 On same horizontal as V4 in anterior axillary line

V6 On same horizontal as V4 in mid-clavicular axillary line

ECG PAPER





Analysis of ECG

- Horizontal axis, the duration of 1 small box square = 0.04 second
- Vertical axis, the height of 1 small box square = 1 mm
- P wave: 2.5 millimeter squares [mm] or less and 0.11 second or less on duration
- QRS: less than 0.12 second, less than 5 mm
- U wave:- the same size as P wave . 2.5 mm.
- P – R interval: Normal interval .12 to .2 second 3 – 5 mm

Estimating Heart Rate

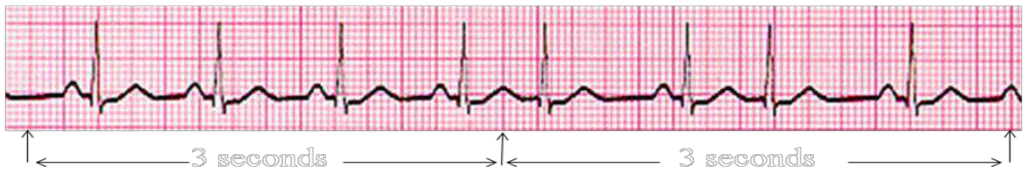
- Heart rate determination for a regular rhythm: 1500 divided by the number of small boxes between two R waves.
- Or Count numbers of large boxes between R-R and divide that into 300
- 1 sec = 5 large boxes = 25 small boxes
- 1 sec = 25 small boxes 60 sec= 25X60=1500



- Example: Heart rate: regular rhythm 1500
- The number of small boxes between two R= 13
- HR: 1500/13= 115 bpm
- If two large box: HR: 300/2= 150 bpm

Estimating Heart Rate

- Heart rate determination if the rhythm is irregular, count the number of RR interval in 6 seconds and multiply the number by 10.



3 seconds 3 seconds

- Heart rate: 10 x 8 = 80bpm

- Rate: how fast the heart is beating

- Normal rate [60-100 beats per minute (bpm)]
- Abnormal rate: Slow (<60 bpm) , Fast (>100 bpm)

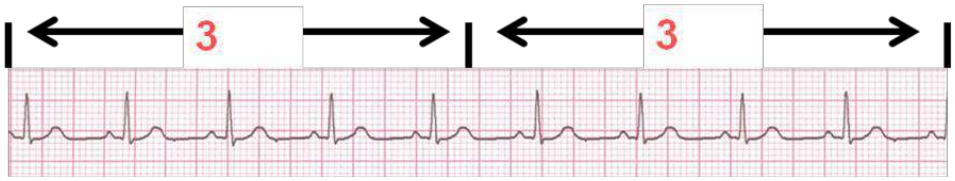
- Rhythm: the pattern of the heart beat, regular or irregular

Rhythm Analysis



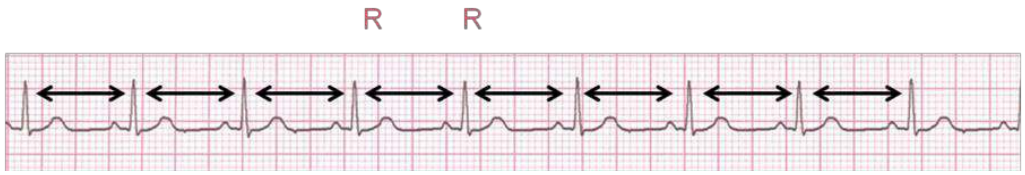
- Step 1: Calculate rate.
- Step 2: Determine regularity.
- Step 3: Assess the P waves.
- Step 4: Determine PR interval.
- Step 5: Determine QRS duration.

Step 1: Calculate Rate



- Count the # of R waves in a 6 second rhythm strip, then multiply by 10.
- Reminder: all rhythm strips in the Modules are 6 seconds in length.
- Interpretation: 9 x 10 = 90 bpm

Step 2: Determine regularity



- Look at the R-R distances (using a caliper or markings on a pen or paper).

- Regular (are they equidistant apart)? Occasionally irregular? Regularly irregular? Irregularly irregular?
- Interpretation? Regular

Step 3: Assess the P waves



- Are there P waves?
- Do the P waves all look alike? & Do the P waves occur at a regular rate?
- Is there one P wave before each QRS?
- Interpretation? Normal P waves with 1 P wave for every QRS

Step 4: Determine PR interval



- Normal: 0.12 - 0.20 seconds (3 - 5 boxes)
- Interpretation? *0.*12 seconds

Step 5: QRS duration



- Normal: 0.04 - 0.12 seconds. (1 - 3 boxes)
- Interpretation? *0.08 seconds*

Rhythm Summary



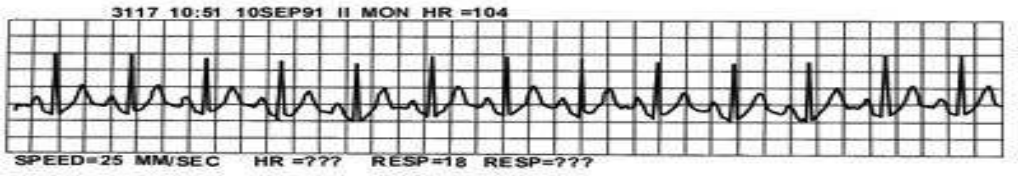
- Rate 90-95 bpm
- Regularity regular
- P waves normal
- PR interval 0.12 s
- QRS duration 0.08 s
- Interpretation? Normal Sinus Rhythm

Sinus Bradycardia



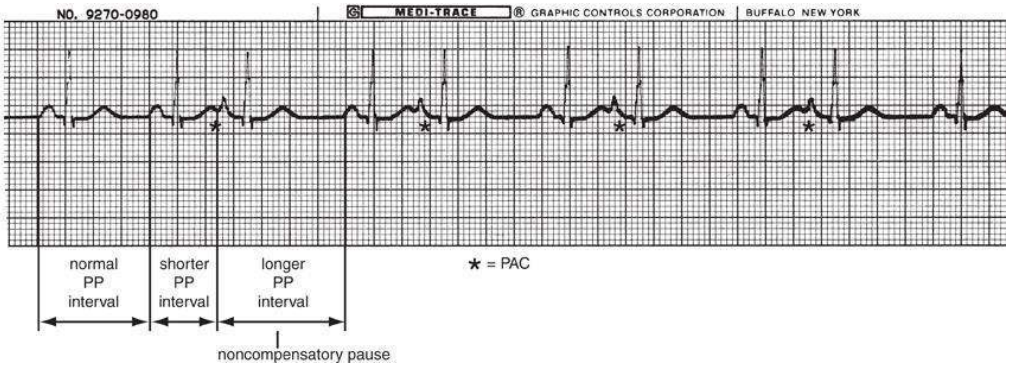
Regularity	**Regular**
Rate	**Less than 60**
P : QRS	**1 : 1**
QRS	**≤ 0.12 sec.**
P-R interval	**Constant, normal 0.12-0.20 sec**
Sinus Bradycardia	

Sinus Tachycardia



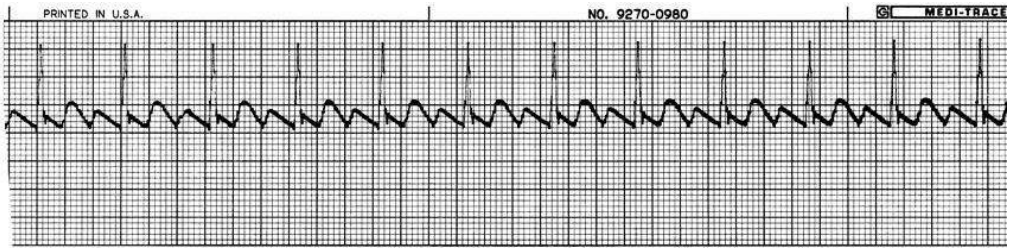
Regularity	**Regular**
Rate	**100 – 160 beats/minute "180"**
P : QRS	**1 : 1**
QRS	**≤ 0.12 sec.**
P-R interval	**Constant, normal 0.12-0.20 sec**
Sinus Tachycardia	

Premature Atrial Complexes (PAC)



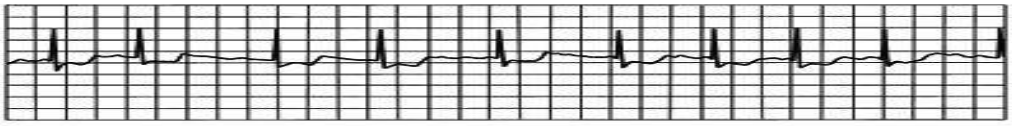
- Occurs when an electrical impulses starts in the atrium before the next normal impulses of the sinus node.
- Ventricular and atrial rhythm: irregular due to P waves, creating a PP interval that is shorter than the others. Sometimes followed by longer than normal PP.
- Causes: caffeine, anxiety, hypokalemia and nicoton.

Atrial flutter



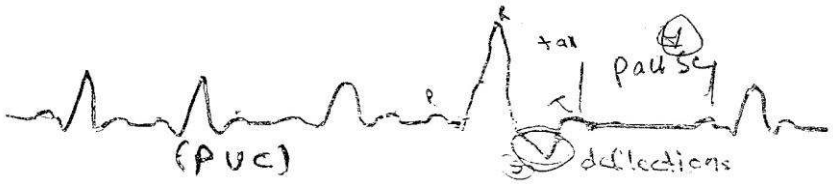
- Occur in the atrium and create impulses at an atrial rate between 250 – 400, because the atrial rate is faster than the AV node can conduct, not all atrial impulses are conducted into the ventricule.
- Rhythm: regular or irregular.
- P wave: atrial flutter is characterized by a series of identical P waves, they described as having the appearance of the Saw toothed shape ▪ T wave: present but may be obscured by flutter waves.

Atrial fibrillation



- Atrial fibrillation causes rapid, disorganized and uncoordinated twitching of atrial musclature, it is the most common dysrhythmia, may occur for a very short time (paroxysmal) or it may be chronic.
- Rate: 350 – 600 beats /min
- P wave: no visible P wave (P wave greater than QRS) ▪ P-R: interval cannot be measured.
- Rhythm: irregular + usually rapid.

Premature ventricular contraction [PVC]



- Is an impulses that starts in a ventricle and is conducted through the ventricles before the next normal sinus impulses.
- PVC originates in one ventricle which there for depolarizes before the other, so the deflection from a P.V.C are very tall + deep.
- PVC has greater deflection than normal QRS complexes.

Ventricular Bigeming



- When a P.V.C becomes coupled with a normal beat, this is called Bigeming.
- Every other complex is a PVC

Ventricular Trigeming



- When a P.V.C coupled with two normal beat, this is called Trigeming.
- A rhythm in which every third complex is a PVC

Ventricular Quadriageming



- P.V.C coupled with three normal beats.
- A rhythm in which every fourth complex is a PVC.

Ventricular tachycardia (V.T)



- It is a three or more PVCs in a row
- Patient is usually unresponsive and pulseless.
- P wave: hidden in the QRS complex ▪ Increased myocardial irritability.
- Rate ventricular: 100 – 200 beats/min.
- It is extremely dangerous & considers an emergency.
- Immediate defibrillation is the action of choice.

Ventricular Fibrillation



- No audible heart sound, no palpable pulse, & no respiration.
- Ineffective quivering of the ventricles ▪ Extremely irregular.
- Ventricular fibrillation is not corrected immediately cardiac arrest + death will be happened.

- P wave: not seen.
- QRS: rapid & irregular.
- Rx: defibrillation.

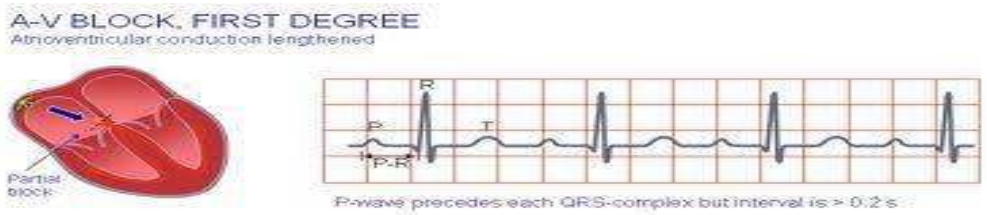
Ventricular A Systole



- Called Flatline, [condition in which the heart no longer beats]
- No QRS complexes
- No heart beat, No palpable pulse and No respiration ▪ Without any treatment ventricular A Systole is fetal.
- Rx: CPR is necessary

Conduction abnormalities 1st – degree AV Block



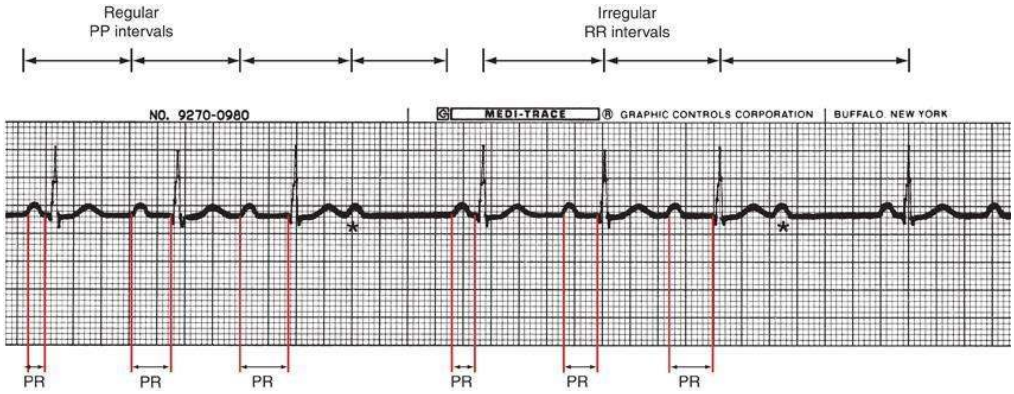


- Heart blocks may occur in any one of the three areas, the SA Node, AV Node or the Bundle Branches.
- Note: you must measure the P – R interval on every ECG, if the P – R interval is longer than one large square, an AV block is present.

- A first degree AV block is present when P – QRS – T sequence is normal but the P – R interval is prolonged.

- P-R interval longer than 0.20 second (Note: normal 3-5 box)

Second degree AV block- Mobitz Type I

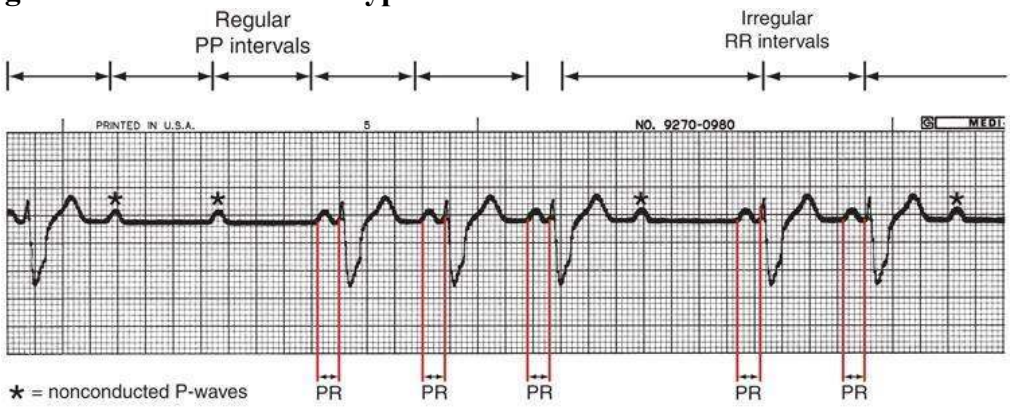


- A second degree AV block is present when it takes 2 or more atrial impulses to stimulate the ventricular response.

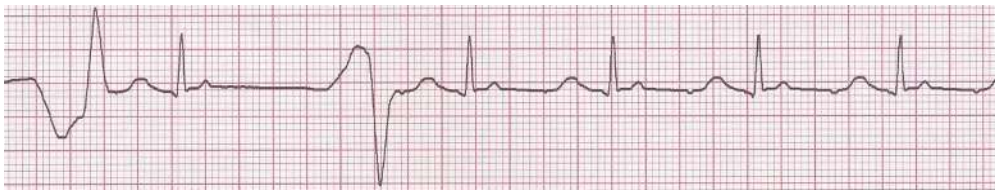
- Two or more P waves before each QRS 2 : 1 or 3 : 1

- Rate : 30 – 55

Second degree AV block - Mobitz Type II

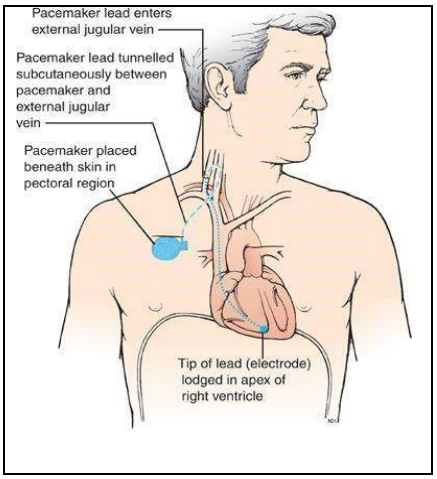


3rd degree heart block



- In 3rd degree block the AV block is complete that is no atrial impulses get through to the AV node [that mean no ventricular contraction] ▪ In this case there is an atrial rate + independent ventricular rate .
- 3rd . AV block called AV dissociation.
- Note : in 3rd block the ventricular rate may be so slow that the blood flow to the brain is diminished, so the patient may lose consciousness.
- [strokes – adamas syndrome]: A state of sudden fainting. It is caused by incomplete heart block

Pacemakers:



- An electronic device that provides electrical stimuli to the heart muscle
- Device used to produce and maintain normal heart rate in patients who have heart block, it consist of a battery that stimulates the heart through an electrode wire attached to the surface of the ventricle epicardium or endocardium .
- Types: Permanent or Temporary

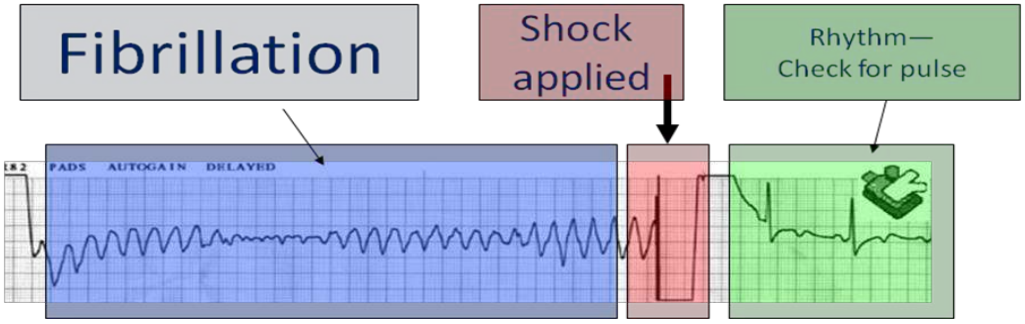
Defibrillation & Cardioversion

- Defibrillation - is the treatment for immediately life-threatening arrhythmias with which the patient does not have a pulse, i.e. Ventricular Fibrillation (VF) or pulseless Ventricular Tachycardia (VT).

- Cardioversion - is any process that aims to convert an arrhythmia back to sinus rhythm. Electrical cardioversion is used when the patient has a pulse but is either unstable, or chemical cardioversion has failed or is unlikely to be successful. It is also used in less urgent cases, e.g. atrial fibrillation (AF) to try to revert the rhythm back to sinus.

Defibrillation

- Treat tachydysrhythmias by delivering an electrical current that depolarizes a critical mass of myocardial cells. When cells repolarize, the sinus node is usually able to recapture its role as heart pacemaker.



Cardioversion:

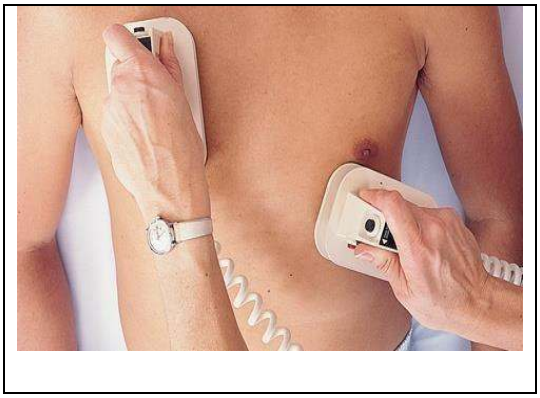
- A method of restoring the normal Rhythm of the heart [terminate dysrhythmias] by giving D/C shock through electrode placed on the chest.
- In cardioversion, the current delivery is synchronized with the patient's ECG.
- In defibrillation, the current delivery is unsynchronized.

A defibrillator

- Is an electronic device used to shock the heart and take it out of fibrillation & an effort to stop ventricular fibrillation.
- With every minute the defibrillator is delayed survival declines 7% to 10%.

Efficient defibrillation requires the following

- Correct paddle size.
- Correct paddle placement.
- Good paddle contact.
- Correct energy selection.



❖ Note: initial defibrillation dose is 2J/Kg.

❖ A subsequent defibrillation dose is 4 J/Kg.

Recommended placement

- The anterior electrode is placed to the right upper part of the sternum below the clavicle and the apex electrode to the left of the nipple with the center of the electrode in the midaxillary line.

Technique of defibrillation

- Apply gel, soap or saline on the site.
- Select the correct paddle.
- Select the energy required.
- Place the paddle on the site and apply firm pressure.
- Deliver the shock.
- Call “clear” three times; follow checks required for clear and ensure that no one is in contact with the patient, bed, or equipment.

Ischemic Heart Disease

- **Definition:** Syndromes causing an imbalance between myocardial oxygen demand and supply
- **Ischemia:** insufficient tissue oxygenation.
- **Infarction:** a zone of tissue deprived of blood supply.

Coronary Atherosclerosis

- Abnormal accumulation of lipid in the blood vessel wall.

Pathophysiology:

- Lipid (insoluble in blood) → combine with protein → lipoprotein (soluble in blood).
- High fat diet → increase lipid → deposit of fat in the wall of arteries → atheromas or plaques → stagnation of blood (high risk for thrombus formation) → obstruction blood flow (partial or complete).

Coronary Atherosclerosis

- Atherosclerosis is the abnormal accumulation of lipid deposits and fibrous tissue within arterial walls and lumen.
- In coronary atherosclerosis, blockages and narrowing of the coronary vessels reduce blood flow to the myocardium.
- Cardiovascular disease is the leading cause of death
- CAD (coronary artery disease) is the most prevalent cardiovascular disease in adults.

Clinical Manifestations

- Symptoms are due to myocardial ischemia.
- Symptoms and complications are related to the location and degree of vessel obstruction.
- Angina pectoris
- Myocardial infarction
- Heart failure
- Sudden cardiac death

Risk factors:

1] Cigarette smoking:

Nicotine → vasoconstriction

2] ↑ Bp: creates very high pressure against left ventricle.

3] High blood cholesterol: can be controlled by diet and exercise, and medication.

Diet: ↓ fat diet.

Exercise: reduce level of cholesterol.

4] Hyperglycemia: ↑ platelet aggregation.

5] Behavior patterns such as stress.

Angina pectoris

- Angina is chest pain caused by inadequate myocardial oxygen supply. Its usually caused by insufficient coronary blood flow

- **categories of Angina:**

1- **In Stable angina:** symptoms are consistent and pain is relieved by rest.

2- **In Unstable angina:** pain is marked by increasing severity, duration, and frequency. Pain responds slowly to nitroglycerin.

3- **In Prinzmetal (variant) angina:** pain is unpredictable and may occur at rest.

Clinical manifestation

- Pain, feeling pressure or compression in the anterior chest, pain radiated to the neck, jaw, or back; usually lasts 3 to 5 minutes. It usually occurs after exertion, and emotional excitement. (Dyspnea, sweating, pallor)
- ECG shows ST segment depression and T wave inversion during angina pain.

Factors can produce anginal pain

1] Physical exertion: can increasing myocardial O2 demands.

2] Exposure to cold can cause vasoconstriction

3] Eating heavy meal.

4] Stress → adrenalin → ↑ B.P.

Diagnostic evaluation:

- Signs & symptoms
- ECG shows ST segment depression and T wave inversion during angina pain.

Management:

- Treatment seeks to decrease myocardial oxygen demand and increase oxygen supply.
- Bed rest and avoid activities .
- O2 5L/min, fowler position .
- Administer medication as prescribed to increase oxygenation and to reduce cardiac workload.
- Reduce risk factors through diet, exercise, weight loss, smoking cessation, and stress reduction.

Treatment of Anginal Pain

- Treatment of anginal pain is a priority nursing concern.
- Patient is to stop all activity and sit or rest in bed.
- Assess the patient while performing other necessary interventions. Assessment includes VS, observation for respiratory distress, and assessment of pain. In the hospital setting, the ECG is assessed or obtained.
- Anxiety: Use a calm manner, Stress-reduction techniques Addressing patient spiritual needs may assist in allaying anxieties, Address both patient and family needs

Medications

- Nitroglycerin: 5mg, direct relaxation of blood vessels and smooth muscles of coronary arteries → lead to ↑ blood supply to myocardium. Pain should be relieved within 3 minutes.
- Beta-adrenergic blocking agents: reduce the heart rate, slow the conduction of impulses through the conduction system, decrease blood pressure, and reduce myocardial contractility to balance the myocardial oxygen needs and the amount of oxygen available.
- Calcium channel blocking agents: relax the blood vessels, increase perfusion.
- Antiplatelet and anticoagulant (Aspirin, Heparin, Ticlid): prevent platelet aggregation and subsequent thrombosis, which impedes blood flow.

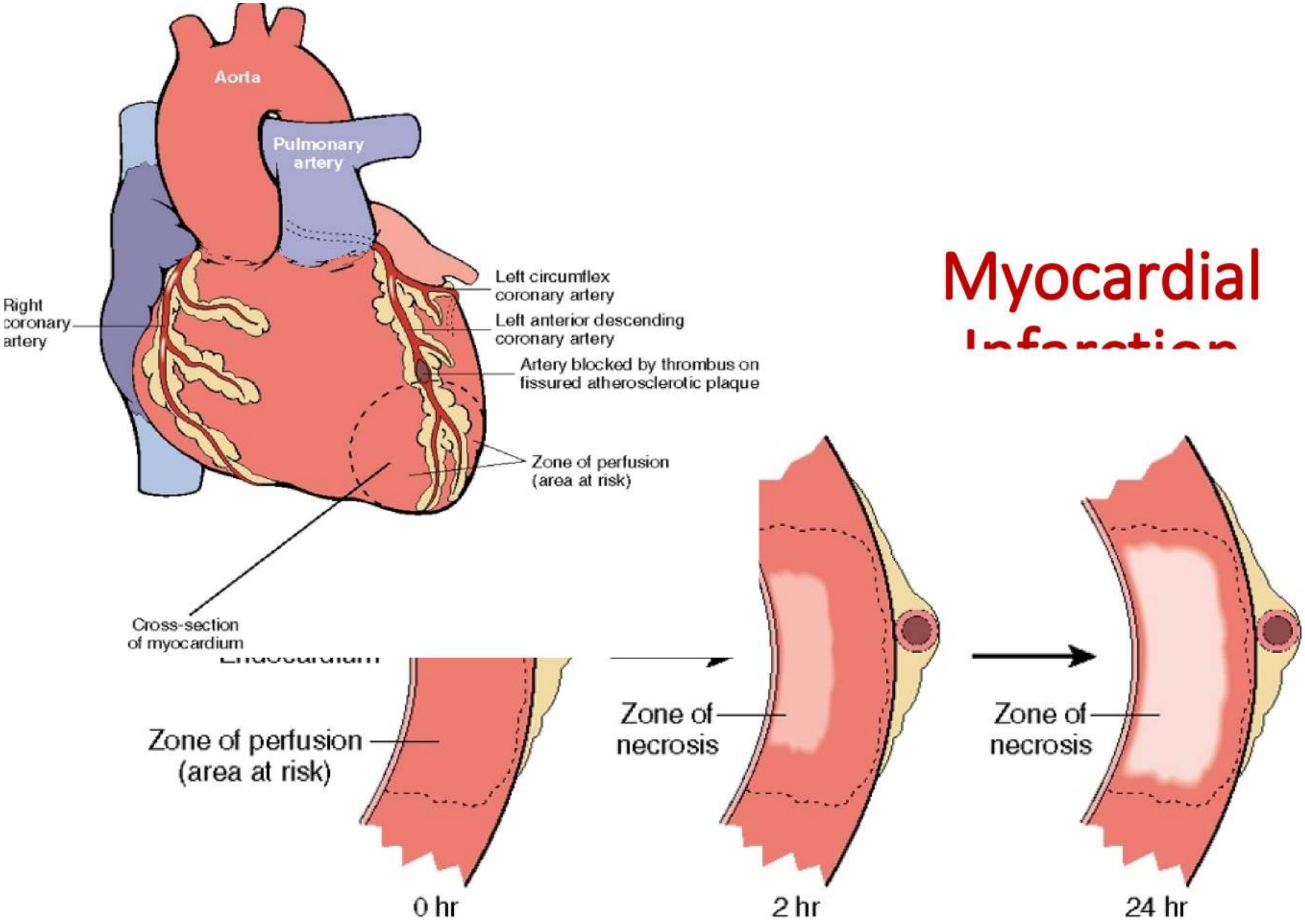
Myocardial infarction

- Refers to the process by which myocardial tissue is destroyed because of reduced coronary blood flow due to Atherosclerosis or complete occlusion of an artery, embolus or thrombus.

Clinical manifestation:

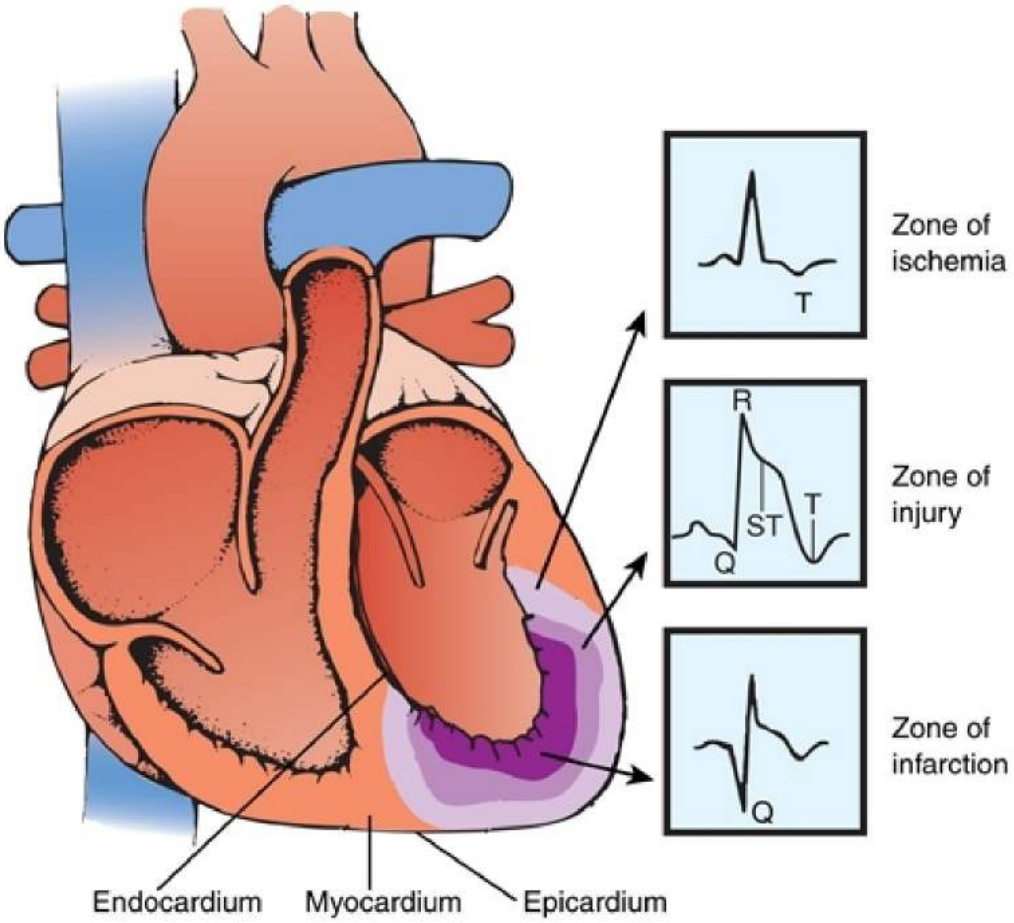
- Crushing chest pain radiate to the shoulder + down the arms [not after effort or emotional upset]
- Dyspnea, pallor, diaphoresis, nausea + vomiting, elevated temprature.
- **Note** : rate of MI greater in ♂ male than ♀ female

Myocardial Infarction



Effects of Ischemia, Injury, and Infarction on ECG

Ischemia causes inversion of the T wave because of alter depolarization. Cardiac muscle injury causes elevation of the ST segment and tall, symmetrical t wave. Later Q waves develop because of the absence of depolarization current from the necrotic tissue



Diagnostic evaluation:

1] A: History of present illness [duration, pain]

B: History of previous illness [↑ BP., D.M]

2] ECG: enlarge Q wave, elevation or depressed ST segment and T wave inversion.

3] Cardiac markers

4] WBC shows increase count.

Management

- Bed rest to ↓ O2 demand.
- Relieving chest pain, analgesics [morphine sulfate].
- Vasodilators Nitroglycerine IV. [dilates both arteries + veins.
- Anticoagulant [heparin].
- Thrombolytic [streptokinase] dissolving thrombus.
- O2 administration.
- ↓ anxiety
- Patient education.

Invasive Coronary Artery Procedures

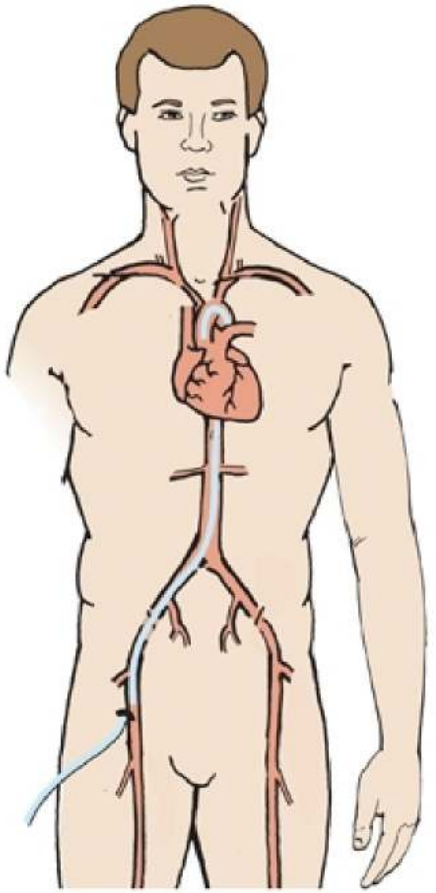
- **Percutaneous Transluminal Coronary Angioplasty (PTCA):**
- **PTCA:** an invasive interventional procedure, a balloon tipped catheter is used to open blocked coronary vessels and resolve ischemia. It is used in patients with angina and MI.

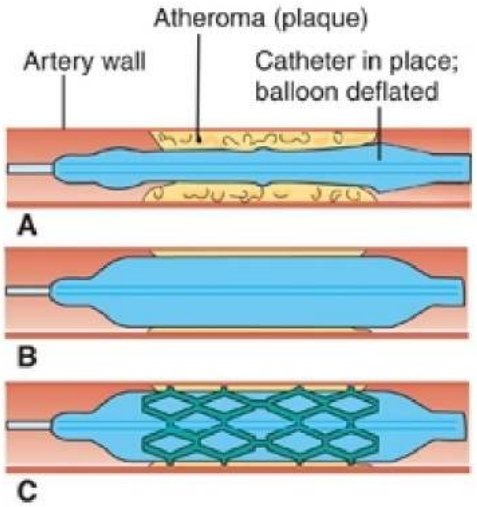
Percutaneous Coronary Intervention

A: A balloon tipped catheter is passed into the affected coronary artery placed across the area of theroma (plaque)

B: the balloon is then rapidly inflated and deflated with controlled pressure.

C: A stent is placed to maintain patency of the artery and the balloon is removed





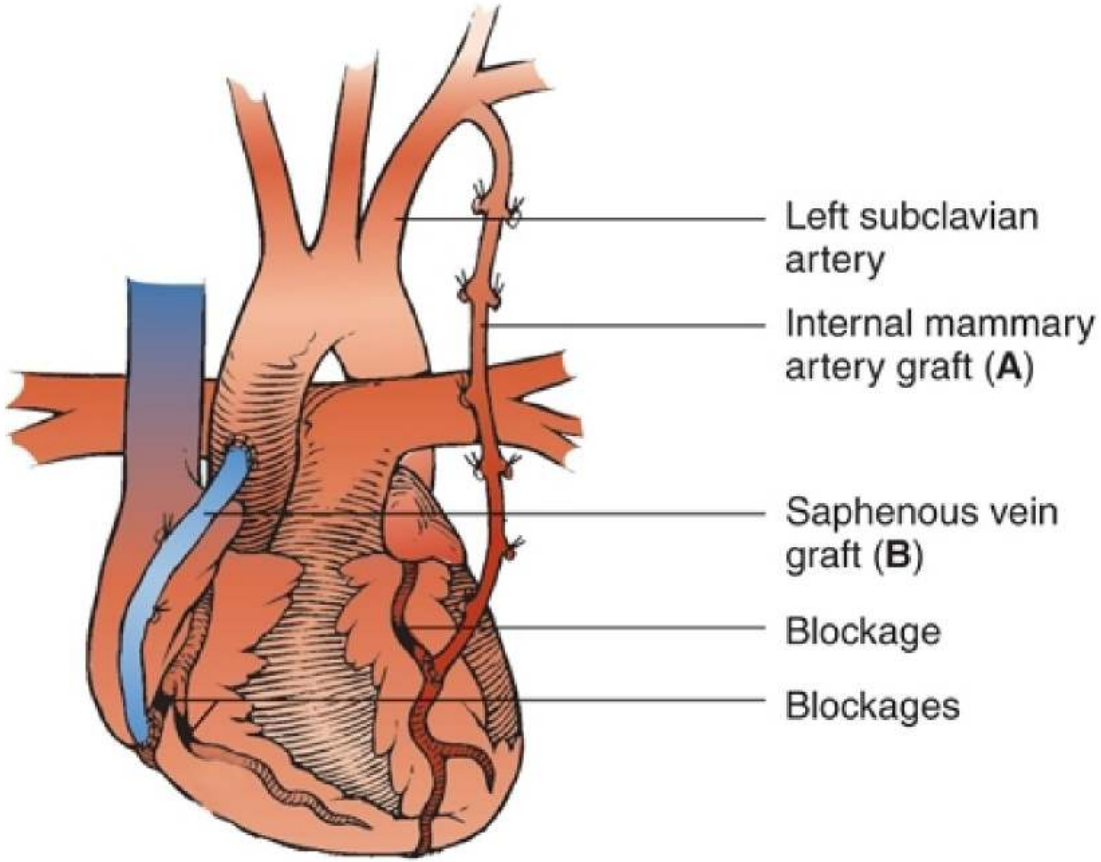
Coronary Artery Bypass Graft (CABG)

- We perform coronary artery bypass grafting (CABG) to bypass blockages or obstructions in the coronary arteries, thereby restoring the flow of oxygen and nutrients to the heart. This process is also known as revascularization.
- CABG, an open heart procedure, requires a median sternotomy, or incision down the front of the chest through the breastbone. This incision gives the surgeon a good view of the heart.

Coronary Artery Bypass Graft (CABG)

- Traditionally, the heart is stopped during the CABG process. The patient is kept alive by a heart-lung machine during the operation. This is known as cardiopulmonary bypass.
- To perform a CABG, arteries from the chest and arm and veins from the leg are used to create a detour around the obstruction in the coronary artery. The artery and/or vein are connected to both the aorta and coronary artery, thereby creating a bypass around the blockage.

Coronary Artery Bypass Grafts



Cardiac markers

Cardiac markers are biomarkers measured to evaluate heart function. They can be useful in the early prediction or diagnosis of disease. Although they are often discussed in the context of myocardial infarction, other conditions can lead to an elevation in cardiac marker level.

Most of the early markers identified were enzymes, and as a result, the term "cardiac enzymes" is sometimes used. However, not all of the markers currently used are enzymes. For example, in formal usage, troponin would not be listed as a cardiac enzyme.

Types

Types of cardiac markers include the following:

Test	Sensitivity and specificity	Approximate peak	Description
Troponin test	The most sensitive and specific test for myocardial damage. Because it has increased specificity compared with CK-MB, troponin is a superior marker for myocardial injury.	12 hours	• Troponin is released during MI from the myocytes. • Isoforms of the protein, T and I, are specific to myocardium. • Differential diagnosis of troponin elevation includes acute infarction, severe pulmonary embolism causing acute right heart overload, heart failure, myocarditis. • After myocyte injury, troponin is released in 2–4 hours and persists for up to 7 days. • Troponin T is less specific than troponin I because troponin T is also found in muscle.
Creatine Kinase (CK-MB) test	It is relatively specific when skeletal muscle damage is not present.	10–24 hours	• The CK-MB isoform of creatine kinase is expressed in heart muscle. • Since it has a short duration, it cannot be used for late diagnosis of acute MI but can be used to suggest infarct extension if levels rise again. • Increase in 4–9 h • This is usually back to normal within 2–3 days.

Lactate dehydrogenase (LDH)	LDH is not as specific as troponin.	72 hours	• LDH-1 isozyme is normally found in the heart muscle and LDH-2 is found predominantly in blood serum. • A high LDH-1 level to LDH-2 suggest MI. • LDH levels are also high in tissue breakdown or hemolysis. • It can mean cancer, meningitis, encephalitis, or HIV. • This is usually back to normal 10–14 days.
Aspartate transaminase (AST)			This was the first used. It is not specific for heart damage, and it is also one of the liver transaminases.
Myoglobin (Mb)	low specificity for myocardial infarction	2 hours	• Myoglobin is used less than the other markers. • Myoglobin is the primary oxygen-carrying pigment of muscle tissue. • It is high when muscle tissue is damaged but it lacks specificity. • It has the advantage of responding very rapidly, rising and falling earlier than CK-MB or troponin. • Increase within 1–4 h

Heart Failure

- HF referred to congestive heart failure (CHF), the inability of the heart to pump sufficient blood to meet the needs of the tissues for oxygen and nutrients
- A **syndrome** characterized by fluid overload or inadequate tissue perfusion
- The term HF indicates myocardial disease, in which there is a problem with the contraction of the heart (systolic failure) or filling of the heart (diastolic failure).
- Some cases are reversible.
- Most HF is a progressive, lifelong disorder managed with lifestyle changes and medications.

Types of Heart Failure

- **Heart failure can occur as left side failure or right side failure.**
- Atherosclerosis of the coronary arteries is the primary cause of heart failure, and coronary artery disease is found in more than 60% of patients with heart failure.

(1)Left heart failure

- Inability of the left heart to pump volume of blood returned to it resulting in pulmonary congestion
- LV cannot pump blood effectively to the systemic circulation. Pulmonary venous pressures increase, resulting in pulmonary congestion with dyspnea, cough, crackles, and impaired oxygen exchange.

Caused by: hypertension, mitral and aortic valve disease, ischemic heart disease, and primary myocardial diseases

- Left side heart failure causes mostly pulmonary symptoms such as shortness of breath, dyspnea, and a moist cough
- Dyspnea on exertion, also occur on rest
- Orthopnea, difficult of breathing when lying flat.
- Paroxysmal nocturnal dyspnea: sudden attach of dyspnea at night
- Oliguria: results in the release of renin, lead to aldosterone secretion (Na & water retention).
- Rest lead to decrease cardiac workload, improve renal perfusion, frequent urination at night (nocturia)

(2)Right heart failure

- Inability of right heart to meet demands of venous blood return resulting in systemic congestion
- RV cannot eject sufficient amounts of blood, and blood backs up in the venous system. This results in peripheral edema, hepatomegaly, ascites, anorexia, nausea, weakness, and weight gain.

- **Caused by:** left heart failure (most common cause), primary vascular disease of the lungs, tricuspid and pulmonary valve disease, congenital or acquired left to right shunts
- Right side heart failure causes systemic symptoms such as edema (pitting edema) and swelling, jugular vein distention, hepatomegaly and weight gain.
- Hepatomegally : result from venous engorgement of the liver lead to pressure within the portal vein which lead to Ascites (accumulation of fluid in the peritoneal cavity).
- Anorexia and nausea (venous stasis within abdominal organs)
- Nocturia : because cardiac output improved with physical rest and that lead to ↑ blood supply to kidney.

Diagnostic Test:

- Chest x-ray shows increase pulmonary congestion and Lt ventricular (Lt H.F.), while Pulmonary congestion, cardiomegaly a and pleural effusion (Rt H.F.)
- ABGs, Blood chemistry, hypertrophy, ECG, Echocardiography, CVP increase.

Medical Management of HF

- Eliminate or reduce etiologic or contributory factors.
- Reduce the workload of the heart by reducing afterload and preload.
- Prevent exacerbations of HF.
- Medications are routinely prescribed for HF.

❑ Treatment options vary according to the severity of the patients condition and may include basic lifestyle changes.

❑ Providing general education and counseling to the patient and family (restriction of dietary sodium, wt reduction, exercise (bed rest), avoid smoking & alcohol)

Medications

- Angiotensin-converting enzyme inhibitors: promote vasodilatation and diuresis by decrease afterload and preload, decrease cardiac workload & improving ventricular emptying , decrease secretion of aldosterone,
- Angiotensin II receptor blockers: (diovan) decrease BP, decrease vascular resistance & improved cardiac output
- Hydrazine & Isosorbide Dintrate : venous dilatation, reduce blood return to heart
- Diuretics
- Digitalis

Other complication: Pulmonary Edema

- Acute event in which the LV cannot handle an overload of blood volume. Pressure increases in the pulmonary vasculature, causing fluid to move out of the pulmonary capillaries and into the interstitial space of the lungs and alveoli, results in hypoxemia
- Is accumulation of water in the lung in the line of alveolar lead to impair gas exchange
- Clinical manifestations: restlessness, anxiety, dyspnea, cool and clammy skin, cyanosis, weak and rapid pulse, cough, lung congestion (moist, noisy respirations), increased sputum production (sputum may be frothy and blood-tinged), decreased level of consciousness

Management of Pulmonary Edema

- Prevention
- Early recognition: monitor lung sounds and for signs of decreased activity tolerance and increased fluid retention
- Place patient upright and dangle legs.
- Minimize exertion and stress.
- Oxygen
- Medications
  - Morphine
  - Diuretic (furosemide)

Management (Damp Moist)

- D: Diuretics
- A: Aminophylline
- M: Morphine
- P: Position
- M: Monitor BP, P
- O: Oxygen
- I: Indwelling catheter
- T: Tourniquet: 3 part, rotate each 15 minute to decrease venous return and pulmonary congestion

HYPERTENSIVE DISORDERS

HYPERTENSION

Hypertension (high BP) is a disease of vascular regulation in which the mechanisms that control arterial pressure within the normal range are altered.
Predominant mechanisms of control are the central nervous system (CNS), the renal pressor system (renin-angiotensin-aldosterone system), and extracellular fluid volume. Why these mechanisms fail is not known. The basic explanation is that BP is elevated when there is increased cardiac output plus increased peripheral vascular resistance.

Pathophysiology and Etiology

Primary or Essential Hypertension

(Approximately 95% of patients with hypertension)

- When the diastolic pressure is 90 mm Hg and/or the systolic pressure is 140 mm Hg or higher and other causes of hypertension are absent, the condition is said to be primary hypertension. More specifically, an individual is considered hypertensive when the average of two or more properly measured, seated BP readings taken at rest on each of two or more office visits exceeds the upper limits of normal.

Classification of Blood Pressure for Adults

BLOOD PRESSURE (BP) CLASSIFICATION	SBP[*] (MM HG)	DBP[*] (MM HG)
Normal	<120	<80
Prehypertension	120-139	80-89
Stage 1 hypertension	140-159	90-99
Stage 2 hypertension	≥160	≥100
DBP: diastolic blood pressure; *SBP:* systolic blood pressure.		
[*] Treatment determined by highest BP category.		

- Cause of essential hypertension is unknown; however, there are several areas of investigation:
  - Hyperactivity of sympathetic vasoconstricting nerves
  - Presence of vasoactive substance released from the arterial endothelial cells, which acts on smooth muscle, sensitizing it to vasoconstriction
  - Increased cardiac output, followed by arteriole constriction
  - Excessive dietary sodium intake, sodium retention, insulin resistance, and hyperinsulinemia play roles that are not clear
  - Familial (genetic) tendency
- Systolic BP elevation in the absence of elevated diastolic BP is termed isolated systolic hypertension and is treated in the same manner.

Secondary Hypertension

- Occurs in approximately 5% of patients with hypertension secondary to other pathology.
- Renal pathology:
    - Congenital anomalies, pyelonephritis, renal artery obstruction, acute and chronic glomerulonephritis, and obstructive uropathy (hydronephrosis).
    - Reduced blood flow to kidney causes release of renin. Renin reacts with serum protein in liver (α2-globulin) to form angiotensin I, which is converted to angiotensin II through the action of angiotensin-converting enzyme in the lungs, leading to vasoconstriction and increased salt and water retention..
- Coarctation of aorta (stenosis of aorta)—blood flow to upper extremities is greater than flow to lower extremities —hypertension of upper part of body.
- Endocrine disturbances:
    - Pheochromocytoma—a tumor of the adrenal gland that causes release of epinephrine and norepinephrine and a rise in BP (extremely rare).
    - Adrenal cortex tumors lead to an increase in aldosterone secretion (hyperaldosteronism) and an elevated BP (rare).
    - Cushing's syndrome leads to an increase in adrenocortical steroids (causing sodium and fluid retention) and hypertension *(Cushing's syndrome is associated with* HTN *in approximately 80% of cases).*
    - Hyperthyroidism causes increased cardiac output.
- Obstructive sleep apnea causes nocturnal hypertension, which leads to sustained daytime hypertension.
- Prescription medications such as estrogens and steroids (cause fluid retention), sympathomimetics (cause vasoconstriction and tachycardia), antidepressants (prevent the breakdown of epinephrine), appetite suppressants (cause tachycardia and vasoconstriction), and NSAIDs (cause fluid retention and can lead to renal insufficiency).
- Nonprescription drugs and substances such as cocaine (cause vasoconstriction and tachycardia); NSAIDs; Antihistamines (cause vasoconstriction and tachycardia); and nicotine (causes vasoconstriction).

Consequences of Hypertension

- Hypertension can cause intimal wall injury in the arteries, which can lead to arteriosclerosis in which smooth muscle cell proliferation, lipid infiltration, and calcium accumulation occur in the vascular endothelium.
- Prolonged hypertension damages small blood vessels in the brain, eyes, heart, and kidneys.
- The major objective in patients with high BP is to prevent target organ damage of the heart (left ventricular hypertrophy, angina, myocardial infarction, heart failure), brain (stroke, transient ischemic attack, dementia), kidneys (chronic kidney disease), eyes (retinopathy, blindness), or vasculature (aneurysm, PAD).
- The Heart Disease and Stroke Statistics 2016 Update reported that compared to individuals with normal BP (120/80 mm Hg), those patients with prehypertension have a 1.5- to 2-fold risk for major cardiovascular events between the ages 60 to 80 years.

Prevalence and Risk Factors

- Hypertension is one of the most prevalent chronic diseases for which treatment is available; however, most patients with hypertension are untreated.
- There are no symptoms; thus, it is termed "the silent killer."
- Increase in incidence is associated with the following risk factors:
    - Age—between 30 and 70
    - Race—black
    - Overweight, sleep apnea
    - Family history
    - Smoking
    - Sedentary lifestyle
    - Diabetes mellitus
    - Metabolic syndrome
    - Lower education/socioeconomic status
    - Psychological stressors
    - Dietary factors (increased consumption of fats, sodium, and alcohol, with lower intake of potassium-rich foods).

Clinical Manifestations

- Usually asymptomatic
- May cause headache, dizziness, blurred vision, chest pain, and shortness of breath when greatly elevated.
- Elevated BP readings taken in a seated position on at least two occasions.

Diagnostic Evaluation

- ECG—to determine effects of hypertension on the heart (left ventricular hypertrophy, ischemia) or presence of underlying heart disease.
- Chest X-ray—may show cardiomegaly.
- Proteinuria, elevated serum blood urea nitrogen (BUN), and creatinine levels—indicate kidney disease as a cause or effect of hypertension; first voided urine microalbumin or spot urine for albumin– creatinine ratio are early indicators.
- Serum potassium—decreased in primary hyperaldosteronism.
- Urine (24-hour) for catecholamines—increased in pheochromocytoma.
- Renal ultrasound to detect renal vascular diseases.
- Renal duplex imaging to identify renal artery stenosis.
- Outpatient ambulatory BP measurements.
- Tests for essential hypertension are generally done first, unless specific signs and symptoms of secondary hypertension are present.

Management

Lifestyle Modifications

- Lose weight if body mass index is greater than or equal to 25.
- Get regular aerobic exercise equivalent to 30 to 45 minutes of brisk walking most days.

- Cut sodium intake to 2.4 g or less per day.
- Include recommended daily allowances of potassium, calcium, and magnesium in diet. This can be accomplished through following the **DASH** diet (Dietary Approaches to Stop Hypertension)—rich in fruits, vegetables, low-fat dairy products, fibers and low in saturated and total fat.
- Smoking cessation.
- Reduce dietary saturated fat and cholesterol.
- Consider reducing coffee intake (5 cups per day has been shown to increase BP in hypertensive men).
- If, despite lifestyle changes, BP remains at or above 140/90 mm Hg (or is not at optimal level in the presence of other cardiovascular risk factors) over 3 to 6 months, single-drug therapy should be initiated.

Drug Therapy

- Considerations in selecting therapy include:
    - Race—Blacks respond well to diuretic therapy; whites respond well to ACE inhibitors.
    - Age—some adverse effects may not be tolerated well by elderly people. Diuretics are typically the first drug prescribed.
    - Concomitant diseases and therapies—some agents have beneficial effects on conditions such as renal insufficiency, while other agents have adverse effects on such conditions as diabetes or asthma.
    - Economic considerations—newer agents very expensive.
    - Doses per day—may be compliance problem.
- **Agents include:**
    - Diuretics—lower BP by promoting urinary excretion of water and sodium to lower blood volume
    - Beta-adrenergic blockers—lower BP by slowing the heart and reducing cardiac output as well as release of renin from the kidneys
    - Alpha-receptor blockers—lower BP by dilating peripheral blood vessels and lowering peripheral vascular resistance. Research has indicated these medications provide little protection against heart failure.
    - Central alpha agonists—lower BP by diminishing sympathetic outflow from the brain, thereby lowering peripheral resistance.
    - Peripheral adrenergic agents—inhibit peripheral adrenergic release of vasoconstricting catecholamines such as norepinephrine.
    - Combined alpha and beta-adrenergic blockers—work through alpha and beta receptors.
    - ACE inhibitors—lower BP by blocking the enzyme that converts angiotensin I to the potent vasoconstrictor, angiotensin II. These drugs also raise the level of bradykinin, a potent vasodilator, and lower aldosterone levels.
    - Angiotensin receptor blockers—similar in action to ACE inhibitors.
    - Calcium antagonists (calcium channel blockers)—stop the movement of calcium into the cells; relax smooth muscle, which causes vasodilation; and inhibit reabsorption of sodium in the renal tubules.
    - Direct vasodilators—smooth muscle relaxants that primarily dilate arteries and arterioles.

    - Direct renin inhibitors—suppress the entire renin-angiotensin-aldosterone system.
- If hypertension is not controlled with the first drug within 1 to 3 months, three options can be considered:
    - If the patient has faithfully taken the drug and not developed adverse effects, the dose of the drug may be increased.
    - If the patient has had adverse effects, another class of drugs can be substituted.
    - A second drug from another class could be added. If adding the second agent lowers the pressure, the first agent can be slowly withdrawn or, if necessary, combination therapy can be continued.
- If the desired BP is still not achieved with the addition of a second drug, a third agent or a diuretic or both (if not already prescribed) could be added.
  75% of individuals with hypertension require two or more antihypertensive agents to reach their BP goal.
- The best management of hypertension is to use the fewest drugs at the lowest doses while encouraging the patient to maintain lifestyle changes.
- Resistant hypertension is defined as an individual that requires three or more antihypertensive agents to control blood pressure.

Complications

- Angina pectoris or MI due to decreased coronary perfusion
- Left ventricular hypertrophy and heart failure due to consistently elevated aortic pressure
- Renal failure due to thickening of renal vessels and diminished perfusion to the glomerulus
- TIAs, stroke, or cerebral hemorrhage due to cerebral ischemia and arteriosclerosis
- Retinopathy
- Accelerated hypertension

Nursing Assessment

Nursing History

Query the patient about the following:

- Family history of high BP
- Previous episodes of high BP
- Dietary habits and salt intake
- Target organ disease or other disease processes that may place the patient in a high-risk group—diabetes, CAD, kidney disease
- Cigarette smoking
- Episodes of headache, weakness, muscle cramp, tingling, palpitations, sweating, vision disturbances
- Medication that could elevate BP:
    - Hormonal contraceptives, steroids
    - NSAIDs
    - Nasal decongestants, appetite suppressants, tricyclic antidepressants
- Other disease processes, such as gout, migraines, asthma, heart failure, and benign prostatic hyperplasia, which may be helped or worsened by particular hypertension drugs.

Physical Examination

- Auscultate heart rate and palpate peripheral pulses; determine respirations.
- If skilled in doing so, perform funduscopic examination of the eyes for the purpose of noting vascular changes. Look for edema, spasm, and hemorrhage of the eye vessels. Refer to ophthalmologist for definitive diagnosis.
- Examine the heart for a shift of the point of maximal impulse to the left, which occurs in heart enlargement.
- Auscultate for bruits over peripheral arteries to determine the presence of atherosclerosis, which may be manifested as obstructed blood flow.
- Determine mental status by asking patient about memory, ability to concentrate, and ability to perform simple mathematical calculations.

Blood Pressure Determination

- Measure the BP of the patient under the same conditions each time.
- Avoid taking BP readings immediately after stressful or taxing situations. Wait 30 minutes after patient has smoked.
- Have the patient assume position of comfort and remain silent. Make sure feet are on the floor or otherwise supported. The patient should be seated and relaxed for at least 5 minutes, and the blood pressure should be taken around the same time each day it is taken to observe for trends in BP.
- Support the bared arm; avoid constriction of arm by a rolled sleeve.
- Use a BP cuff of the correct size.
    - The bladder within the cuff should encircle at least 80% of the patient's arm.
    - Many adults will require a large cuff.
    - Two or more readings separated by 2 minutes should be averaged.
- Be aware that falsely elevated BPs may be obtained with a cuff that is too narrow; falsely low readings may be obtained with a cuff that is too wide.
- Auscultate and record precisely the systolic and diastolic pressures based on Korotkoff sounds.
    - Systolic—the pressure within the cuff indicated by the level of the mercury column at the moment when the first clear, rhythmic pulsatile sound is heard (phase 1).
    - First diastolic—the pressure within the cuff indicated by the level of the mercury column at the moment when the sound becomes muffled (phase 4).
    - Second diastolic—the pressure within the cuff at the moment the sound disappears (ie, the onset of silence) (phase 5).
    - Phases 2 and 3 are less distinct sounds produced between systolic and first diastolic and are not identified clinically or recorded.

Nursing Diagnoses

- Deficient Knowledge regarding the relationship between the treatment regimen and control of the disease process
- Ineffective Therapeutic Regimen Management related to medication adverse effects and difficult lifestyle adjustments

Nursing Interventions

Providing Basic Education

- Explain the meaning of high BP, risk factors, and their influences on the cardiovascular, cerebral, and renal systems.
- Stress that there can never be total cure, only control, of essential hypertension; emphasize the consequences of uncontrolled hypertension.
- Stress the fact that there may be no correlation between high BP and symptoms; the patient cannot tell by the way he or she feels whether BP is normal or elevated.
- Have the patient recognize that hypertension is chronic and requires persistent therapy and periodic evaluation. Effective treatment improves life expectancy; therefore, follow-up health care visits are mandatory.
- Present a coordinated and complementary plan of guidance.
    - Inform the patient of the meaning of the various diagnostic and therapeutic activities to minimize anxiety and to obtain cooperation.
    - Solicit the assistance of the patient's spouse, family, and friends—provide information regarding the total treatment plan.
    - Be aware of the dietary plan developed for this particular patient.
- Explain the pharmacologic control of hypertension.
    - Explain that antihypertensive drugs affect people differently, and side effects such as anorexia, fatigue, nausea, or lightheadedness often occur during the first few days or weeks of therapy until the body develops tolerance to them.
    - Warn the patient of the possibility that orthostatic hypotension may occur initially with some drug therapy.
        - Instruct the patient to get up slowly to offset the feeling of dizziness.
        - Encourage the patient to sit or lie down immediately if feeling faint.
    - If constipation occurs, encourage high-fiber diet.
    - If ankle swelling occurs, elevate legs periodically.
    - If dry cough develops, dosage or drug may need to be changed.
    - Advise the patient to notify the health care provider if side effects are significant, dosage may be lowered or a substitute medication may be ordered. However, the patient should not discontinue medication without notifying the health care provider.
- Educate the patient to be aware of serious adverse effects (fainting, palpitations, shortness of breath) and report them immediately so that adjustments can be made in individual pharmacotherapy.
- Warn the patient on vasodilating drugs to use caution in certain circumstances that produce vasodilation—a hot bath, hot weather, febrile illness, consumption of alcohol —which may exacerbate BP reduction.
- Warn patients that BP is often decreased when circulating blood volume is reduced—as in dehydration, diarrhea, hemorrhage—so BP should be monitored closely and treatment adjusted.

Encouraging Self-Management

- Enlist the patient's cooperation in redirecting lifestyle in keeping with the guidelines of therapy, acknowledge the difficulty, and provide support and encouragement.
- Develop a plan of instruction for medication self-management.
    - Plan the patient's medication schedule so that they are given at proper and convenient times; set up a daily checklist on which the patient can record the medication taken.
    - Be sure the patient knows the generic and brand names for all medications and throws away old medications and dosages so they will not be mixed up with current medications.
- Instruct the patient regarding proper method of taking BP at home and at work if health care provider so desires. Inform patient of desired range and the readings that are to be reported.
- Determine recommended dietary plans and provide dietary education as appropriate.
    - Teach patient and family members how to read food labels to monitor sodium intake.
    - Refer to dietitian for thorough dietary education and planning, as needed.
    - Encourage the patient to keep follow-up appointments as directed by health care provider, have blood work drawn to monitor serum creatinine and electrolytes, and schedule eye exam by an ophthalmologist yearly.

ACCELERATED HYPERTENSION

Accelerated hypertension, also called malignant hypertension, occurs when the BP elevates extremely rapidly, threatening one or more of the target organs: brain, kidneys, or heart. Malignant hypertension is defined as blood pressure ≥180/120 mm Hg or isolated diastolic BP greater than 130 mm Hg and is a medical emergency.

Pathophysiology

- Elevated diastolic pressure → strain on arterial wall → thickening and calcification of arterial media (sclerosis) → narrowed blood vessel lumen.
- Sclerosis of vessels → increased wall permeability → deposits placed on intima and media of vessels → cerebral, myocardial, or renal ischemia.

Clinical Manifestations

- Brain effects:
  - Encephalopathy
  - Stroke
  - Progressive headache, stupor, seizures
- Kidney effects:
  - Decreased blood flow, vasoconstriction
  - BUN elevated
  - Plasma renin activity increased
  - Urine specific gravity lowered
  - Proteinuria
  - Renal failure
- Cardiac effects:
  - Left-sided heart failure
  - Right-sided heart failure
  - Acute MI

Management

Goal is to lower BP to reduce the probability of permanent damage to a target organs.

- If diastolic BP exceeds 115 to 130 mm Hg, clinical condition is assessed very carefully.
- Immediate hospitalization and treatment if the following are present:
  - Seizures
  - Abnormal neurologic signs
  - Severe occipital headache
  - Pulmonary edema
- The patient is hemodynamically monitored in the ICU.
- Antihypertensive agents are administered parenterally.

- **Agents include:**
    - Nitroprusside—a rapidly acting arteriolar and venous dilator, given as an IV infusion. Initial dose: 0.25 to 0.5 mcg/kg/minute; maximum dose: 8 to 10 mcg/kg/minute. Avoid in neurologic emergencies. Preferred in perioperative HTN.
    - Nitroglycerin—a rapidly acting venous and, to a lesser degree, arteriolar dilator, given as an IV infusion. Initial dose: 5 mcg/minute; maximum dose: 200 mcg/minute. Preferred perioperative.
    - Labetalol—an alpha- and β-adrenergic blocker, given as an IV bolus or infusion. Bolus: 20 mg initially, followed by 20 to 80 mg every 10 minutes to a total dose of 300 mg. Infusion: 0.5 to 2 mg/minute. Preferred in acute ischemic stroke.
    - Esmolol—antagonizes beta-1 adrenergic receptors, given as bolus 1,000 mcg/kg IV, then 150 mcg/kg/minute IV prn, max 300 mcg/kg/minute, may continue 50 mcg/kg/minute IV q4min prn, taper to d/c.
    - Nicardipine—a calcium channel blocker, given as an IV infusion. Initial dose: 5 mg/hour; maximum dose: 15 mg/hour. Preferred in acute ischemic stroke.
    - Clevidipine—a calcium channel blocker. Initial dose: 1 to 2 mg/hour; 32 mg/hour maximum dose: 16 mg/hour.
    - Fenoldopam—a peripheral dopamine-1 receptor agonist, given as an IV infusion. Initial dose: 0.01 mcg/kg/minute; the dose is titrated at 15-minute intervals, depending on the blood pressure response normal dosing: 0.01 to 1.6 mcg/kg/minute.
    - Hydralazine—an arteriolar dilator, given as an IV bolus. Initial dose: 10 mg given every 20 to 30 minutes; maximum dose: 20 mg. Avoid in neurologic emergencies.
    - Propranolol—a β-adrenergic blocker, given as an IV infusion and then followed by oral therapy. Dose: 1 to 10 mg load, followed by 3 mg/hour.
    - Phentolamine—an alpha-adrenergic blocker, given as an IV bolus. Dose: 5 to 10 mg every 5 to 15 minutes.
    - Enalaprilat—an ACE inhibitor, given as an IV bolus. Dose: 1.25 to 5 mg every 6 hours.
- Diuretics may be administered to maintain a sodium diuresis when the arterial pressure falls.
- Vasopressor agents should be available if the BP responds too vigorously to antihypertensive agents.

Nursing Interventions

- Record BP frequently or monitor BP via an intra-arterial line or electronically controlled cuff. Some drugs necessitate the taking of BP readings every 5 minutes or more frequently while titrating drug therapies.
- Monitor for adverse effects of medications—headache, tachycardia, orthostatic hypotension.
- Measure urine output accurately.
- Observe for hypokalemia, especially if patient is placed on diuretic therapy. Monitor for ventricular dysrhythmias.
- Observe for CNS complications.
    - Note signs of confusion, irritability, lethargy, and disorientation.
    - Listen for complaints of headache, difficulty with vision.
    - Check for evidence of nausea or vomiting.
    - Be alert for signs of seizure activity. Provide a safe environment—padded side rails. Keep bed in lowest position.

- Reduce activity and provide quiet environment.
- Monitor ECG continuously.
- Maintain constant vigilance until BP is decreased and stable and then begin a hypertension teaching program.

NURSING ALERT

BP should be reduced gradually and wide pressure variations avoided because lowered BP may not be adequate to perfuse vital organs.

Assessment and Management of Patients With Hematological Disorders

Anemia

- A low hemoglobin or red blood cell (RBC) count results in decreased oxygen carrying capability of the blood.
- Anemia is a symptom of something else happening.
- The cause of the anemia needs to be determined in order to correct the anemia and its symptoms.

Classification of Anemias

Type of Anemia	Laboratory Findings
Hypoproliferative (Resulting From Defective RBC Production)	
Iron deficiency	Decreased,iron, ferritin, MCV; increased total iron-binding capacity TIBC
Vitamin B_{12} deficiency (megaloblastic)	Decreased vitamin B_{12} level; increased MCV
Folate deficiency (megaloblastic)	Decreased folate level; increased MCV
Decreased erythropoietin production	Decreased erythropoietin level; normal MCV and MCH; increased creatinine level.
Cancer/inflammation	Normal MCV, MCH
Bleeding (Resulting From RBC Loss)	
Bleeding from gastrointestinal tract, menorrhagia, epistaxis, trauma.	Normal Hgb and Hct if measured soon after bleeding starts, but levels decrease thereafter.
Hemolytic (Resulting From RBC Destruction)	
Sickle cell anemia, thalassemia.	Decreased MCV; fragmented RBCs
Hypersplenism (hemolysis)	Increased MCV
Drug-induced anemia	Increased spherocyte level
Autoimmune anemia	Increased spherocyte level
Mechanical heart valve–related anemia	Fragmented red cells

Signs and Symptoms

- Fatigue due to hypoxia from less oxygen being available to the tissues
- Weakness due to hypoxia
- Pallor due to less oxygen being available to the surface tissues
- Tachycardia as the body attempts to compensate for less available oxygen by beating more rapidly to increase blood supply
- Systolic murmur due to increased turbulence of blood flow
- Dyspnea or shortness of breath due to hypoxia as body attempts to get more oxygen
- Angina as the myocardium is not getting enough oxygen
- Headache due to hypoxia
- Lightheadedness due to hypoxia
- Bone pain due to increased erythropoiesis as body attempts to correct anemia
- Jaundice in hemolytic anemia due to increased levels of bilirubin

Diagnostic evaluation

- Hemoglobin level low.
- Hematocrit level low.
- RBC count low.
- MCV (mean corpuscular volume) shows size of cell—normal (normocytic), microcytic (low), or macrocytic (high).
- MCH (mean corpuscular hemoglobin) shows color of cell—normal (normochromic), hypochromic (low).
- RDW (red cell distribution width) elevated—shows the variation of the cell sizes; there is greater variation in cell size when body is attempting to compensate for anemia.
- Reticulocyte count elevated when RBC cell production is increased to compensate for the anemia.

Treatment

- Correction of the underlying cause is necessary.
- Treatment may include dietary modifications and supplementations.

Nursing intervention

- Check vital signs for changes.
- Monitor CBC—hemoglobin, RBC, MCV, MCH, RDW.
- Plan nursing care based on patient tolerance of activity.
- Monitor for angina.

Aplastic Anemia (Pancytopenia)

- The bone marrow stops producing a sufficient amount of RBC, WBC, and platelets, thereby increasing the risk of infection and hemorrhage.
- The red cells remaining in circulation are normal in size and color.
- This may be due to chemical exposure, high dose radiation exposure, or exposure to toxins.
- Cancer treatments such as radiation therapy and chemotherapeutic agents may suppress bone marrow function, which will result in anemia (low RBC), thrombocytopenia (low platelets), and leukopenia (low WBC).
- The cause may also be unknown or idiopathic.

Signs and symptoms

- Fatigue due to hypoxemia
- Weakness due to tissue hypoxia
- Pallor due to lack of oxygen reaching superficial tissues due to anemia
- Infections due to low white blood cell production, causing decreased ability to fight infection
- Bruising (ecchymosis), and tiny subcutaneous (SC) hemorrhages (petechiae) due to decrease in platelets, altering clotting ability
- Bleeding from mucous membranes (GI tract, mouth, nose, vagina)

Diagnostic evaluation

- Low hemoglobin.
- Low hematocrit.
- Low RBC count.
- Thrombocytopenia—low platelet count.
- Leukopenia—low WBC.
- Reticulocyte count low.
- Positive fecal occult blood test.
- Decreased cell counts in bone marrow biopsy as body stops producing.

Treatment

- Administer hematopoietic growth factor to correct anemia in patients with low erythropoietin levels:
  - Erythropoietin, epoetin alfa (erythropoietin) by SC injection or IV.
- Administer human granulocyte colony-stimulating factor (G-CSF) to correct low WBC levels:
  - Filgrastim by SC injection or IV
  - Granulocyte-macrophage colony-stimulating factor (GM-CSF) sargramostim by IV infusion
- Packed RBC transfusions when anemia is symptomatic.
- Platelet transfusion for severe bleeding.
- Bone marrow transplant replaces functioning stem cells.
- Administer immunosuppressive drugs, antithymocyte globulin, and corticosteroids.
- Splenectomy when spleen is enlarged and destroying RBCs.

Nursing intervention

- Monitor vital signs for changes.
- Record intake and output of fluids.
- Protect patient from falls.

- Avoid IM injections due to altered clotting ability.
- Explain to the patient:
  - No aspirin due to effect on platelet aggregation (clotting ability).
  - Plan to take rest periods during activities due to fatigue.
  - Only use an electric razor to decrease risk of bleeding due to decreased platelet count.
  - Call your physician, nurse practitioner, or physician assistant for signs of bleeding or bruising.

Iron Deficiency Anemia

- A lower-than-normal amount of iron in blood serum results in decreased formation of hemoglobin and a decreased ability for the blood to carry oxygen.
- Iron stores are typically depleted first, followed by serum iron levels.
- Iron deficiency may be due to blood loss, dietary deficiency, or increased demand due to pregnancy or lactation.
- As red blood cells age, the body breaks them down and the iron is released.
- This iron is reused for the production of new blood cells.
- A small amount of iron is lost daily through the GI tract, necessitating dietary replacement.
- When RBCs are produced without a sufficient amount of iron, the cells are smaller and paler than usual.

Signs and symptoms

- Weakness due to anemia and tissue hypoxia
- Pallor due to decreased amount of oxygen getting to surface tissues
- Fatigue due to anemia and hypoxemia
- Koilonychia—thin, concave nails raised at edges, also called spoon nails
- Tachycardia and tachypnea on exertion due to increased demand for oxygen

Diagnostic evaluation

- Decrease in serum hemoglobin as fewer RBCs are made.
- Serum ferritin is low.

- Mean corpuscular volume (MCV) initially normal, then low—microcytic anemia.
- Mean corpuscular hemoglobin (MCH) initially normal, then low—hypochromic anemia.
- Serum iron level is low.
- Serum iron-binding capacity is increased.
- Peripheral blood smear shows poikilocytosis (red blood cells of different shapes).

Treatment

- Iron replacement therapy is continued to correct the deficiency and replace the lost stores of iron in the body.
- The typical timeframe for oral therapy is to continue for 3 to 6 months after the anemia has been corrected.
- Oral replacement in split doses (three times a day):
  - Ferrous sulfate
  - Ferrous gluconate
- Parenteral iron replacement for those who cannot tolerate or do not respond to oral therapy, have gastrointestinal illness, or continued bleeding:
  - Iron dextran given deep IM or IV
  - Iron sodium gluconate given IV
- IM injection of iron using Z-track method.
- Anaphylactic reactions to iron dextran may occur; so a smaller test dose initially prior to the initiation of treatments.
- Increase dietary intake of iron.

Nursing intervention

- Monitor intake and output.
- Monitor vital signs for tachycardia or tachypnea.
- Monitor for reactions to parenteral iron therapy.

- Explain to the patient:
  - Check for bleeding.
  - Increase iron in diet.
  - Teach dietary sources of iron

Pernicious Anemia

- The body is unable to absorb Vitamin B12, which is needed to make RBC, resulting in a decreased RBC count.
- The intrinsic factor is normally secreted by the parietal cells of the gastric mucosa and are necessary to allow intestinal absorption of vitamin B12.
- Destruction of the gastric mucosa due to an autoimmune response results in loss of parietal cells within the stomach.
- The ability of vitamin B12 to bind with intrinsic factor is lost, decreasing the amount that is absorbed.
- Typical onset is between the ages of 40 and 60.

Signs and symptoms

- Pallor due to anemia
- Weakness and fatigue due to anemia
- Tingling in hands and feet— "stocking-glove paresthesia"—due to bilateral demyelination of dorsal and lateral columns of spinal cord nerves
- Diminished vibratory and position sense
- Poor balance due to effect on cerebral function
- Dementia appears later in the disease
- Atrophic glossitis—beefy red tongue
- Nausea may lead to anorexia and weight loss
- Premature graying of hair

Diagnostic evaluation

- Decreased hemoglobin due to decreased production of RBCs.
- Increased MCV—macrocytic anemia.
- Positive Schilling test due to decrease in intrinsic factor.
- Decreased amount of hydrochloric acid in the stomach (hypochlorhydria) due to changes within the parietal cells of the gastric mucosa.
- Positive Romberg test due to ataxia and neurologic changes.
- Diminished sensation when testing for vibration, position sense, or propioception of extremities.

Treatment

- Lifelong replacement with vitamin B12 will correct the anemia and improve the neurologic changes that have occurred.
- Initially the patient is given weekly injections of B12 to combat the deficiency.
- The injections eventually become monthly for lifelong maintenance.
- Oral supplementation is not effective in these patients because they cannot adequately absorb vitamin B12 due to insufficient intrinsic factor.
    - Administer vitamin B12 by IM injection.
    - Transfusion of packed RBC if anemia is severe.

Nursing intervention

- Prevent injuries.
- Explain to the patient:
    - Use soft toothbrush due to oral changes.
    - Avoid activities that could lead to injury due to paresthesias or changes in balance.
    - Inspect feet each day for injury due to paresthesia.

Disseminated Intravascular Coagulation (DIC)

- Blood coagulates through the entire body within the vascular compartment.
- This depletes platelets and the body's ability to coagulate, resulting in an increased risk of hemorrhage.
- It occurs as a complication of some other condition.
- The coagulation sequence is activated causing many microthrombi to develop throughout the body.
- The clots that form are the result of coagulation proteins and platelets, resulting in the risk of bleeding or severe hemorrhage.
- It is often due to obstetric complications, posttrauma, sepsis, cancer, or shock.

Signs and symptoms

- Unexpected bleeding—oozing from puncture sites (venipuncture, IVs, surgical wounds)
- Petechiae as clotting factors are lost
- Severe hemorrhage as clotting factors are lost
- Uncontrolled postpartum bleeding
- Tissue hypoxia from microemboli
- Hemolytic anemia, as cells are destroyed trying to pass through partially blocked vessels

Diagnostic evaluation

- PT and PTT—prolonged.
- Platelet count low—thrombocytopenia.
- Fibrin degradation products elevated:
    - D-dimer may be elevated.

Treatment

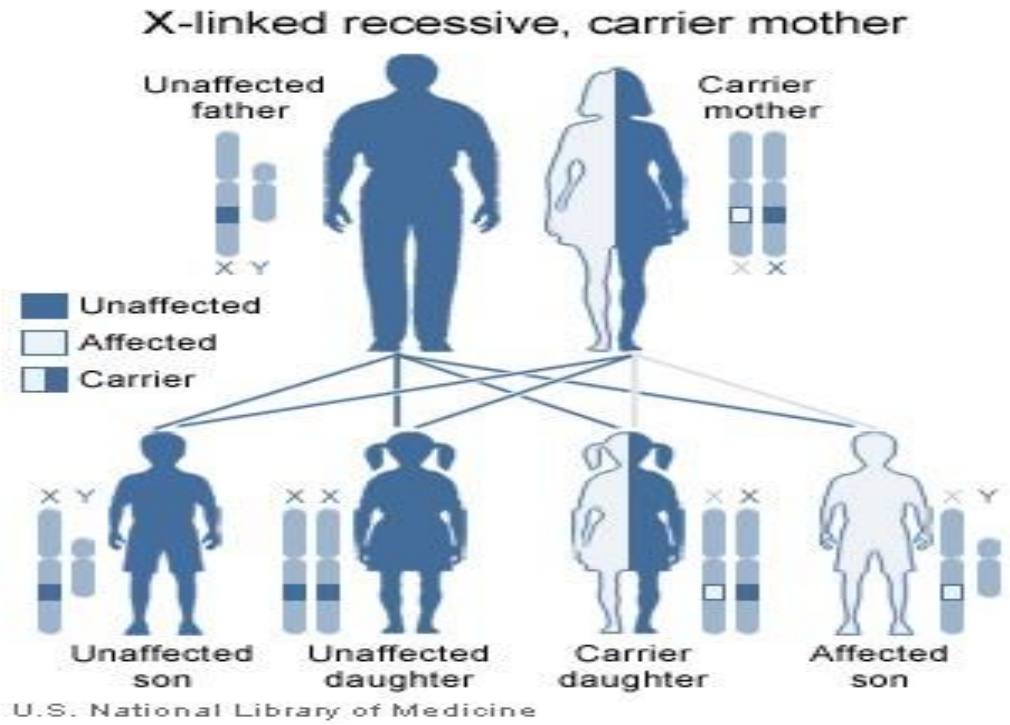
- Treatment needs to decrease coagulation ability (to prevent further clot development) and replace clotting components (to prevent further bleeding).
- Other interventions may be necessary depending on the locations of clot development and any compromise of body system function due to clot formation.
- Transfusion:
  - Packed RBC to replace what has been lost due to bleeding.
  - Fresh frozen plasma—replaces coagulation factor deficiency.
  - Platelets—replaces needed cells.
  - Cryoprecipitate—replaces fibrinogen.
- Administer anticoagulant drugs to decrease coagulation; not done in all patients: Heparin
- Bed rest.

Nursing intervention

- Monitor for bleeding from obvious sites (wounds, suture lines, venipuncture,etc.) and occult sites (GI, urine).
- Avoid cleaning clots from exposed areas—may start bleeding from the site and not have sufficient clotting factors to stop.
- Explain to the patient:
  - Avoid situations that might cause bleeding—use electric razor, soft toothbrush, don't floss between teeth.

Hemophilia

- The patient is missing a coagulation factor that is essential for normal blood clotting and as a result the blood does not clot when the patient bleeds.
- It is an X-linked recessive inherited disorder, passed on so that it presents symptoms in males, and rarely in females.
- Hemophilia A is the result of missing clotting factor VIII.
- Hemophilia B is the result of missing clotting factor IX and is also known as Christmas disease.



Signs and symptoms

- Tender joints due to bleeding
- Swelling of knees, ankles, hips, and elbows due to bleeding
- Blood in stool (tarry stool) due to GI blood loss
- Blood in the urine (hematuria)

Diagnostic evaluation

- PTT prolonged.
- Fibrinogen level normal.
- Decrease in clotting factor VIII found in blood serum in Hemophilia A.
- Decrease in clotting factor IX found in blood serum in Hemophilia B.

Treatment

- Avoid aspirin.
- For hemophilia A administer factor VIII concentrates.
- Cryoprecipitate.
- DDAVP (Desmopressin) which causes the release of factor VIII from the endothelial stores for patients with mild deficiency.
- For hemophilia B administer factor IX concentrates.

Nursing intervention

- No IM injections.
- No aspirin.
- To stop bleeding:
  - Elevate site.
  - Apply direct pressure to the site.
- Explain to the patient:
  - Wear a medical alert identification.
  - Contact physician for any injury.
  - Avoid situations where injury might occur.

Idiopathic Thrombocytopenic Purpura (ITP)

- An autoimmune disorder in which antibodies are developed to the patient's own platelets.
- Antibodies attach to the platelets and macrophages within the spleen.
- The body destroys the platelets within the spleen.
- ITP is typically more common in women and becomes chronic in adults who are in early to mid-adulthood.
- Problems for the patient are most likely the result of bleeding due to inadequate platelets.
- Prednisone can control the majority of cases of ITP.

Signs and symptoms

- Bleeding in mucous membranes or skin due to low platelet count:
  - Epistaxis
  - Oral bleeding
  - Menorrhagia (heavy menstrual bleeding)
  - Petechiae

Diagnostic evaluation

- Thrombocytopenia—low platelet count.
- Mild anemia—usually secondary to bleeding.
- PT normal.
- PTT normal.

Treatment

- The use of prednisone in patients with ITP is to decrease the body's action on the antibody-tagged platelets.
- Most patients will be on long-term maintenance doses of prednisone.
- Bleeding will stop even before platelet count begins to rise.
- Splenectomy provides complete or partial remission.
- High-dose IV immunoglobulin.

- Immunosuppressive therapy:
  - vincristine, azathioprine, cyclosporine
- Stem cell transplantation.

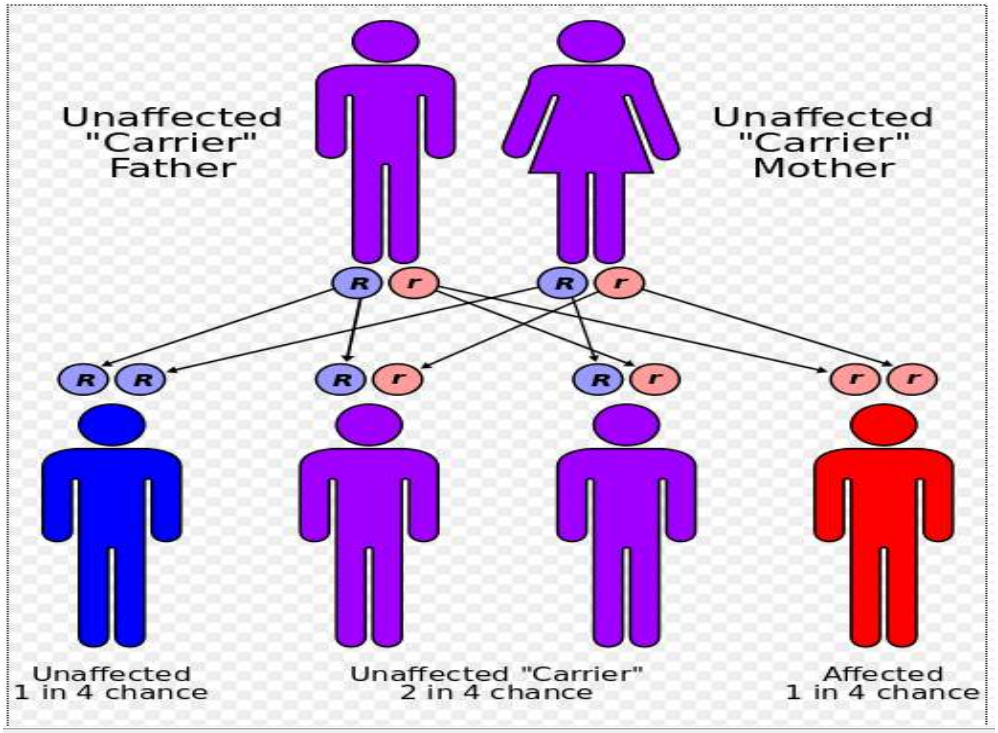
Nursing interventions

- Monitor vital signs for changes.
- Monitor for signs of bleeding or bruising.
- Decrease chance of bleeding:
  - Use soft toothbrushes, no flossing, only electric razors.
- Protect from potential infection, sick visitors, etc.
- Encourage patient to discuss feelings about illness.

Thalassemia

- The thalassemias are a group of hereditary anemias characterized by hypochromia,extreme microcytosis, destruction of blood elements (hemolysis), and variable degrees of anemia.
- The thalassemias occur worldwide, but the highest prevalence is found in people of Mediterranean, African, and Southeast Asian
- Thalassemias are associated with defective synthesis of the hemoglobin chain; the production of one or more globulin chains within the hemoglobin molecule is reduced.
- When this occurs, the imbalance in the configuration of the hemoglobin causes it to precipitate in the erythroid precursors or the erythrocytes themselves.
- This increases the rigidity of the erythrocytes and thus the premature destruction of these cells.
- Thalassemias are classified into two major groups according to which hemoglobin chain is diminished: alpha or beta.
- The alpha-thalassemias occur mainly in people from Asia and the Middle East

- The beta-thalassemias are most prevalent in people from Mediterranean regions but also occur in those from the Middle East or Asia.
- The alpha-thalassemias are milder than the beta forms and often occur without symptoms; the erythrocytes are extremely microcytic, but the anemia, if present, is mild.
- Bone marrow transplant (BMT) offers a chance of cure, but when this is not possible, the disease is usually treated with transfusion of PRBCs.
- Patients may survive into their 20s and 30s.



Sickle Cell Anemia

- This is an autosomal recessive disorder in which an abnormal gene causes damage to the RBC membrane.
- The abnormal hemoglobin within the red blood cell is called hemoglobin S.
- Dehydration or drying of the RBC makes it more vulnerable to sickling (forming a crescent-like shape), as do hypoxemia and acidosis.
- Hemolytic anemia results as RBCs are destroyed due to the damage to the outer membrane.
- The sickled cells can also clump together, causing difficulty getting through the smaller vessels.
- Sickle cell anemia will become a chronic multisystem disease.
- Causes of death in these patients are usually related to organ failure.
- Patients may also inherit a single gene for sickle cell.
- These patients may develop sickle cell trait, in which symptoms would only be present in the setting of extreme circumstances (vigorous exercise at high altitude, especially with rapid ascent).

Signs and symptoms

- Acute pain (especially back, chest, and long bones) from vascular occlusion of the small vessels as the sickled cells clump.
- Fever as body responds to acute sickling episode and accompanying provoking event.
- Painful, swollen joints due to vaso-occlusive process.
- Fatigue due to chronic anemia.
- Stroke (cerebrovascular accident) due to vaso-occlusive process.
- Enlarged liver (hepatomegaly).
- Enlarged heart and systolic murmur.

Diagnostic evaluation

- Low RBC count due to chronic hemolytic anemia; the RBCs have a shorter lifespan.
- Elevated WBCs.
- Increased reticulocytes.
- Presence of Howell-Jolly bodies and target cells.
- Sickle cells appear in blood smear.
- Indirect bilirubin level elevated.
- Hemoglobin electrophoresis shows majority hemoglobin S (80 to 98 percent).

Treatment

- During acute episodes pain control, hydration, and oxygenation are the focus of treatment.
- The underlying cause that sent the patient into crisis will also need to be treated concurrently.
- Administer analgesics to alleviate the pain associated with the vaso-occlusive process:
  - narcotic pain control necessary when pain is severe
- Warm compresses on joint.
- Blood transfusion of packed RBC when anemia indicates.
- Supplemental oxygen if hypoxic.
- Adequate hydration, using IV fluids.
- Treat infections.

Nursing intervention

- Increase fluid intake.
- Monitor IV fluids.
- Monitor pain control.
- Record fluid intake and output to monitor renal function.
- Administer supplemental O2 to increase available oxygen.

- Explain to the patient:
  - Avoid the cold.
  - No cold compresses.
  - Plan for rest periods during the day.

Leukemia

- Replacement of bone marrow by abnormal cells results in unregulated proliferation of immature white blood cells entering the circulatory system.
- These leukemic cells may also enter the liver, spleen, or lymph nodes, causing these areas to enlarge.
- Leukemia is classified according to the type of cell it is derived from, lymphocytic or myelocytic, and as either acute or chronic.
- Lymphocytic leukemias involve immature lymphocytes originating in the bone marrow and typically infiltrate the spleen, lymph nodes, or central nervous system.
- Myelogenous or myelocytic leukemia involves the myeloid stem cells in the bone marrow and interferes with the maturation of all blood cell types (granulocytes, erythrocytes, thrombocytes).
- The exact cause of leukemia is unknown.
- There is a higher incidence in:
  - People who have been exposed to high levels of radiation
  - Who have had exposure to benzene
  - Who have a history of aggressive chemotherapy
  - Genetic predisposition
  - Down's syndrome
  - Fanconi's anemia
  - Family history of leukemia

Signs and symptoms

- **Acute patients:**
  - Fatigue and weakness due to anemia
  - Fever due to increased susceptibility to infection
  - Bleeding, petechiae, ecchymosis (bruising), epistaxis (nosebleed), gingival (gum) bleeding—due to decreased platelet count
  - Bone pain due to bone infiltration and marrow expansion
  - Lymph nodes (lymphadenopathy) enlarged as leukemic cells invade nodes
  - Liver (hepatomegaly) and spleen (splenomegaly) enlarged as leukemic cells invade
  - Headache, nausea, vomiting, and weight loss
  - Papilledema, cranial nerve palsies, seizure if there is central nervous system involvement
- **Chronic patients:**
  - Fatigue due to anemia
  - Weight loss due to chronic disease process and loss of appetite
  - Poor appetite
  - Enlarged lymph nodes (lymphadenopathy) due to infiltration of lymph nodes
  - Enlarged spleen (splenomegaly) due to involvement of the spleen

Diagnostic evaluation

- Low RBC count, low hemoglobin—anemia.
- Low platelet count—thrombocytopenia.
- Elevated WBC count—leukocytosis.
- Abnormal amount of immature WBC shown in bone marrow biopsy.

Treatment

- Chemotherapy
- Administer antibiotics for infections.

- Bone marrow transplant.
  - Administer immunosuppressives to avoid transplant rejection.
- Transfusion if hemolytic anemia or bleeding:
  - Packed RBCs.
  - Whole blood.
  - Platelets.
- High protein diet.

Nursing intervention

- Monitor for bleeding—platelet count may be decreased.
- Monitor for infection—patients have increased susceptibility to infection.
- Monitor pain control.
- Small, frequent meals.
- Teach patients about infection control:
  - Avoid others with infection.
  - Report signs of infection, sore throat, fevers, etc.
- Explain to the patient:
  - Use an electric razor.
  - Use soft toothbrush.
  - Watch for bleeding or bruising.